PROBLEM STATEMENT

The project problem is struggling to understand and predict the consumer buying pattern due to different customer preferences. and product choosen by customer.

DATASET DESCRIPTION

Dataset Information	Description
Dataset Name	Multiclass Clothing Sales Dataset
Format	Csv
Type	Mixed (Nominal 9 & Numeric 11)
Instance	100 000
Attribute	20
Missing Value	Yes
Method Use	Classification
Target Category	Sales Category (Low ,Medium ,High)

DATA SOURCE DETAILS: HTTPS://WWW.KAGGLE.COM/DATASETS/RAYZEM/DYNAMIC-APPAREL-SALES-DATASET-WITH-ANOMALIES/DATA

A2. DATA PREPARATION – SELECTION AND CLEANING

1. NUMERIC ATTRIBUTE

Attribute Name	Statistic	Value
Selling_price	Minimum	-613.616
	Maximum	19989.216
	Mean	1554.03
	Standard Devition	974.709
Cost_Price	Minimum	-726.935
	Maximum	14992.757
	Mean	1041.472
	Standard Devition	734.619
Discount_Percentage	Minimum	0
	Maximum	50
	Mean	24.994
	Standard Devition	14.431
Quantity_Sold	Minimum	1
	Maximum	199
	Mean	5.349
	Standard Devition	7.215

Attribute Name	Statistic	Value
Total_Sales	Minimum	-37155.409
	Maximum	141640.192
	Mean	50051.437
	Standard Devition	20076.412
Stock_Availability	Minimum	0
	Maximum	499
	Mean	250.33
	Standard Devition	144.473
Customer_Age	Minimum	18
	Maximum	64
	Mean	41.067
	Standard Devition	13.567
Purchase_Frequency	Minimum	-2.096
	Maximum	49
	Mean	2.081
	Standard Devition	1.897

Attribute Name	Statistic	Value
Store_Rating	Minimum	1.002
	Maximum	5
	Mean	3.995
	Standard Devition	0.586
Return_Rate	Minimum	0
	Maximum	29.999
	Mean	14.957
	Standard Devition	8.652

2. NOMINAL ATTRIBUTE

Attribute Name	Statistic	Value
Product_Category	Traditional Wear	19981
	Athleisure	20187
	Outerwear	19732
	Bottoms	20082
	Tops	20018

2. NOMINAL ATTRIBUTE

Attribute Name	Statistic	Value
Brand	Forever 21	5075
	Ralph Lauren	5102
	Nike	5045
	Zara	5007
	Wrangleer	5134
	Balenciaga	4851
	Uniqlo	4985
	Reebok	4922
	Gucci	4953
	Louis Vuitton	4980
	Tommy Hilfiger	4997
	Under Armour	4993
	Calvin Klein	4994
	Puma	4877
	Diesel	4980
	Versace	5010
	Levia	5036
	Adidas	5011
	H&M	5064
	GAP	4984

2. NOMINAL ATTRIBUTE

Attribute Name	Statistic	Value
Product_Name	Tops	5054
	casual Shirt	5050
	Blazer	4945
	Leggings	5025
	Dress	5090
	Jeans	4940
	Hoodie	4915
	Jumpsuit	4989
	Trousers	5019
	Jacket	5062
	Skirt	4859
	T-shirt	4969
	Saree	5038
	Kurta	5029
	Shorts	5049
	Polo Shirt	4991
	Palazzo	5062
	Sweatshirt	4932
	Formal Shirts	5065
	Track Pants	4917

2. NOMINAL ATTRIBUTE

Attribute Name	Statistic	Value
Gender	Women	49825
	Men	50175
Size	S	19953
	XXL	20037
	M	19983
	XL	19963
	L	20064

2. NOMINAL ATTRIBUTE

Attribute Name	Statistic	Value
Color	White	11272
	Yellow	10921
	Green	11200
	Orange	11090
	Purple	10994
	Black	11199
	Pink	11002
	Red	11245
	Blue	11077
Season	winter	33488
	All-Season	33225
	Summer	33287

2. NOMINAL ATTRIBUTE

Attribute Name	Statistic	Value
Payment_method	Card	25144
	UPI	24957
	Net Banking	24810
	Cash	25089
Customer_Type	New	50255
	Returning	49745

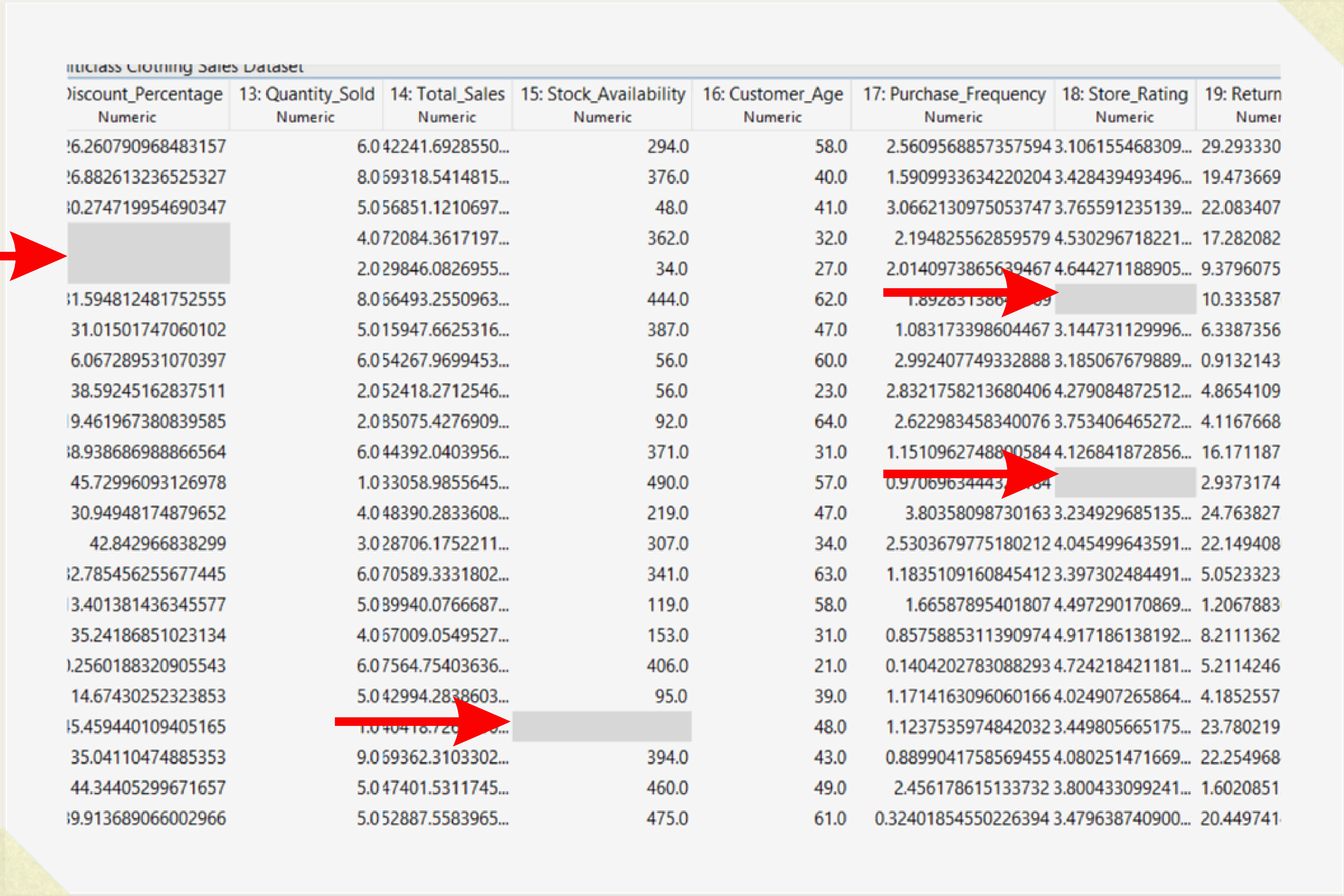
Why we need to remove Missing values?

Based on the dataset can identify there is the some attributes that have missing values:

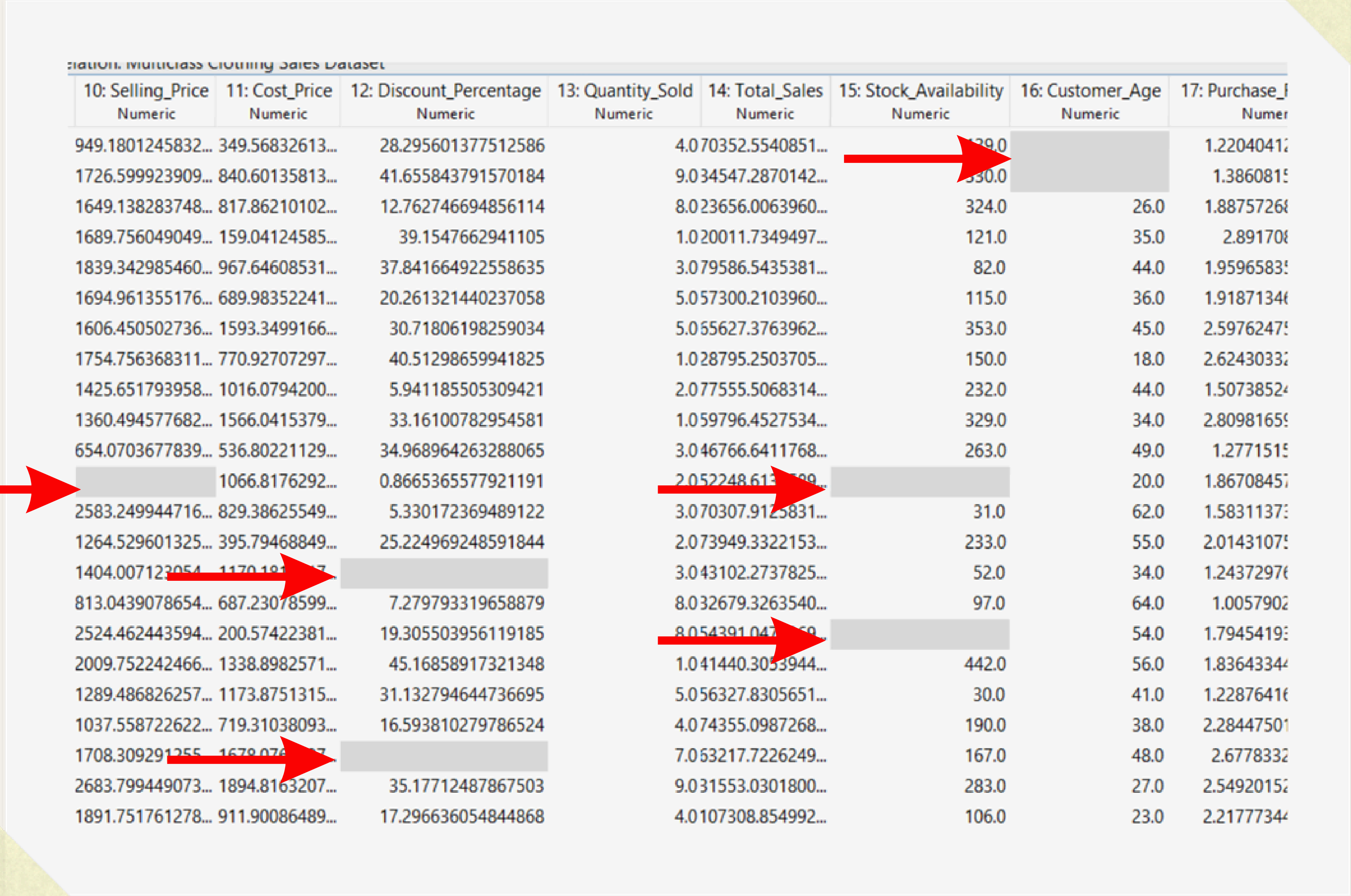
- Selling_Price
- Discount_Percentage
- Stock_Availability
- Customer_Age
- Store_Rating

Meaning that this dataset has problem incomplete dataset that need to apply filter missing value

There is some attributes that have the missing value



iscount_Percentage Numeric	13: Quantity_Sold Numeric	14: Total_Sales Numeric	15: Stock_Availability Numeric	16: Customer_Age Numeric	17: Purchase_Frequency Numeric	18: Store_Rating Numeric	19: Return Numer
6.260790968483157	6.0	42241.6928550...	294.0	58.0	2.5609568857357594	3.106155468309...	29.293330
6.882613236525327	8.0	69318.5414815...	376.0	40.0	1.5909933634220204	3.428439493496...	19.473669
0.274719954690347	5.0	56851.1210697...	48.0	41.0	3.0662130975053747	3.765591235139...	22.083407
	4.0	72084.3617197...	362.0	32.0	2.194825562859579	4.530296718221...	17.282082
	2.0	29846.0826955...	34.0	27.0	2.0140973865639467	4.644271188905...	9.3796075
1.594812481752555	8.0	66493.2550963...	444.0	62.0	[illegible]		10.333587
31.01501747060102	5.0	15947.6625316...	387.0	47.0	1.083173398604467	3.144731129996...	6.3387356
6.067289531070397	6.0	54267.9699453...	56.0	60.0	2.992407749332888	3.185067679889...	0.9132143
38.59245162837511	2.0	52418.2712546...	56.0	23.0	2.8321758213680406	4.279084872512...	4.8654109
9.461967380839585	2.0	35075.4276909...	92.0	64.0	2.622983458340076	3.753406465272...	4.1167668
8.938686988866564	6.0	44392.0403956...	371.0	31.0	1.1510962748800584	4.126841872856...	16.171187
45.72996093126978	1.0	33058.9855645...	490.0	57.0	[illegible]		2.9373174
30.94948174879652	4.0	48390.2833608...	219.0	47.0	3.80358098730163	3.234929685135...	24.763827
42.842966838299	3.0	28706.1752211...	307.0	34.0	2.5303679775180212	4.045499643591...	22.149408
2.785456255677445	6.0	70589.3331802...	341.0	63.0	1.1835109160845412	3.397302484491...	5.0523323
3.401381436345577	5.0	39940.0766687...	119.0	58.0	1.66587895401807	4.497290170869...	1.2067883
35.24186851023134	4.0	57009.0549527...	153.0	31.0	0.8575885311390974	4.917186138192...	8.2111362
.2560188320905543	6.0	7564.75403636...	406.0	21.0	0.1404202783088293	4.724218421181...	5.2114246
14.67430252323853	5.0	42994.2838603...	95.0	39.0	1.1714163096060166	4.024907265864...	4.1852557
5.459440109405165	[illegible]	[illegible]		48.0	1.1237535974842032	3.449805665175...	23.780219
35.04110474885353	9.0	69362.3103302...	394.0	43.0	0.8899041758569455	4.080251471669...	22.254968
44.34405299671657	5.0	47401.5311745...	460.0	49.0	2.456178615133732	3.800433099241...	1.6020851
9.913689066002966	5.0	52887.5583965...	475.0	61.0	0.32401854550226394	3.479638740900...	20.449741



10: Selling_Price Numeric	11: Cost_Price Numeric	12: Discount_Percentage Numeric	13: Quantity_Sold Numeric	14: Total_Sales Numeric	15: Stock_Availability Numeric	16: Customer_Age Numeric	17: Purchase_ Numer
949.1801245832...	349.56832613...	28.295601377512586	4.0	70352.5540851...	[illegible]		1.2204041
1726.599923909...	840.60135813...	41.655843791570184	9.0	34547.2870142...	[illegible]		1.386081
1649.138283748...	817.86210102...	12.762746694856114	8.0	23656.0063960...	324.0	26.0	1.8875726
1689.756049049...	159.04124585...	39.1547662941105	1.0	20011.7349497...	121.0	35.0	2.891708
1839.342985460...	967.64608531...	37.841664922558635	3.0	79586.5435381...	82.0	44.0	1.9596583
1694.961355176...	689.98352241...	20.261321440237058	5.0	57300.2103960...	115.0	36.0	1.9187134
1606.450502736...	1593.3499166...	30.71806198259034	5.0	55627.3763962...	353.0	45.0	2.5976247
1754.756368311...	770.92707297...	40.51298659941825	1.0	28795.2503705...	150.0	18.0	2.6243033
1425.651793958...	1016.0794200...	5.941185505309421	2.0	77555.5068314...	232.0	44.0	1.5073852
1360.494577682...	1566.0415379...	33.16100782954581	1.0	59796.4527534...	329.0	34.0	2.8098165
654.0703677839...	536.80221129...	34.968964263288065	3.0	46766.6411768...	263.0	49.0	1.2771515
	1066.8176292...	0.8665365577921191	2.0	[illegible]		20.0	1.8670845
2583.249944716...	829.38625549...	5.330172369489122	3.0	70307.9125831...	31.0	62.0	1.5831137
1264.529601325...	395.79468849...	25.224969248591844	2.0	73949.3322153...	233.0	55.0	2.0143107
[illegible]	[illegible]		3.0	43102.2737825...	52.0	34.0	1.2437297
813.0439078654...	687.23078599...	7.279793319658879	8.0	32679.3263540...	97.0	64.0	1.0057902
2524.462443594...	200.57422381...	19.305503956119185	8.0	[illegible]		54.0	1.7945419
2009.752242466...	1338.8982571...	45.16858917321348	1.0	41440.3053944...	442.0	56.0	1.8364334
1289.486826257...	1173.8751315...	31.132794644736695	5.0	56327.8305651...	30.0	41.0	1.2287641
1037.558722622...	719.31038093...	16.593810279786524	4.0	74355.0987268...	190.0	38.0	2.2844750
[illegible]	[illegible]		7.0	63217.7226249...	167.0	48.0	2.6778332
2683.799449073...	1894.8163207...	35.17712487867503	9.0	31553.0301800...	283.0	27.0	2.5492015
1891.751761278...	911.90086489...	17.296636054844868	4.0	107308.854992...	106.0	23.0	2.2177734

Missing Value

Selling_price

Selected attribute

Name: Selling_Price		Type: Numeric
Missing: 5000 (5%)	Distinct: 95000	Unique: 95000 (95%)

Discount_Percentage

Selected attribute

Name: Discount_Percentage		Type: Numeric
Missing: 5000 (5%)	Distinct: 95000	Unique: 95000 (95%)

Stock_Availability

Selected attribute

Name: Stock_Availability		Type: Numeric
Missing: 5000 (5%)	Distinct: 500	Unique: 0 (0%)

Customer_Age

Selected attribute

Name: Customer_Age		Type: Numeric
Missing: 5000 (5%)	Distinct: 47	Unique: 0 (0%)

Store_Rating

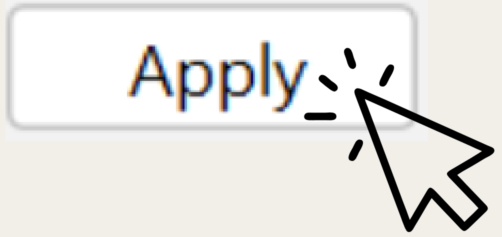
Selected attribute

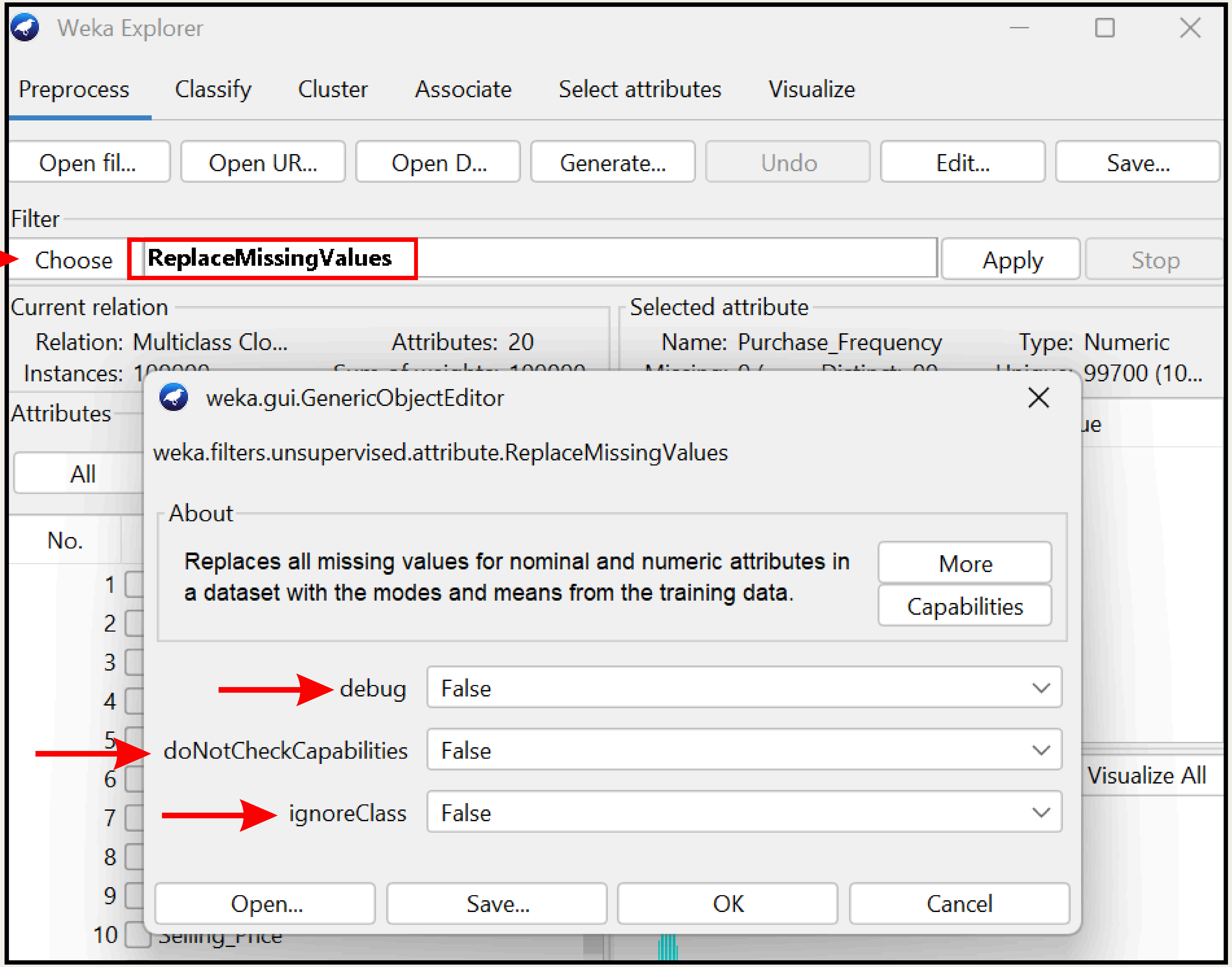
Name: Store_Rating		Type: Numeric
Missing: 5000 (5%)	Distinct: 95000	Unique: 95000 (95%)

There are missing value in some attributes. For example, Selling_Price, Discount_Percentage, Stock_Availability, Customer_age, and Store_Rating

How to Apply filter

- In Weka, go to **preprocess** and choose **unsupervised › attribute › ReplaceMissingValues.** Then choose the default setting, which is **debug = “False,” doNotCheckCapabilities = “False,”** and **ignoreClass = “False,”** then click **“OK”** and **”Apply”**.





AFTER APPLY FILTER REPLACEMISSINGVALUE

ALL THE MISSING VALUE FOR THE ALL ATTRIBUTES HAS BEEN REPLACE

SELLING_PRICE

Selected attribute

Name: Selling_Price		Type: Numeric
Missing: 0 (0%)	Distinct: 95001	Unique: 95000 (95%)

DISCOUNT_PERCENTAGE

Selected attribute

Name: Discount_Percentage		Type: Numeric
Missing: 0 (0%)	Distinct: 95001	Unique: 95000 (95%)

STOCK_AVAILABILITY

Selected attribute

Name: Stock_Availability		Type: Numeric
Missing: 0 (0%)	Distinct: 501	Unique: 0 (0%)

CUSTOMER_AGE

Selected attribute

Name: Customer_Age		Type: Numeric
Missing: 0 (0%)	Distinct: 48	Unique: 0 (0%)

STORE_RATING

Selected attribute

Name: Store_Rating		Type: Numeric
Missing: 0 (0%)	Distinct: 95001	Unique: 95000 (95%)

This the attributes after apply the filter ReplaceMissingValues

Before

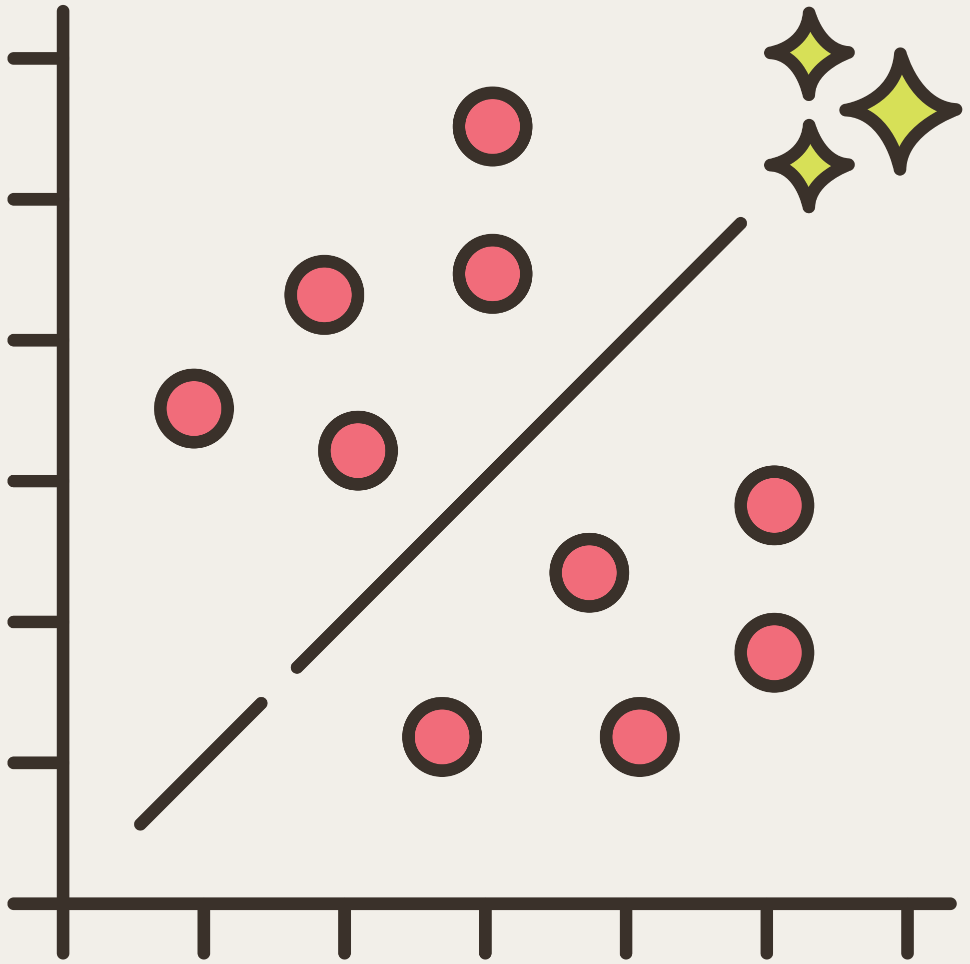
12: Discount_Percentage Numeric	13: Quantity_Sold Numeric	14: Total_Sales Numeric	15: Stock_Availability Numeric	16: Customer_Age Numeric	17: Purchase_Frequency Numeric	18: Store_Rating Numeric
26.260790968483157	6.0	42241.6928550...	294.0	58.0	2.5609568857357594	3.106155468309...
26.882613236525327	8.0	69318.5414815...	376.0	40.0	1.5909933634220204	3.428439493496...
30.274719954690347	5.0	56851.1210697...	48.0	41.0	3.0662130975053747	3.765591235139...
	4.0	72084.3617197...	362.0	32.0	2.194825562859579	4.530296718221...
	2.0	29846.0826955...	34.0	27.0	2.0140973865639467	4.644271188905...
31.594812481752555	8.0	66493.2550963...	444.0	62.0	1.89283138641569	
31.01501747060102	5.0	15947.6625316...	387.0	47.0	1.083173398604467	3.144731129996...
6.067289531070397	6.0	54267.9699453...	56.0	60.0	2.992407749332888	3.185067679889...
38.59245162837511	2.0	52418.2712546...	56.0	23.0	2.8321758213680406	4.279084872512...
19.461967380839585	2.0	85075.4276909...	92.0	64.0	2.622983458340076	3.753406465272...
38.938686988866564	6.0	44392.0403956...	371.0	31.0	1.1510962748800584	4.126841872856...
45.72996093126978	1.0	33058.9855645...	490.0	57.0	0.9706963444326184	
30.94948174879652	4.0	48390.2833608...	219.0	47.0	3.80358098730163	3.234929685135...
42.842966838299	3.0	28706.1752211...	307.0	34.0	2.5303679775180212	4.045499643591...
32.785456255677445	6.0	70589.3331802...	341.0	63.0	1.1835109160845412	3.397302484491...
13.401381436345577	5.0	89940.0766687...	119.0	58.0	1.66587895401807	4.497290170869...
35.24186851023134	4.0	67009.0549527...	153.0	31.0	0.8575885311390974	4.917186138192...
0.2560188320905543	6.0	7564.75403636...	406.0	21.0	0.1404202783088293	4.724218421181...
14.67430252323853	5.0	42994.2838603...	95.0	39.0	1.1714163096060166	4.024907265864...
45.459440109405165	1.0	[illegible]		48.0	1.1237535974842032	3.449805665175...

After

12: Discount_Percentage Numeric	13: Quantity_Sold Numeric	14: Total_Sales Numeric	15: Stock_Availability Numeric	16: Customer_Age Numeric	17: Purchase_Frequency Numeric	18: Store_Rating Numeric
26.260790968483157	6.0	42241.6928550...	294.0	58.0	2.5609568857357594	3.106155468309...
26.882613236525327	8.0	69318.5414815...	376.0	40.0	1.5909933634220204	3.428439493496...
30.274719954690347	5.0	56851.1210697...	48.0	41.0	3.0662130975053747	3.765591235139...
24.99438861981897	4.0	72084.3617197...	362.0	32.0	2.194825562859579	4.530296718221...
24.99438861981897	2.0	29846.0826955...	34.0	27.0	2.0140973865639467	4.644271188905...
31.594812481752555	8.0	66493.2550963...	444.0	62.0	1.89283138641569	3.995121038040...
31.01501747060102	5.0	15947.6625316...	387.0	47.0	1.083173398604467	3.144731129996...
6.067289531070397	6.0	54267.9699453...	56.0	60.0	2.992407749332888	3.185067679889...
38.59245162837511	2.0	52418.2712546...	56.0	23.0	2.8321758213680406	4.279084872512...
19.461967380839585	2.0	35075.4276909...	92.0	64.0	2.622983458340076	3.753406465272...
38.938686988866564	6.0	44392.0403956...	371.0	31.0	1.1510962748800584	4.126841872856...
45.72996093126978	1.0	33058.9855645...	490.0	57.0	0.9706963444326184	3.995121038040...
30.94948174879652	4.0	48390.2833608...	219.0	47.0	3.80358098730163	3.234929685135...
42.842966838299	3.0	28706.1752211...	307.0	34.0	2.5303679775180212	4.045499643591...
32.785456255677445	6.0	70589.3331802...	341.0	63.0	1.1835109160845412	3.397302484491...
13.401381436345577	5.0	39940.0766687...	119.0	58.0	1.66587895401807	4.497290170869...
35.24186851023134	4.0	57009.0549527...	153.0	31.0	0.8575885311390974	4.917186138192...
0.2560188320905543	6.0	7564.75403636...	406.0	21.0	0.1404202783088293	4.724218421181...
14.67430252323853	5.0	42994.2838603...	95.0	39.0	1.1714163096060166	4.024907265864...
45.459440109405165	1.0	40418.7262010...	250.32984210526317	48.0	1.1237535974842032	3.449805665175...

- For a **numeric attributes** it will replace all the missing value in the column with the mean(average)
- The mean value for Discount_Percentage =24.99438861981897 and Store_Rating = 3.995121038040468
- For a **nominal attributes** (Categorical) it will replace the missing values with the mode value (most frequent value)

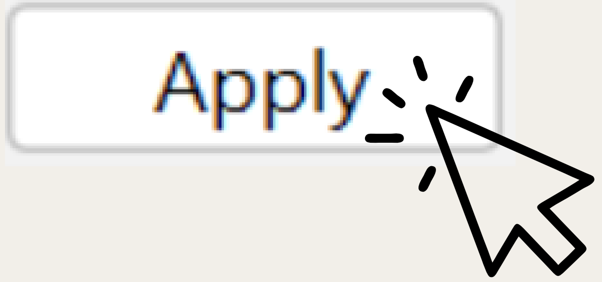
Why we need to remove Outliers & Extreme Values

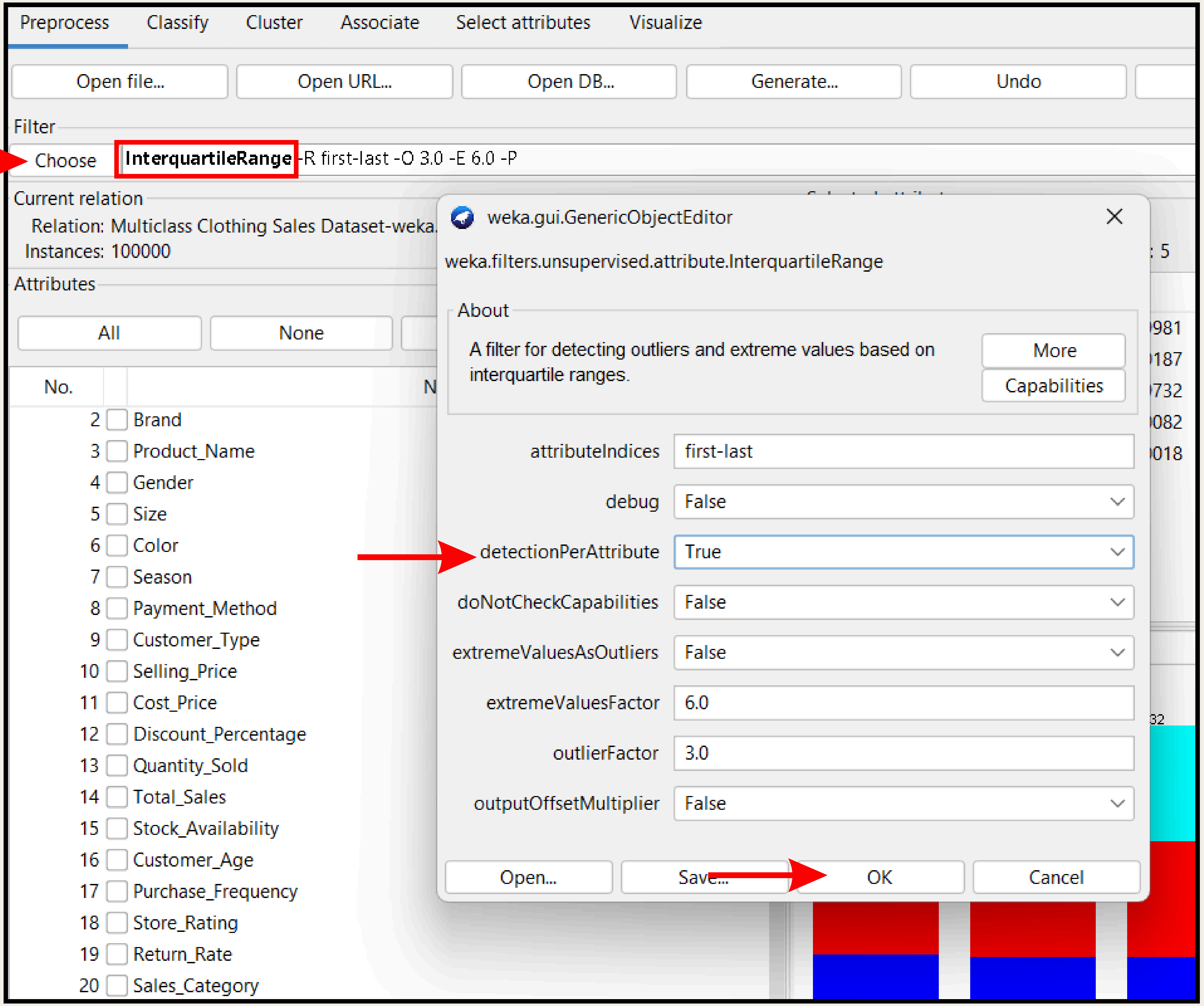


- Unrealistic Selling Prices: From the dataset, some attributes record contains unusually high or negative value in the dataset. For example, selling_prices have record that exceed RM19 000 and some below RM0 This can lead to incorrect conclusion about the pricing trends.

How to apply Filter InterquartileRange for Each Attributes

- In Weka, go to **preprocess** and **choose › unsupervised › attribute › InterquartileRange.** Change the **detectionPerAttribute = "TRUE"**. This changes will output the outlier for each attributes. Then click **"OK"** and **"Apply"**





Outliers for each attributes

Selling_Price_Outlier

Selected attribute

Name: Selling_Price_Outlier Type: Nominal
Missing: 0 (0%) Distinct: 2 Unique: 0 (0%)

No.	Label	Count	Weight
1	no	99981	99981
2	yes	19	19

Cost_Price_Outlier

Selected attribute

Name: Cost_Price_Outlier Type: Nominal
Missing: 0 (0%) Distinct: 2 Unique: 0 (0%)

No.	Label	Count	Weight
1	no	99935	99935
2	yes	65	65

Quantity_Sold_Outlier

Selected attribute

Name: Quantity_Sold_Outlier Type: Nominal
Missing: 0 (0%) Distinct: 1 Unique: 0 (0%)

No.	Label	Count	Weight
1	no	100000	100000
2	yes	0	0

Purchase_Frequency_Outlier

Selected attribute

Name: Purchase_Frequency_Outlier Type: Nominal
Missing: 0 (0%) Distinct: 2 Unique: 0 (0%)

No.	Label	Count	Weight
1	no	99995	99995
2	yes	5	5

Extreme Value for each attributes

Selling_Price_ExtremeValue

Selected attribute

Name: Selling_Price_ExtremeValue Type: Nominal
Missing: 0 (0%) Distinct: 2 Unique: 0 (0%)

No.	Label	Count	Weight
1	no	99536	99536
2	yes	464	464

Cost_Price_ExtremeValue

Selected attribute

Name: Cost_Price_ExtremeValue Type: Nominal
Missing: 0 (0%) Distinct: 2 Unique: 0 (0%)

No.	Label	Count	Weight
1	no	99565	99565
2	yes	435	435

Quantity_Sold_ExtremeValue

Selected attribute

Name: Quantity_Sold_ExtremeValue Type: Nominal
Missing: 0 (0%) Distinct: 2 Unique: 0 (0%)

No.	Label	Count	Weight
1	no	99700	99700
2	yes	300	300

Purchase_Frequency_ExtremeValue

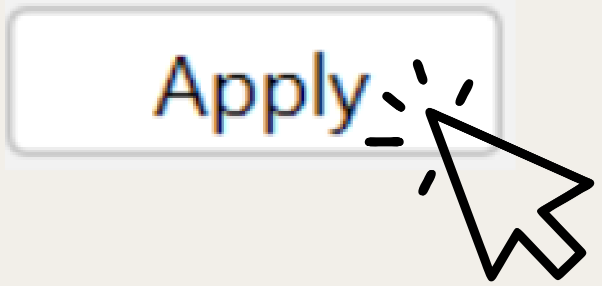
Selected attribute

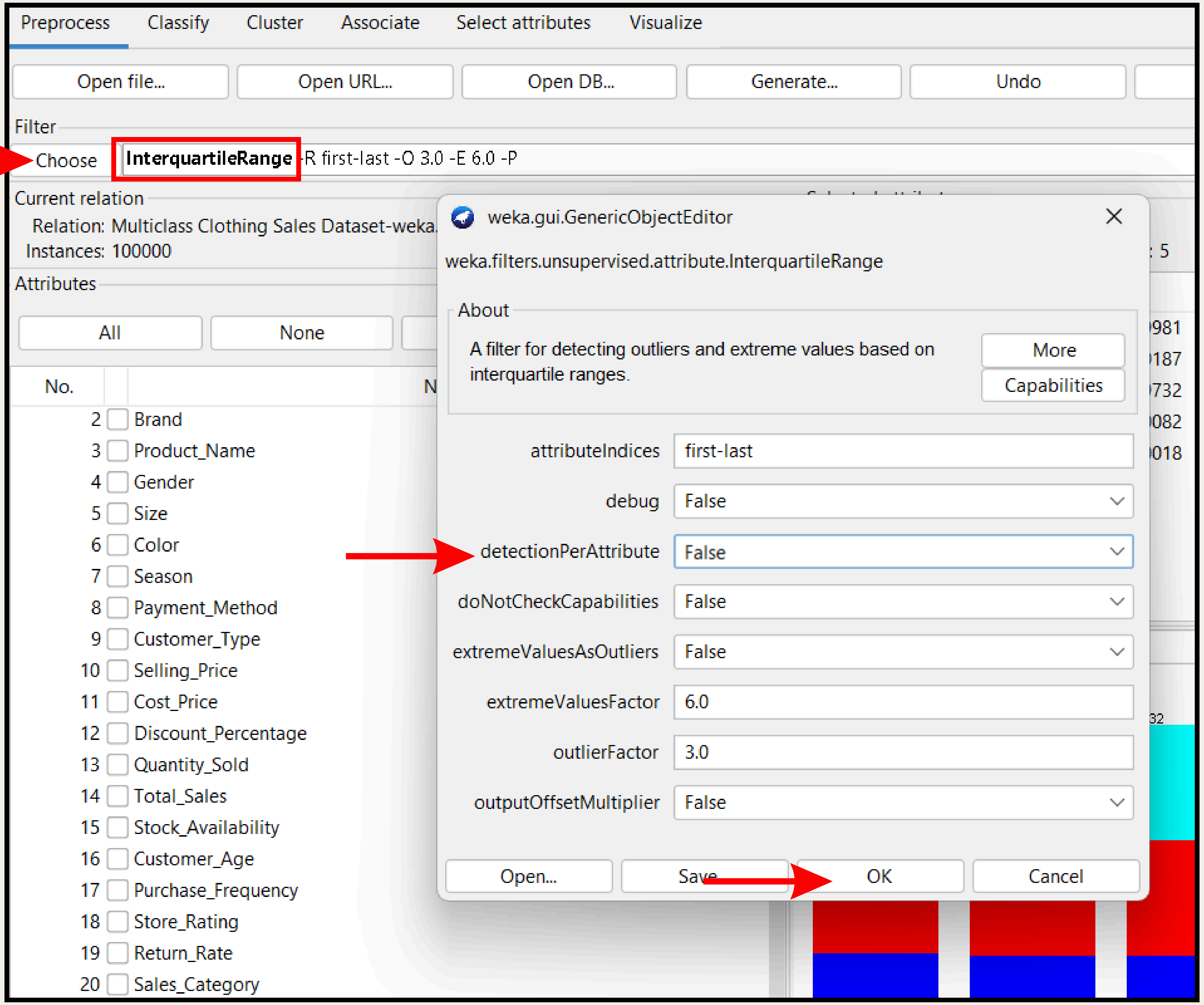
Name: Purchase_Frequency_ExtremeValue Type: Nominal
Missing: 0 (0%) Distinct: 2 Unique: 0 (0%)

No.	Label	Count	Weight
1	no	99705	99705
2	yes	295	295

How to apply Filter InterquartileRange for Overall Attributes

- In Weka, go to **preprocess** and **choose › unsupervised › attribute › InterquartileRange**. Change the **detectionPerAttribute = "FALSE"**. This changes will output the outlier for each attributes. Then click **"OK"** and **"Apply"**





Outliers and Extreme value for overall attributes

Outliers (21st attribute)

Selected attribute

Name: Outlier Type: Nominal

Missing: 0 (0%) Distinct: 2 Unique: 0 (0%)

No.	Label	Count	Weight
1	no	99914	99914
2	yes	86	86

Extereme Value (22st attribute)

Selected attribute

Name: ExtremeValue Type: Nominal

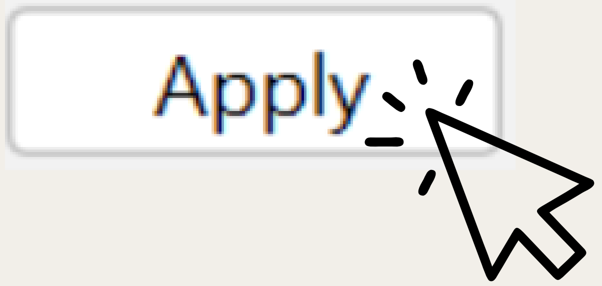
Missing: 0 (0%) Distinct: 2 Unique: 0 (0%)

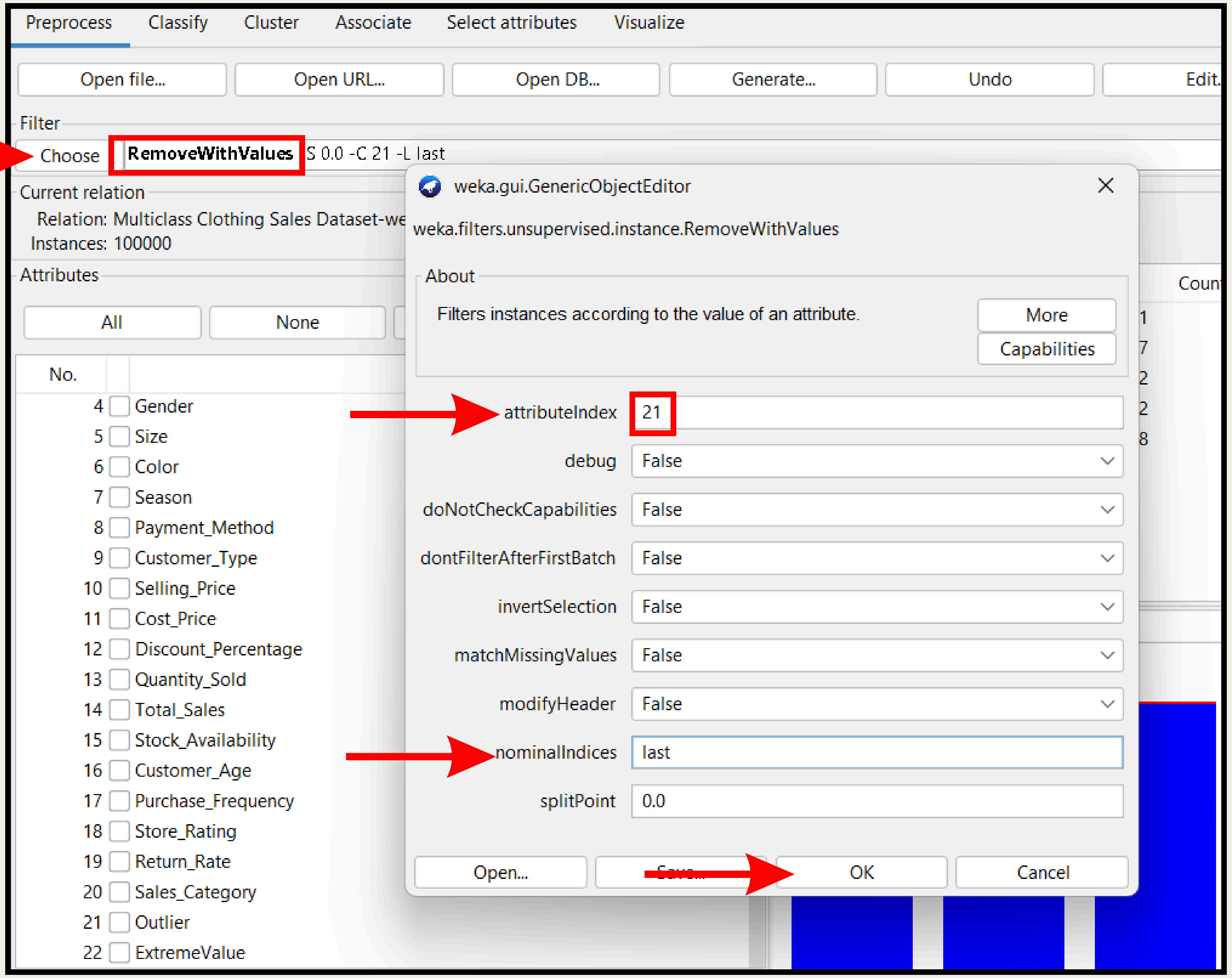
No.	Label	Count	Weight
1	no	99501	99501
2	yes	499	499

Apply filter RemoveWithValues to remove the Outliers and Extreme Values

Remove the Outliers

- Go to Preprocess > Choose > Filters > unsupervised > instance > RemoveWithValues and click the default setting.
- Change the attributeIndex = 21
- Change the nominalIndices = Last
- Click "OK" and "Apply"

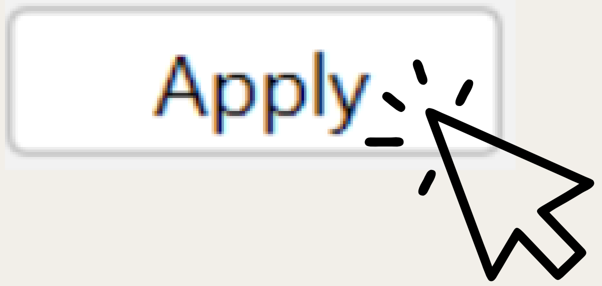


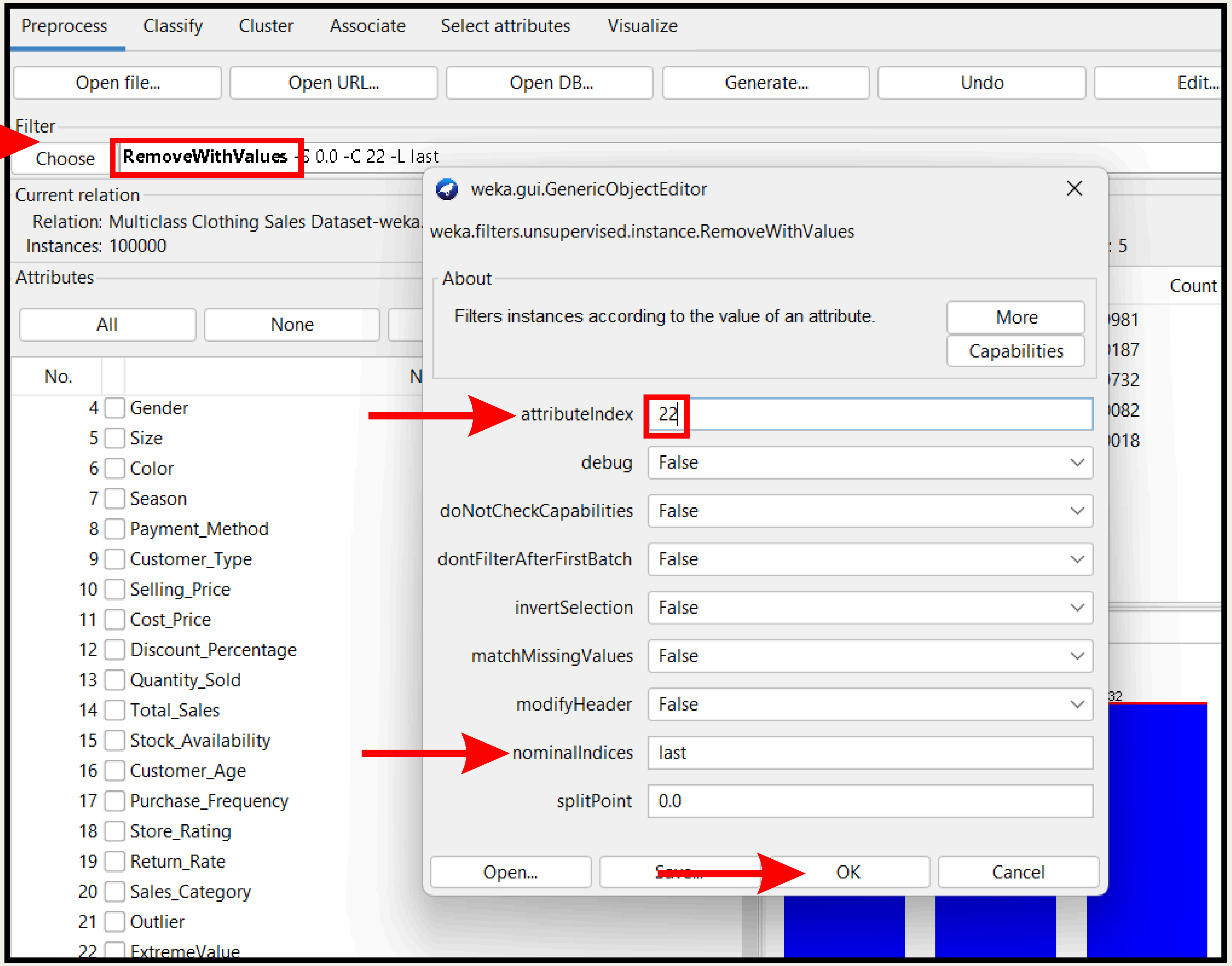


Apply filter RemoveWithValues to remove the Outliers and Extreme Values

Remove the Extreme Value

- Go to Preprocess > Choose > Filters > unsupervised > instance > RemoveWithValues and click the default setting.
- Change the attributeIndex = 22
- Change the nominalIndices = Last
- Click "OK" and "Apply"





Before and After Apply filter RemoveWithValue

Outlier & Extreme Value

Before

Selected attribute
Name: Outlier — Type: Nominal
Missing: 0 (0%) — Distinct: 2 — Unique: 0 (0%)

No.	Label	Count	Weight
1	no	99914	99914
2	yes	86	86

After

Selected attribute
Name: Outlier — Type: Nominal
Missing: 0 (0%) — Distinct: 1 — Unique: 0 (0%)

No.	Label	Count	Weight
1	no	99499	99499
2	yes	0	0

Before

Selected attribute
Name: ExtremeValue — Type: Nominal
Missing: 0 (0%) — Distinct: 2 — Unique: 0 (0%)

No.	Label	Count	Weight
1	no	99501	99501
2	yes	499	499

After

Selected attribute
Name: ExtremeValue — Type: Nominal
Missing: 0 (0%) — Distinct: 1 — Unique: 0 (0%)

No.	Label	Count	Weight
1	no	99499	99499
2	yes	0	0

After Remove Outliers and Extreme Values From the Dataset

Improve Data Quality
After outlier removal, average **Selling_Price, Total_Sales**, and **Quantity_Sold** become **more representative** of **typical transactions**, allowing for **fair benchmarking** and **promotions**.

Improve Model Accuracy
Predictive models (e.g., for demand, churn, return rates) will train on **cleaner**, **less volatile data**, **enhancing performance** and **generalization**.

Enable Fair Feature Scaling
Techniques like Min-Max scaling or normalization **won't be dominated** by **extreme values**, **improving** model **convergence** and **reducing feature bias.**

A3. DATA PREPARATION – TRANSFORMATION

NORMALIZATION

Why we need to apply this ?

The Multiclass Clothing Sales Dataset contains numerical features with drastically different value ranges, which creates imbalanced and misleading analysis when apply clasification

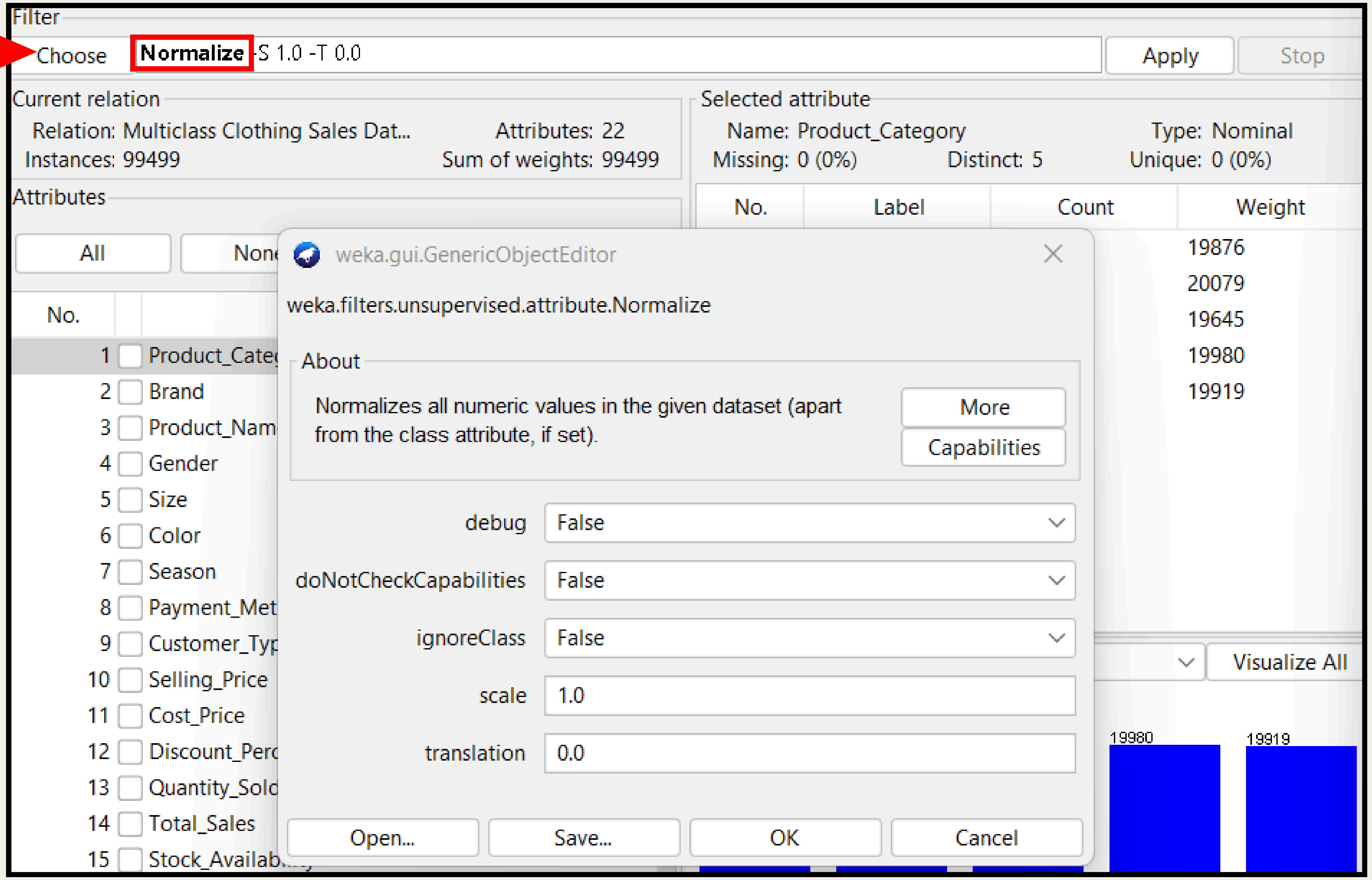
Algorithms perform poorly if one feature dominates due to scale. For example:

- Total_Sales and Selling_Price can overpower smaller attributes like Store_Rating or Purchase_Frequency.

How Apply filter

- In Weka, go to **preprocess** and choose **unsupervised › attribute › normalize**. Then choose the default setting, which is :

- **debug = "false",**
- **doNotCheckCapabilities = "false"**
- **ignoreClass = "false"**
- **scale = 1.0**
- **translation = 0.0**



After apply filter Normalization

All numeric features are scaled between 0 and 1.
For example:
Selling price of 19989 becomes 1 & Quantity sold of 199 becomes 1.

Machine learning models in Weka become more stable and accurate because:

- They no longer favor high-scale attributes.
- Models converge faster and produce better generalization.

AFTER APPLY FILTER NORMALIZE

Attribute Name	Statistic	Value
Selling_price	Minimum	0
	Maximum	1
	Mean	0.505
	Standard Devition	0.117
Cost_Price	Minimum	0
	Maximum	1
	Mean	0.484
	Standard Devition	0.112
Discount_Percentage	Minimum	0
	Maximum	1
	Mean	0.5
	Standard Devition	0.281
Quantity_Sold	Minimum	0
	Maximum	1
	Mean	0.5
	Standard Devition	0.323

Attribute Name	Statistic	Value
Total_Sales	Minimum	0
	Maximum	1
	Mean	0.488
	Standard Devition	0.112
Stock_Availability	Minimum	0
	Maximum	1
	Mean	0.502
	Standard Devition	0.282
Customer_Age	Minimum	0
	Maximum	1
	Mean	0.501
	Standard Devition	0.287
Purchase_Frequency	Minimum	0
	Maximum	1
	Mean	0.472
	Standard Devition	0.116

Attribute Name	Statistic	Value
Store_Rating	Minimum	0
	Maximum	1
	Mean	0.5
	Standard Devition	0.282
Return_Rate	Minimum	0
	Maximum	1
	Mean	0.499
	Standard Devition	0.288
Markup	Minimum	0
	Maximum	1
	Mean	0.49
	Standard Devition	0.111

After apply filter Normalize

Selling_Price

Selected attribute
Name: Selling_Price Type: Numeric
Missing: 0 (0%) Distinct: 94518 Unique: 94517 (95%)

Statistic	Value
Minimum	0
Maximum	1
Mean	0.505
StdDev	0.117

Cost_Price

Selected attribute
Name: Cost_Price Type: Numeric
Missing: 0 (0%) Distinct: 99499 Unique: 99499 (100%)

Statistic	Value
Minimum	0
Maximum	1
Mean	0.484
StdDev	0.112

Discount_Percentage

Selected attribute
Name: Discount_Percentage Type: Numeric
Missing: 0 (0%) Distinct: 94515 Unique: 94514 (95%)

Statistic	Value
Minimum	0
Maximum	1
Mean	0.5
StdDev	0.281

Quantity_Sold

Selected attribute
Name: Quantity_Sold Type: Numeric
Missing: 0 (0%) Distinct: 9 Unique: 0 (0%)

Statistic	Value
Minimum	0
Maximum	1
Mean	0.5
StdDev	0.323

Total_Sales

Selected attribute
Name: Total_Sales Type: Numeric
Missing: 0 (0%) Distinct: 99499 Unique: 99499 (100%)

Statistic	Value
Minimum	0
Maximum	1
Mean	0.488
StdDev	0.112

Stock_Availability

Selected attribute
Name: Stock_Availability Type: Numeric
Missing: 0 (0%) Distinct: 501 Unique: 0 (0%)

Statistic	Value
Minimum	0
Maximum	1
Mean	0.502
StdDev	0.282

After apply filter Normalize

Customer_Age

Selected attribute
Name: Customer_Age — Type: Numeric
Missing: 0 (0%) — Distinct: 48 — Unique: 0 (0%)

Statistic	Value
Minimum	0
Maximum	1
Mean	0.501
StdDev	0.287

Purchase_Frequency

Selected attribute
Name: Purchase_Frequency — Type: Numeric
Missing: 0 (0%) — Distinct: 98032 — Unique: 96577 (97%)

Statistic	Value
Minimum	0
Maximum	1
Mean	0.472
StdDev	0.116

Markup

Selected attribute
Name: markup — Type: Numeric
Missing: 0 (0%) — Distinct: 99497 — Unique: 99495 (100%)

Statistic	Value
Minimum	0
Maximum	1
Mean	0.49
StdDev	0.111

Store_Rating

Selected attribute
Name: Store_Rating — Type: Numeric
Missing: 0 (0%) — Distinct: 92354 — Unique: 90229 (91%)

Statistic	Value
Minimum	0
Maximum	1
Mean	0.5
StdDev	0.282

Return_Rate

Selected attribute
Name: Return_Rate — Type: Numeric
Missing: 0 (0%) — Distinct: 99313 — Unique: 99127 (100%)

Statistic	Value
Minimum	0
Maximum	1
Mean	0.499
StdDev	0.288

- **all of the numeric data has been through min-max normalization,**
- **min = 0**
- **max = 1**

Dataset Before Normalization

10: Selling_Price Numeric	11: Cost_Price Numeric	12: Discount_Percentage Numeric	13: Quantity_Sold Numeric	14: Total_Sales Numeric	15: Stock_Availability Numeric	16: Customer_Age Numeric	17: Purchase_Frequency Numeric	18: Store_Rating Numeric	19: Return_Rate Numeric	20: markup Numeric	21: **Sales_Category** Nominal
1817.433878	816.027585	26.260791	6.0	42241.692855	294.0	58.0	2.560957	3.106155	29.293331	1001.40629...	High Sales
1191.442375	1159.158217	31.015017	5.0	15947.662532	387.0	47.0	1.083173	3.144731	6.338736	32.2841580...	High Sales
1546.977432	1331.422136	19.461967	2.0	85075.427691	92.0	64.0	2.622983	3.753406	4.116767	215.555296	High Sales
1096.725802	1082.336297	45.729961	1.0	33058.985565	490.0	57.0	0.970696	3.995121	2.937317	14.3895049...	High Sales
1352.939214	1480.667077	30.949482	4.0	48390.283361	219.0	47.0	3.803581	3.23493	24.763827	-127.72786...	High Sales
1784.431124	301.023951	42.842967	3.0	28706.175221	307.0	34.0	2.530368	4.0455	22.149408	1483.40717...	High Sales
638.737526	784.131804	24.994389	5.0	62560.290378	250.329842	20.0	1.631012	3.281522	20.127198	-145.39427...	High Sales
1201.833496	1844.108272	49.400047	9.0	54665.702354	407.0	41.066821	1.927622	3.018543	22.844977	-642.27477...	High Sales
1055.08197	1787.492426	35.480239	1.0	109504.997359	11.0	52.0	2.2117	3.555703	26.085221	-732.41045...	High Sales
2028.174773	1834.425739	20.890168	5.0	47031.077084	256.0	41.0	-1.353534	4.821565	16.998307	193.749033...	High Sales
1327.317565	980.323371	44.257372	9.0	65896.27069	260.0	37.0	2.477502	4.344357	6.912793	346.994194...	High Sales
1658.180881	1767.113108	7.653502	7.0	76994.882648	374.0	59.0	3.383548	3.655794	13.633796	-108.93222...	High Sales
1531.836443	1634.118732	24.994389	9.0	25903.786518	292.0	49.0	1.771729	4.32675	19.12302	-102.28228...	High Sales
1624.320751	703.835557	5.190348	4.0	10653.584835	463.0	32.0	1.285273	3.539136	22.067912	920.485194	High Sales
1475.871165	790.020245	10.118179	7.0	44852.146291	344.0	38.0	0.662593	3.368923	27.936154	685.85092	High Sales
1508.537212	1295.223983	5.2997	5.0	20149.726228	318.0	35.0	2.102152	4.850788	0.80506	213.313228...	High Sales
1274.805159	854.564383	35.980741	5.0	58257.134933	38.0	53.0	0.692892	3.109198	11.397131	420.240776	High Sales
1388.423293	740.809895	40.276711	9.0	72100.179332	53.0	44.0	3.968596	4.307213	12.963548	647.613398...	High Sales
882.876795	771.942849	25.128378	6.0	54903.539468	63.0	23.0	2.881804	3.833923	11.489	110.933945...	High Sales
1008.226174	1398.743746	12.496279	7.0	41628.778625	273.0	56.0	0.83253	4.322663	28.993626	-390.51757...	High Sales

Dataset After Normalization

10: Selling_Price Numeric	11: Cost_Price Numeric	12: Discount_Percentage Numeric	13: Quantity_Sold Numeric	14: Total_Sales Numeric	15: Stock_Availability Numeric	16: Customer_Age Numeric	17: Purchase_Frequency Numeric	18: Store_Rating Numeric	19: Return_Rate Numeric	20: markup Numeric	21: **Sales_Category** Nominal
0.580168939464...	0.4320283987...	0.5252192044726265	0.625	0.4440663051...	0.5891783567134269	0.86956521739130...	0.5368505709259619	0.053077924623...	0.97647402369...	0.57821226...	High Sales
0.430776347346...	0.5281047260...	0.6203047970381106	0.5	0.2970043512...	0.7755511022044088	0.63043478260869...	0.3665065836969426	0.072366078928...	0.21128476559...	0.40854250...	High Sales
0.515624629630...	0.5763384980...	0.38924119064063045	0.125	0.6836344756...	0.1843687374749499	1.0	0.5440003006238522	0.376706013648...	0.13721563867...	0.44062883...	High Sales
0.408172279577...	0.5065946443...	0.9146069967669236	0.0	0.3927076180...	0.9819639278557114	0.84782608695652...	0.3535413731847443	0.497564480515...	0.09789878422...	0.40540958...	High Sales
0.469317495719...	0.6181269729...	0.6189940822532478	0.375	0.4784552414...	0.4388777555110220...	0.63043478260869...	0.680087692715393	0.117465939727...	0.82548343530...	0.38052828...	High Sales
0.572292846988...	0.2878277560...	0.8568664654537304	0.25	0.3683624421...	0.6152304609218436	0.34782608695652...	0.5333245805069639	0.522754182033...	0.73833202466...	0.66259891...	High Sales
0.298873582416...	0.4230976033...	0.49989087876911253	0.5	0.5577077901...	0.5016630100200401	0.04347826086956...	0.4296559205550752	0.140762126097...	0.67092184514...	0.37743532...	High Sales
0.433256183724...	0.7198902430...	0.9880095447476648	1.0	0.5135535267...	0.8156312625250501	0.50145263043478...	0.46384612024054983	0.009271574172...	0.76151875163...	0.29044361...	High Sales
0.398233997701...	0.7040378473...	0.7096102444035568	0.0	0.8202685406...	0.0220440881763527...	0.73913043478260...	0.496591757765415	0.277853722829...	0.86953197959...	0.27466304...	High Sales
0.630462159769...	0.7171791408...	0.4178055328464105	0.5	0.4708532295...	0.5130260521042084	0.5	0.08562765545786179	0.910789786318...	0.56662055893...	0.43681108...	High Sales
0.463202900648...	0.4780310993...	0.8851548845470976	1.0	0.5763658572...	0.5210420841683366	0.41304347826086...	0.5272307229277443	0.672183877471...	0.23042090151...	0.46364059...	High Sales
0.542163282981...	0.6983316531...	0.15306922662087627	0.75	0.6384401585...	0.749498997995992	0.89130434782608...	0.631670538389885	0.327899623196...	0.45446490226...	0.38381894...	High Sales
0.512011239380...	0.6610933246...	0.49989087876911253	1.0	0.3526887415...	0.5851703406813628	0.67391304347826...	0.44587635253356867	0.663380307042...	0.63744766628...	0.38498319...	High Sales
0.534082578106...	0.4006147125...	0.10380559092706565	0.375	0.2673946869...	0.9278557114228457	0.30434782608695...	0.3898026268392319	0.269570156561...	0.73561536319...	0.56404494...	High Sales
0.498655150824...	0.4247463625...	0.20236332265827955	0.75	0.4586665158...	0.6893787575150301	0.43478260869565...	0.3180263767460073	0.184462975703...	0.93123265647...	0.52296618...	High Sales
0.506450888326...	0.5662030415...	0.10599265559715512	0.5	0.3205064043...	0.6372745490981964	0.36956521739130...	0.48396417273745107	0.925401403211...	0.02682019835...	0.44023630...	High Sales
0.450670838451...	0.4428186741...	0.7196203973180818	0.5	0.5336403330...	0.0761523046092184...	0.76086956521739...	0.3215189388992395	0.054599436795...	0.37990588556...	0.47646429...	High Sales
0.477785755296...	0.4109675003...	0.8055407664998461	1.0	0.6110641867...	0.1062124248496994	0.56521739130434...	0.6991089522855309	0.653611728893...	0.43212224843...	0.51627171...	High Sales
0.357137309734...	0.4196847047...	0.5025706889973719	0.625	0.5148837447...	0.1262525050100200...	0.10869565217391...	0.5738345667402818	0.416964835718...	0.38296833000...	0.42231217...	High Sales
0.387051882950...	0.5951884998...	0.2499258591636913	0.75	0.4406382888...	0.5470941883767535	0.82608695652173...	0.3376149946751078	0.661336790694...	0.96648338579...	0.33452018...	High Sales

DISCRETIZATION

Why we need to apply this ?

The dataset have several continuos attributes that take many different decimal values. It are not suitable for algorithm that expect categorical.

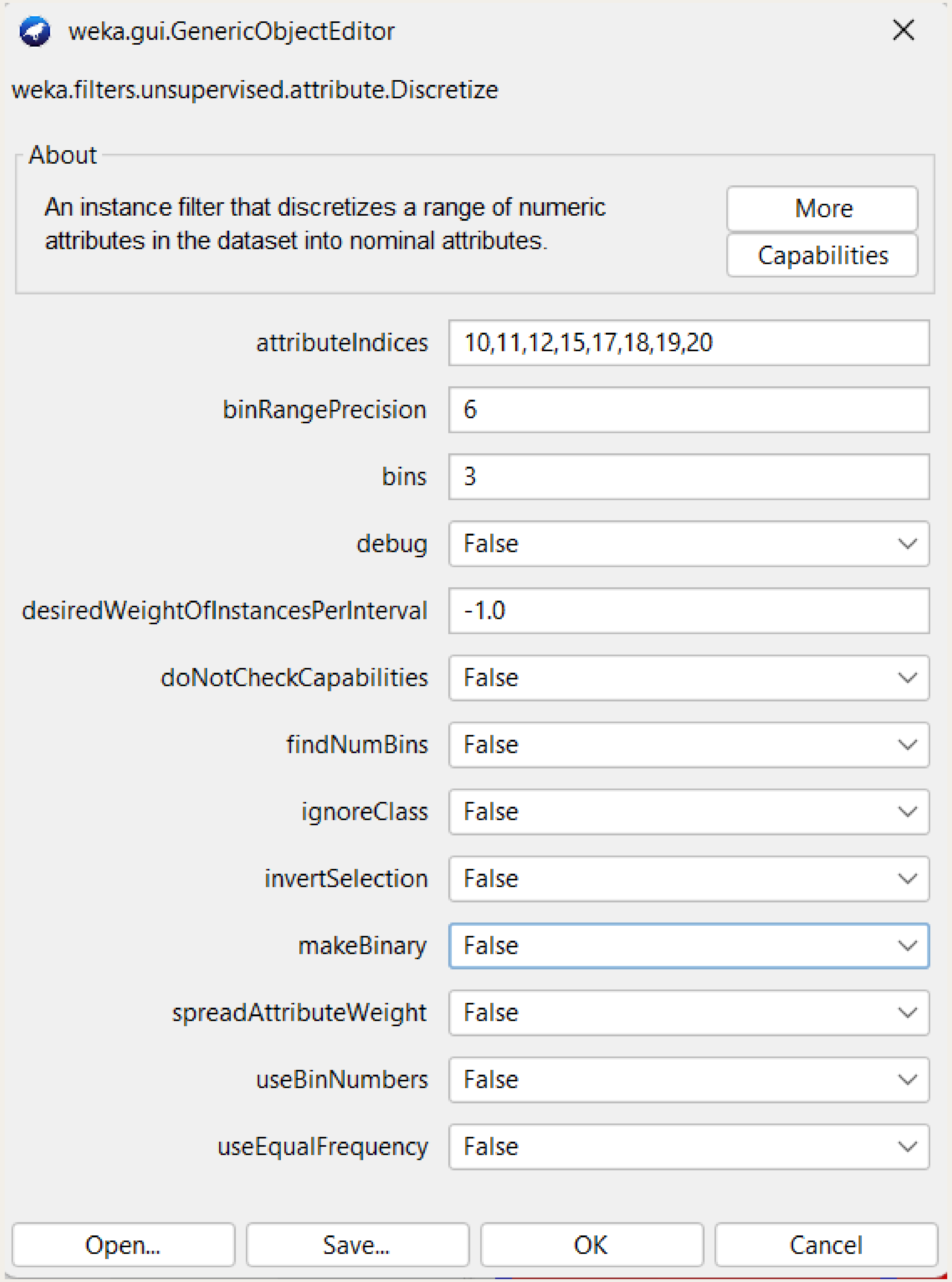
DISCRETIZE

All numerical attributes that have been normalized earlier need to be discretized to form categorical attributes. The nominal attributes don't need to be discretized because it's already categorical.

3 BINS DISCRETIZATION

How Apply filter

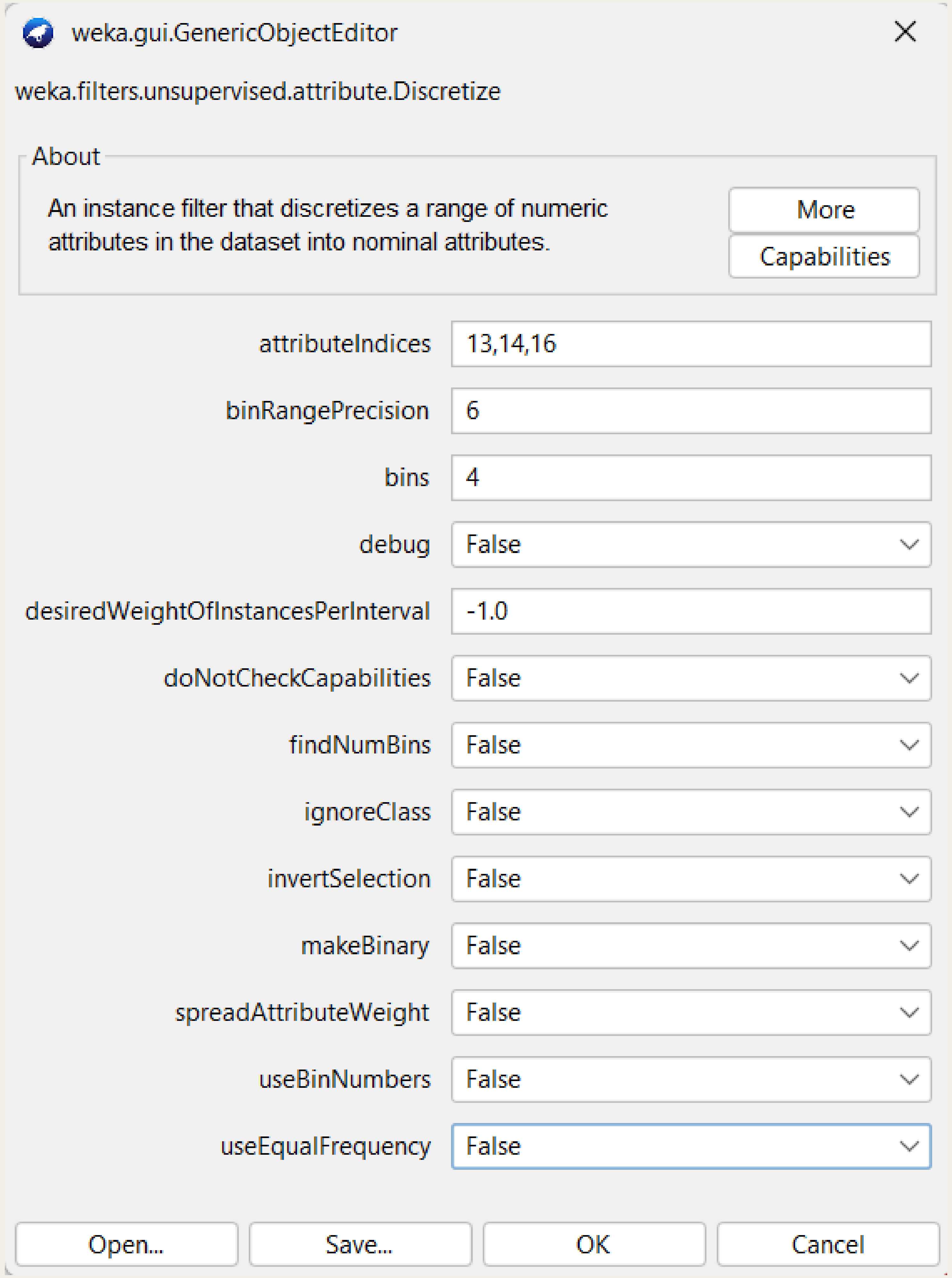
- In Weka, go to preprocess and choose unsupervised > attribute > discretize. Then change some of the setting :
- attributeIndeces = the selected attribute
- bins = 3
- debug = “false”
- desiredWeightOfInstancesPerInterval = -1.0
- doNotCheckCapabilities = “false”
- findNumBins = “false”
- ignoreClass = “false”
- invertSelection = “false”
- makeBinary = “false”
- spreadAttributeWeight = “false”
- useBinNumbers = “false”
- useEqualFrequency = “false”



4 BINS DISCRETIZATION

How Apply filter

- In Weka, go to preprocess and choose unsupervised > attribute > discretize. Then change some of the setting :
- attributeIndeces = the selected attribute
- bins = 4
- debug = “false”
- desiredWeightOfInstancesPerInterval = -1.0
- doNotCheckCapabilities = “false”
- findNumBins = “false”
- ignoreClass = “false”
- invertSelection = “false”
- makeBinary = “false”
- spreadAttributeWeight = “false”
- useBinNumbers = “false”
- useEqualFrequency = “false”



The bins values

3 Bins

- Selling_Price : Low, Medium, High
- Cost_Price : Low, Medium, High
- Discount_Percentage : Small, Moderate, Large
- Stock_Availability : Low, Medium, High
- Purchase_Frequency : Infrequent, Moderate, Frequent
- Store_Rating : Poor, Average, Excellent
- Return_Rate : Low, Medium, High
- Markup : Low, Medium, High

4 Bins

- Quantity_Sold : Very Low, Low, Medium, High
- Total_Sales : Low, Medium, High, Very High
- Customer_Age : Teen, Adult, Senior, Elder

The values later will be inserted using Excel after the discretization

After apply filter Discretize

Selling_Price

Selected attribute

Name: Selling_Price | Type: Nominal

Missing: 0 (0%) | Distinct: 3 | Unique: 0 (0%)

No.	Label	Count	Weight
1	'(0.333333-0.6666..	84103	84103
2	'(-inf-0.333333]'	7222	7222
3	'(0.666667-inf)'	8174	8174

Cost_Price

Selected attribute

Name: Cost_Price | Type: Nominal

Missing: 0 (0%) | Distinct: 3 | Unique: 0 (0%)

No.	Label	Count	Weight
1	'(0.333333-0.6666..	85622	85622
2	'(-inf-0.333333]'	8832	8832
3	'(0.666667-inf)'	5045	5045

Discount_Percentage

Selected attribute

Name: Discount_Percentage | Type: Nominal

Missing: 0 (0%) | Distinct: 3 | Unique: 0 (0%)

No.	Label	Count	Weight
1	'(0.333333-0.6666..	36514	36514
2	'(0.666667-inf)'	31501	31501
3	'(-inf-0.333333]'	31484	31484

Quantity_Sold

Selected attribute

Name: Quantity_Sold | Type: Nominal

Missing: 0 (0%) | Distinct: 4 | Unique: 0 (0%)

No.	Label	Count	Weight
1	'(0.5-0.75]'	21984	21984
2	'(0.25-0.5]'	22239	22239
3	'(-inf-0.25]'	33096	33096
4	'(0.75-inf)'	22180	22180

Total_Sales

Selected attribute

Name: Total_Sales | Type: Nominal

Missing: 0 (0%) | Distinct: 4 | Unique: 0 (0%)

No.	Label	Count	Weight
1	'(0.25-0.5]'	52231	52231
2	'(0.5-0.75]'	44591	44591
3	'(0.75-inf)'	962	962
4	'(-inf-0.25]'	1715	1715

Stock_Availability

Selected attribute

Name: Stock_Availability | Type: Nominal

Missing: 0 (0%) | Distinct: 3 | Unique: 0 (0%)

No.	Label	Count	Weigh
1	'(0.333333-0.6666..	36357	36357
2	'(0.666667-inf)'	31853	31853
3	'(-inf-0.333333]'	31289	31289

After apply filter Discretize

Customer_Age

Selected attribute
Name: Customer_Age — Type: Nominal
Missing: 0 (0%) — Distinct: 4 — Unique: 0 (0%)

No.	Label	Count	Weight
1	'(0.75-inf)'	24290	24290
2	'(0.5-0.75]'	27317	27317
3	'(0.25-0.5]'	23937	23937
4	'(-inf-0.25]'	23955	23955

Purchase_Frequency

Selected attribute
Name: Purchase_Frequency — Type: Nominal
Missing: 0 (0%) — Distinct: 3 — Unique: 0 (0%)

No.	Label	Count	Weight
1	'(0.333333-0.6666..	83426	83426
2	'(0.666667-inf)'	4571	4571
3	'(-inf-0.333333]'	11502	11502

Markup

Selected attribute
Name: Markup — Type: Nominal
Missing: 0 (0%) — Distinct: 3 — Unique: 0 (0%)

No.	Label	Count	Weight
1	'(0.333333-0.6666..	86173	86173
2	'(-inf-0.333333]'	7785	7785
3	'(0.666667-inf)'	5541	5541

Store_Rating

Selected attribute
Name: Store_Rating — Type: Nominal
Missing: 0 (0%) — Distinct: 3 — Unique: 0 (0%)

No.	Label	Count	Weight
1	'(-inf-0.333333]'	31550	31550
2	'(0.333333-0.6666..	36464	36464
3	'(0.666667-inf)'	31485	31485

Return_Rate

Selected attribute
Name: Return_Rate — Type: Nominal
Missing: 0 (0%) — Distinct: 3 — Unique: 0 (0%)

No.	Label	Count	Weight
1	'(0.666667-inf)'	32940	32940
2	'(-inf-0.333333]'	33426	33426
3	'(0.333333-0.6666..	33133	33133

- **all the numeric attributes turn into nominal (categorical) attributes**
- **the range values will then be replace with the bins values using excel**
- **this ensures the actual values can be comprehend by user**

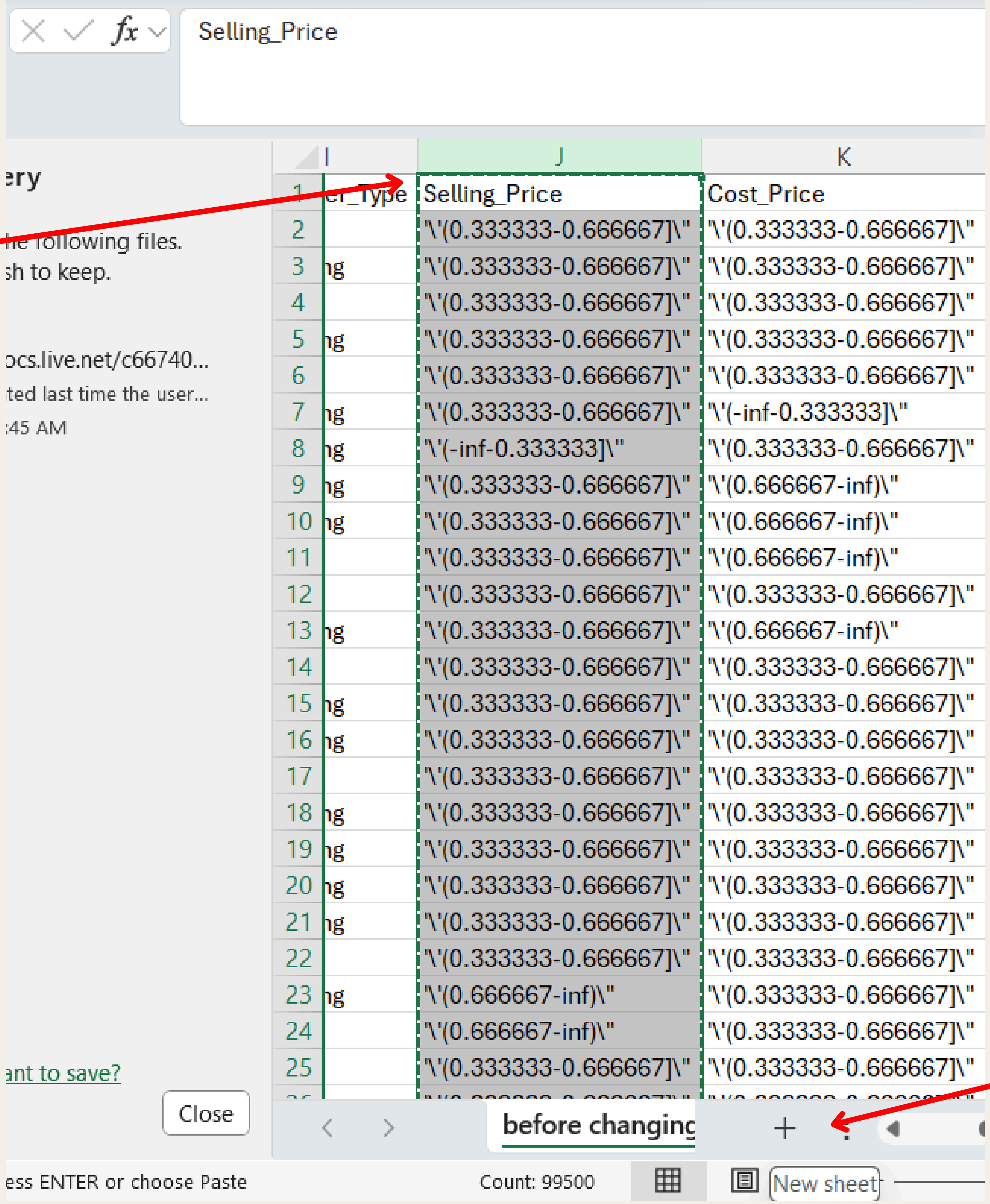
Dataset After Discretization

10: Selling_Price Nominal	11: Cost_Price Nominal	12: Discount_Percentage Nominal	13: Quantity_Sold Nominal	14: Total_Sales Nominal	15: Stock_Availability Nominal	16: Customer_Age Nominal	17: Purchase_Frequency Nominal	18: Store_Rating Nominal	19: Return_Rate Nominal	20: Markup Nominal	21: **Sales_Category** Nominal
'(0.333333-0.666..	'(0.333333-0.6...	'(0.333333-0.666667]'	'(0.5-0.75]'	'(0.25-0.5]'	'(0.333333-0.666667]'	'(0.75-inf)'	'(0.333333-0.666667]'	'(-inf-0.333333]'	'(0.666667-inf)'	'(0.333333-...	High Sales
'(0.333333-0.666..	'(0.333333-0.6...	'(0.333333-0.666667]'	'(0.25-0.5]'	'(0.25-0.5]'	'(0.666667-inf)'	'(0.5-0.75]'	'(0.333333-0.666667]'	'(-inf-0.333333]'	'(-inf-0.333333]'	'(0.333333-...	High Sales
'(0.333333-0.666..	'(0.333333-0.6...	'(0.333333-0.666667]'	'(-inf-0.25]'	'(0.5-0.75]'	'(-inf-0.333333]'	'(0.75-inf)'	'(0.333333-0.666667]'	'(0.333333-0.666..	'(-inf-0.333333]'	'(0.333333-...	High Sales
'(0.333333-0.666..	'(0.333333-0.6...	'(0.666667-inf)'	'(-inf-0.25]'	'(0.25-0.5]'	'(0.666667-inf)'	'(0.75-inf)'	'(0.333333-0.666667]'	'(0.333333-0.666..	'(-inf-0.333333]'	'(0.333333-...	High Sales
'(0.333333-0.666..	'(0.333333-0.6...	'(0.333333-0.666667]'	'(0.25-0.5]'	'(0.25-0.5]'	'(0.333333-0.666667]'	'(0.5-0.75]'	'(0.666667-inf)'	'(-inf-0.333333]'	'(0.666667-inf)'	'(0.333333-...	High Sales
'(0.333333-0.666..	'(-inf-0.33333...	'(0.666667-inf)'	'(-inf-0.25]'	'(0.25-0.5]'	'(0.333333-0.666667]'	'(0.25-0.5]'	'(0.333333-0.666667]'	'(0.333333-0.666..	'(0.666667-inf)'	'(0.333333-...	High Sales
'(-inf-0.333333]'	'(0.333333-0.6...	'(0.333333-0.666667]'	'(0.25-0.5]'	'(0.5-0.75]'	'(0.333333-0.666667]'	'(-inf-0.25]'	'(0.333333-0.666667]'	'(-inf-0.333333]'	'(0.666667-inf)'	'(0.333333-...	High Sales
'(0.333333-0.666..	'(0.666667-inf)'	'(0.666667-inf)'	'(0.75-inf)'	'(0.5-0.75]'	'(0.666667-inf)'	'(0.5-0.75]'	'(0.333333-0.666667]'	'(-inf-0.333333]'	'(0.666667-inf)'	'(-inf-0.333...	High Sales
'(0.333333-0.666..	'(0.666667-inf)'	'(0.666667-inf)'	'(-inf-0.25]'	'(0.75-inf)'	'(-inf-0.333333]'	'(0.5-0.75]'	'(0.333333-0.666667]'	'(-inf-0.333333]'	'(0.666667-inf)'	'(-inf-0.333...	High Sales
'(0.333333-0.666..	'(0.666667-inf)'	'(0.333333-0.666667]'	'(0.25-0.5]'	'(0.25-0.5]'	'(0.333333-0.666667]'	'(0.25-0.5]'	'(-inf-0.333333]'	'(0.666667-inf)'	'(0.333333-0.66...	'(0.333333-...	High Sales
'(0.333333-0.666..	'(0.333333-0.6...	'(0.666667-inf)'	'(0.75-inf)'	'(0.5-0.75]'	'(0.333333-0.666667]'	'(0.25-0.5]'	'(0.333333-0.666667]'	'(0.666667-inf)'	'(-inf-0.333333]'	'(0.333333-...	High Sales
'(0.333333-0.666..	'(0.666667-inf)'	'(-inf-0.333333]'	'(0.5-0.75]'	'(0.5-0.75]'	'(0.666667-inf)'	'(0.75-inf)'	'(0.333333-0.666667]'	'(-inf-0.333333]'	'(0.333333-0.66...	'(0.333333-...	High Sales
'(0.333333-0.666..	'(0.333333-0.6...	'(0.333333-0.666667]'	'(0.75-inf)'	'(0.25-0.5]'	'(0.333333-0.666667]'	'(0.5-0.75]'	'(0.333333-0.666667]'	'(0.333333-0.666..	'(0.333333-0.66...	'(0.333333-...	High Sales
'(0.333333-0.666..	'(0.333333-0.6...	'(-inf-0.333333]'	'(0.25-0.5]'	'(0.25-0.5]'	'(0.666667-inf)'	'(0.25-0.5]'	'(0.333333-0.666667]'	'(-inf-0.333333]'	'(0.666667-inf)'	'(0.333333-...	High Sales
'(0.333333-0.666..	'(0.333333-0.6...	'(-inf-0.333333]'	'(0.5-0.75]'	'(0.25-0.5]'	'(0.666667-inf)'	'(0.25-0.5]'	'(-inf-0.333333]'	'(-inf-0.333333]'	'(0.666667-inf)'	'(0.333333-...	High Sales
'(0.333333-0.666..	'(0.333333-0.6...	'(-inf-0.333333]'	'(0.25-0.5]'	'(0.25-0.5]'	'(0.333333-0.666667]'	'(0.25-0.5]'	'(0.333333-0.666667]'	'(0.666667-inf)'	'(-inf-0.333333]'	'(0.333333-...	High Sales
'(0.333333-0.666..	'(0.333333-0.6...	'(0.666667-inf)'	'(0.25-0.5]'	'(0.5-0.75]'	'(-inf-0.333333]'	'(0.75-inf)'	'(-inf-0.333333]'	'(-inf-0.333333]'	'(0.333333-0.66...	'(0.333333-...	High Sales
'(0.333333-0.666..	'(0.333333-0.6...	'(0.666667-inf)'	'(0.75-inf)'	'(0.5-0.75]'	'(-inf-0.333333]'	'(0.5-0.75]'	'(0.666667-inf)'	'(0.333333-0.666..	'(0.333333-0.66...	'(0.333333-...	High Sales
'(0.333333-0.666..	'(0.333333-0.6...	'(0.333333-0.666667]'	'(0.5-0.75]'	'(0.5-0.75]'	'(-inf-0.333333]'	'(-inf-0.25]'	'(0.333333-0.666667]'	'(0.333333-0.666..	'(0.333333-0.66...	'(0.333333-...	High Sales
'(0.333333-0.666..	'(0.333333-0.6...	'(-inf-0.333333]'	'(0.5-0.75]'	'(0.25-0.5]'	'(0.333333-0.666667]'	'(0.75-inf)'	'(0.333333-0.666667]'	'(0.333333-0.666..	'(0.666667-inf)'	'(0.333333-...	High Sales

Need to change all range values in Excel

We choose Selling_Price as the guide :

1.Cut the whole column

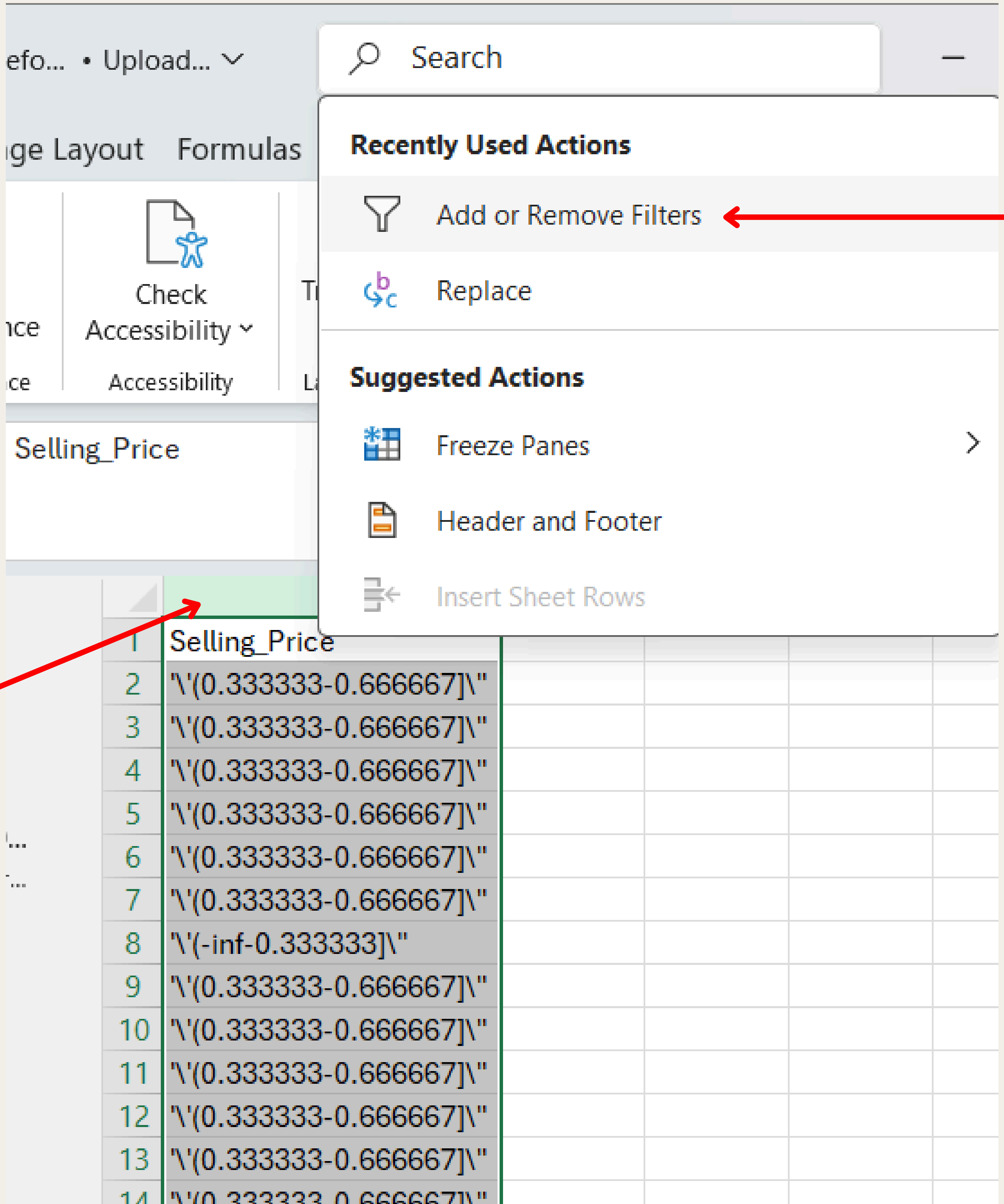


	I	J	K
1	er_Type	Selling_Price	Cost_Price
2		'\'(0.333333-0.666667]\"	'\'(0.333333-0.666667]\"
3	ng	'\'(0.333333-0.666667]\"	'\'(0.333333-0.666667]\"
4		'\'(0.333333-0.666667]\"	'\'(0.333333-0.666667]\"
5	ng	'\'(0.333333-0.666667]\"	'\'(0.333333-0.666667]\"
6		'\'(0.333333-0.666667]\"	'\'(0.333333-0.666667]\"
7	ng	'\'(0.333333-0.666667]\"	'\'(-inf-0.333333]\"
8	ng	'\'(-inf-0.333333]\"	'\'(0.333333-0.666667]\"
9	ng	'\'(0.333333-0.666667]\"	'\'(0.666667-inf)\"
10	ng	'\'(0.333333-0.666667]\"	'\'(0.666667-inf)\"
11		'\'(0.333333-0.666667]\"	'\'(0.666667-inf)\"
12		'\'(0.333333-0.666667]\"	'\'(0.333333-0.666667]\"
13	ng	'\'(0.333333-0.666667]\"	'\'(0.666667-inf)\"
14		'\'(0.333333-0.666667]\"	'\'(0.333333-0.666667]\"
15	ng	'\'(0.333333-0.666667]\"	'\'(0.333333-0.666667]\"
16	ng	'\'(0.333333-0.666667]\"	'\'(0.333333-0.666667]\"
17		'\'(0.333333-0.666667]\"	'\'(0.333333-0.666667]\"
18	ng	'\'(0.333333-0.666667]\"	'\'(0.333333-0.666667]\"
19	ng	'\'(0.333333-0.666667]\"	'\'(0.333333-0.666667]\"
20	ng	'\'(0.333333-0.666667]\"	'\'(0.333333-0.666667]\"
21	ng	'\'(0.333333-0.666667]\"	'\'(0.333333-0.666667]\"
22		'\'(0.333333-0.666667]\"	'\'(0.333333-0.666667]\"
23	ng	'\'(0.666667-inf)\"	'\'(0.333333-0.666667]\"
24		'\'(0.666667-inf)\"	'\'(0.333333-0.666667]\"
25		'\'(0.333333-0.666667]\"	'\'(0.333333-0.666667]\"

2. create new pages

Need to change all range values in Excel
We choose Selling_Price as the guide :

4.add filter

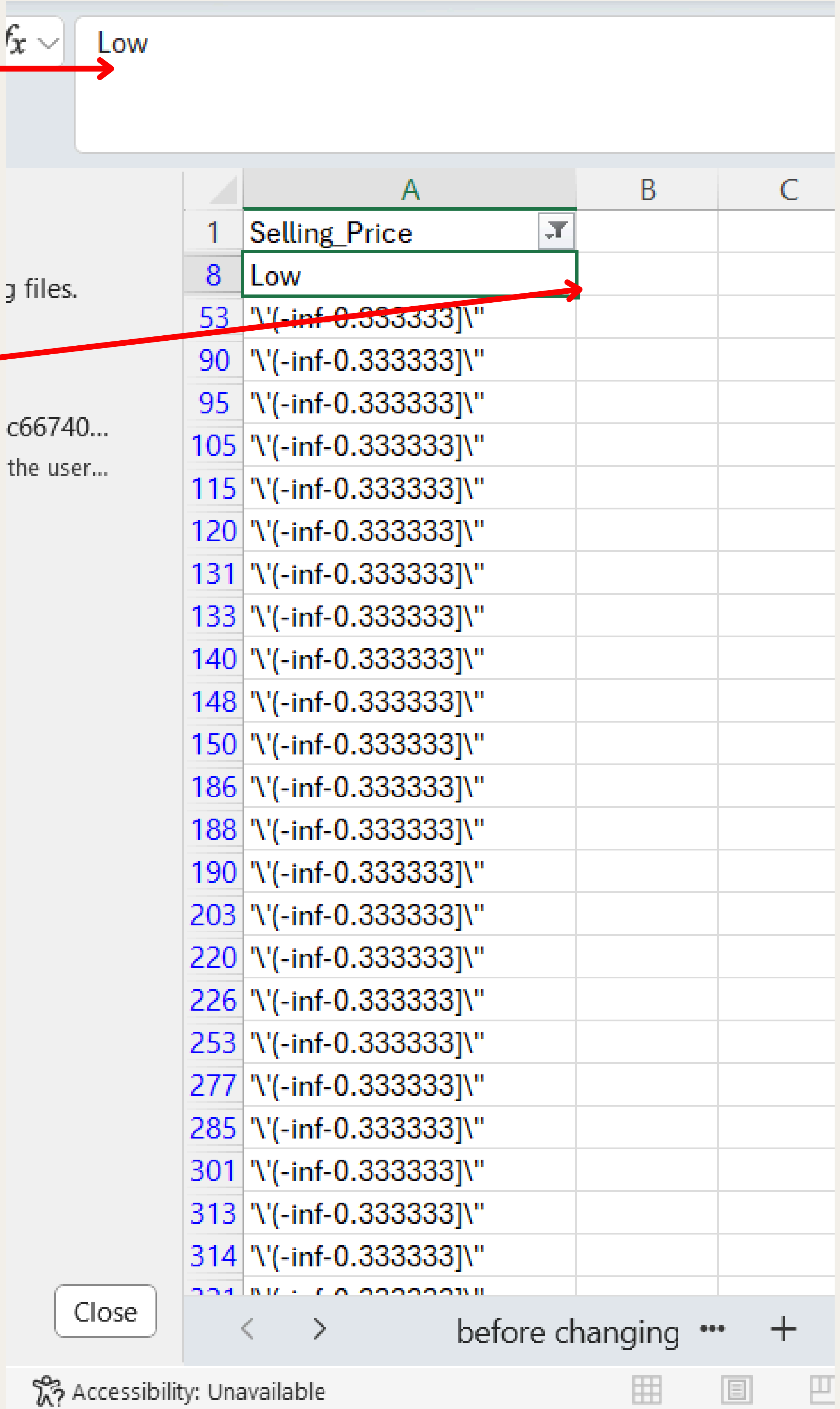


3.paste the copied column

Need to change all range values in Excel

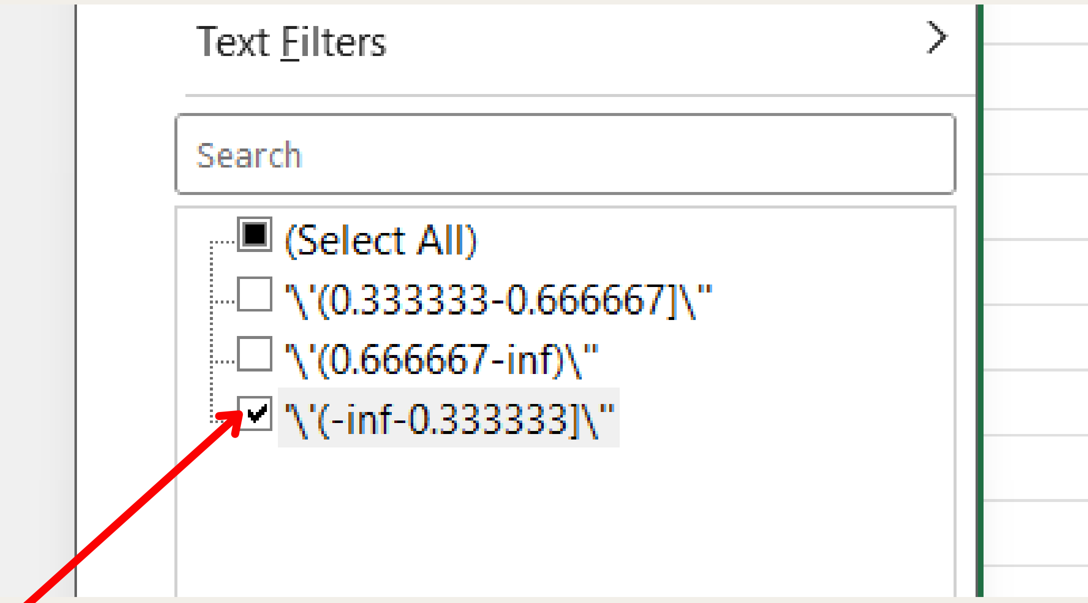
We choose Selling_Price as the guide :

6.place the value



7.double click on this highlighted corner. This will change all the value available in the filtered column into choosen values

8.repeat step 5-7 for the following values

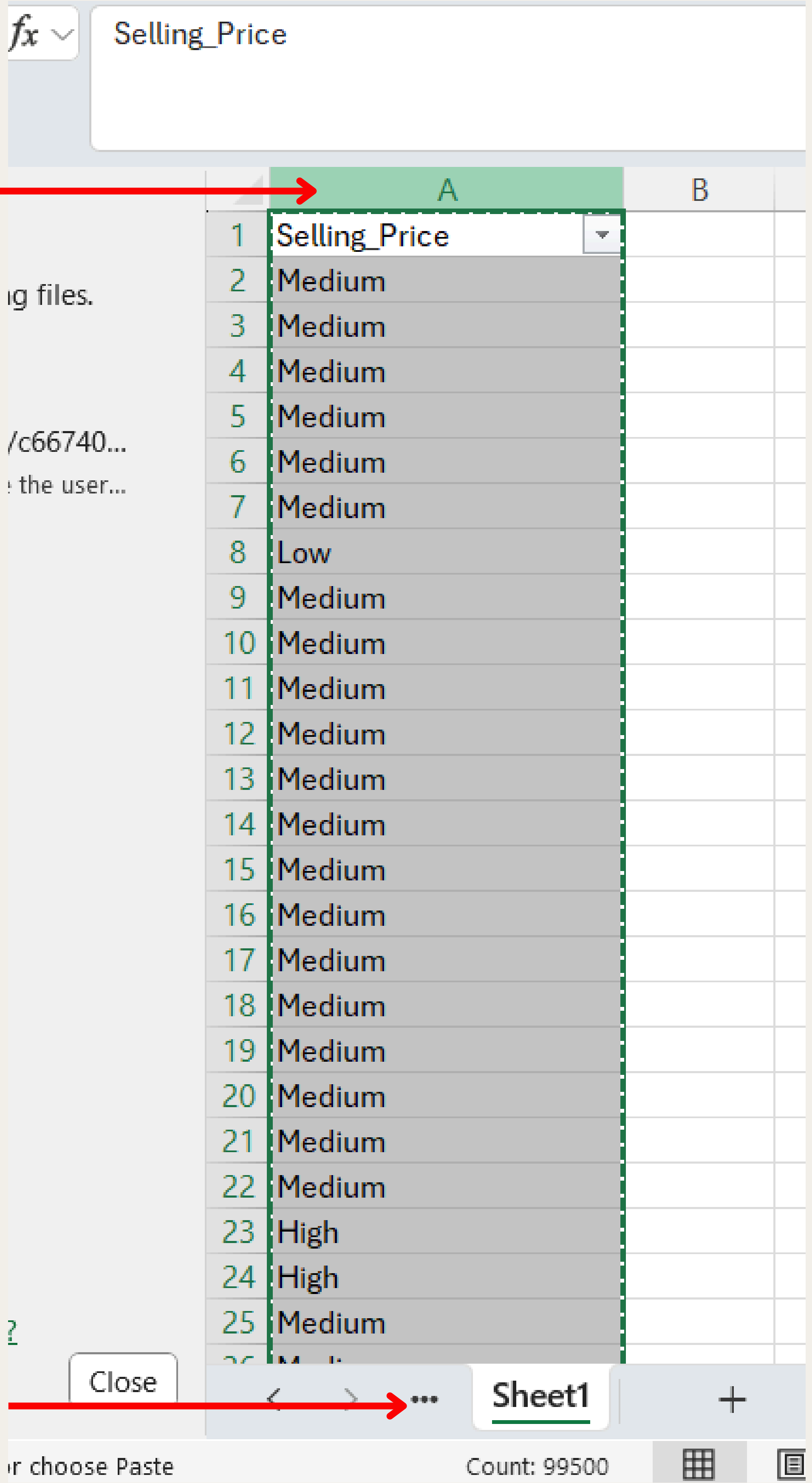


5.select the range values to change

Need to change all range values in Excel
We choose Selling_Price as the guide :

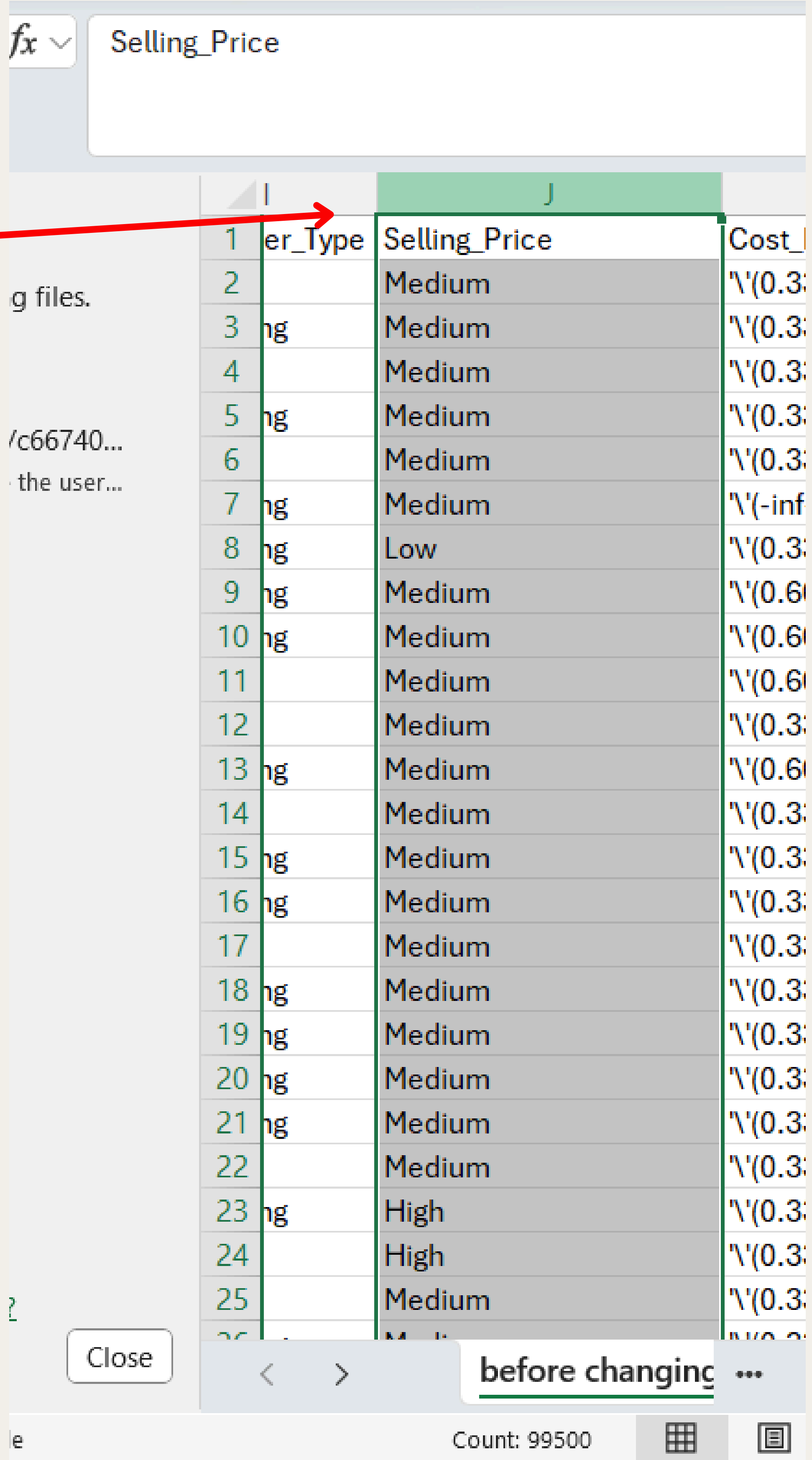
9.cut the column

10.change to the main sheet



11.paste the column

12. repeat the whole process for the next attributes



Dataset After Changing Values

	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
1	Season	Payment_I	Customer_	Selling_Price	Cost_Price	Discount_Percentage	Quantity_Sold	Total_Sales	Stock_Availability	Customer_Age	Purchase_Frequency	Store_Rating	Return_Rate	Markup	Sales_Category
2	Winter	Card	New	Medium	Medium	Moderate	Medium	Medium	Medium	Elder	Moderate	Poor	High	Medium	'High Sales'
3	All-Season	UPI	Returning	Medium	Medium	Moderate	Low	Medium	High	Senior	Moderate	Poor	Low	Medium	'High Sales'
4	Winter	Cash	New	Medium	Medium	Moderate	Very Low	High	Low	Elder	Moderate	Average	Low	Medium	'High Sales'
5	All-Season	Card	Returning	Medium	Medium	Large	Very Low	Medium	High	Elder	Moderate	Average	Low	Medium	'High Sales'
6	All-Season	Card	New	Medium	Medium	Moderate	Low	Medium	Medium	Senior	Frequent	Poor	High	Medium	'High Sales'
7	Winter	Card	Returning	Medium	Low	Large	Very Low	Medium	Medium	Adult	Moderate	Average	High	Medium	'High Sales'
8	Winter	Card	Returning	Low	Medium	Moderate	Low	High	Medium	Teen	Moderate	Poor	High	Medium	'High Sales'
9	Summer	Cash	Returning	Medium	High	Large	High	High	High	Senior	Moderate	Poor	High	Low	'High Sales'
10	Winter	Card	Returning	Medium	High	Large	Very Low	Very High	Low	Senior	Moderate	Poor	High	Low	'High Sales'
11	Summer	UPI	New	Medium	High	Moderate	Low	Medium	Medium	Adult	Infrequent	Excellent	Medium	Medium	'High Sales'
12	All-Season	'Net Bankir	New	Medium	Medium	Large	High	High	Medium	Adult	Moderate	Excellent	Low	Medium	'High Sales'
13	Winter	UPI	Returning	Medium	High	Small	Medium	High	High	Elder	Moderate	Poor	Medium	Medium	'High Sales'
14	All-Season	'Net Bankir	New	Medium	Medium	Moderate	High	Medium	Medium	Senior	Moderate	Average	Medium	Medium	'High Sales'
15	Winter	'Net Bankir	Returning	Medium	Medium	Small	Low	Medium	High	Adult	Moderate	Poor	High	Medium	'High Sales'
16	Winter	'Net Bankir	Returning	Medium	Medium	Small	Medium	Medium	High	Adult	Infrequent	Poor	High	Medium	'High Sales'
17	Winter	UPI	New	Medium	Medium	Small	Low	Medium	Medium	Adult	Moderate	Excellent	Low	Medium	'High Sales'
18	Winter	'Net Bankir	Returning	Medium	Medium	Large	Low	High	Low	Elder	Infrequent	Poor	Medium	Medium	'High Sales'
19	Summer	'Net Bankir	Returning	Medium	Medium	Large	High	High	Low	Senior	Frequent	Average	Medium	Medium	'High Sales'
20	Summer	Cash	Returning	Medium	Medium	Moderate	Medium	High	Low	Teen	Moderate	Average	Medium	Medium	'High Sales'
21	Winter	Card	Returning	Medium	Medium	Small	Medium	Medium	Medium	Elder	Moderate	Average	High	Medium	'High Sales'
22	Winter	Card	New	Medium	Medium	Large	Very Low	Medium	Medium	Teen	Moderate	Average	Medium	Medium	'High Sales'
23	Winter	Cash	Returning	High	Medium	Large	Medium	Low	Low	Senior	Moderate	Average	High	High	'High Sales'
24	All-Season	Cash	New	High	Medium	Large	High	Medium	High	Teen	Moderate	Poor	Medium	Medium	'High Sales'
25	Summer	Card	New	Medium	Medium	Large	Very Low	High	High	Teen	Moderate	Average	Medium	Medium	'High Sales'

after changing values

ADDING ATTRIBUTES

Why we need to apply this ?

The dataset lack of adding attribute . This results in a feature that more accurately captures the underlying pattern in the data and is more powerful, meaningful, or straightforward. This process can greatly **increase** your **model's performance** and **accuracy**.

ADDEXPRESSION

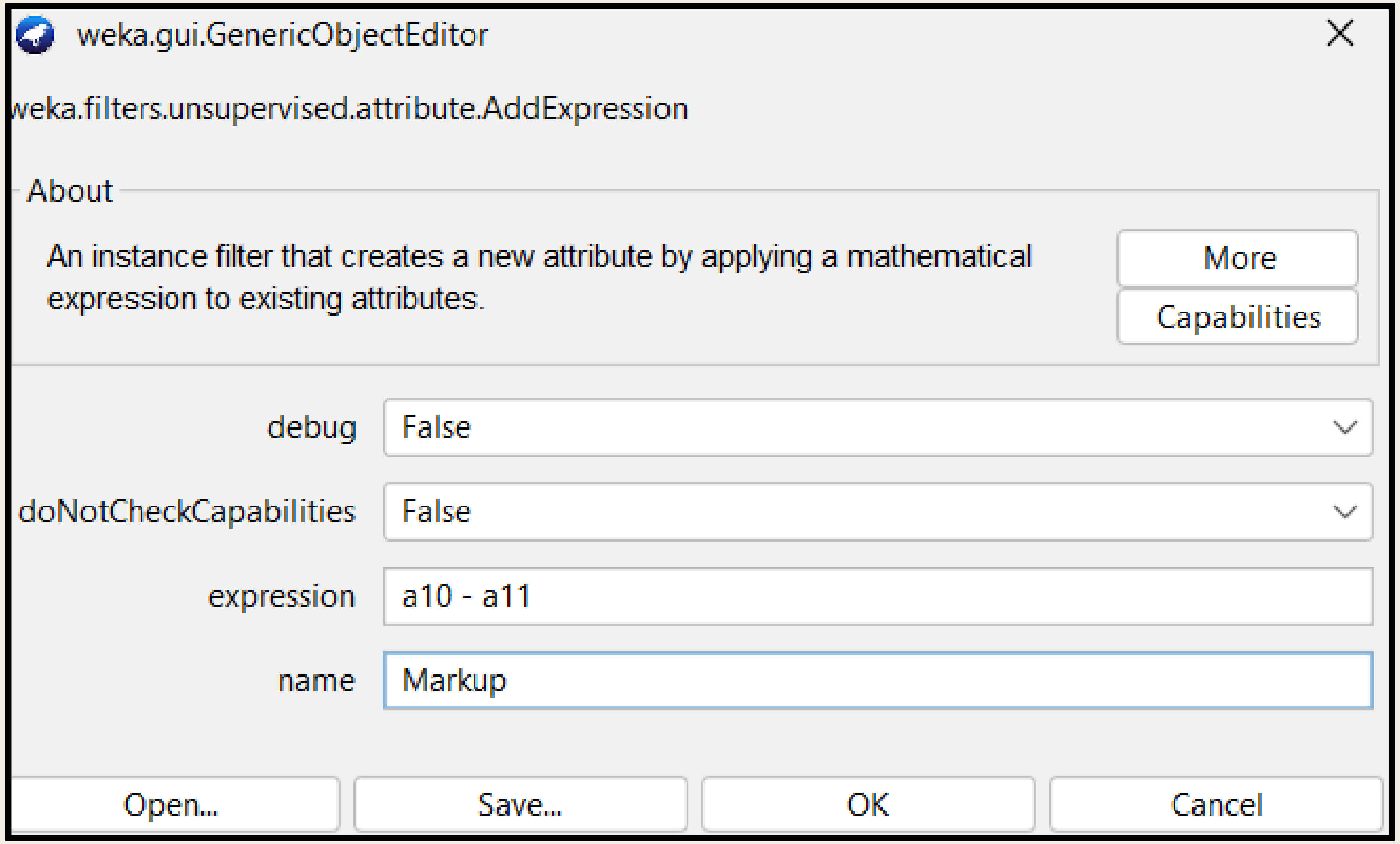
Before normalization and discretization occurs, we want to add two column together by subtracting **Selling_Price** with **Cost_Price** to **create Markup**.

How Apply filter

- In Weka, go to **preprocess** and choose **unsupervised › attribute › AddExpression**.

Then change the setting :

- **debug = “false”**
- **doNotCheckCapabilities = “false”**
- **expression = a10 – a11 (Selling_Price is a10, Cost_Price is a11)**
- **name = “Markup”**



After apply filter AddExpression

Markup

Selected attribute

Name: Markup | Type: Numeric

Missing: 0 (0%) | Distinct: 99497 | Unique: 99495 (100%)

Statistic	Value
Minimum	-2301.235
Maximum	3410.579
Mean	499.705
StdDev	631.577

- The markup attribute exists in the dataset that can help later in classify process.
- Markup value refers to the amount added to the cost price of a product to determine its selling price basically profit.

RESAMPLE

Why do we need to apply this?

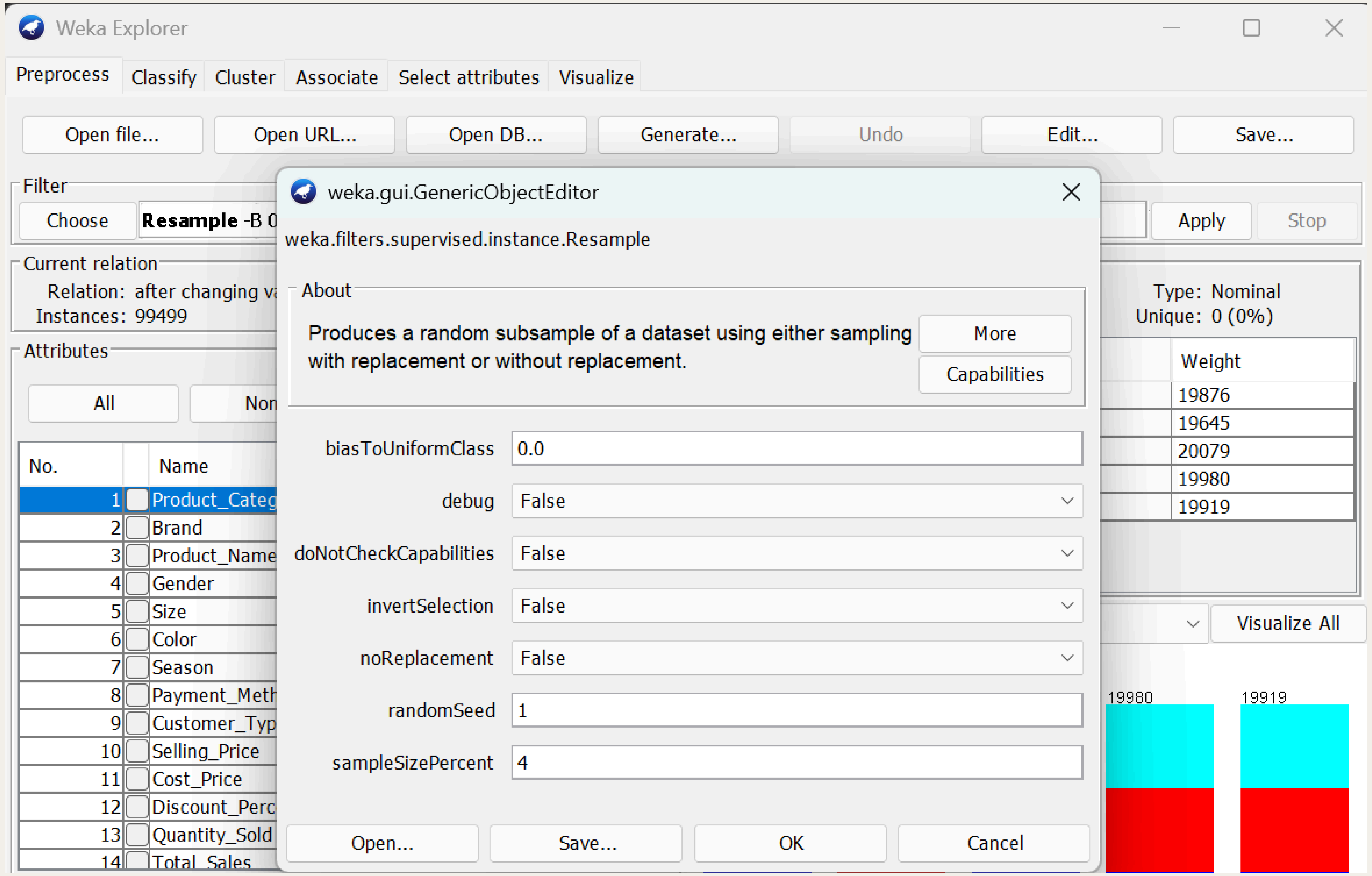
Resampling is necessary since our dataset, which contains **100,000 data points**, contains a lot of information that could throw it off balance. The **tree** will be **difficult to see** and **tricky** to create.

When there are more instances in our dataset, processing time for any method also increases.

RESAMPLE

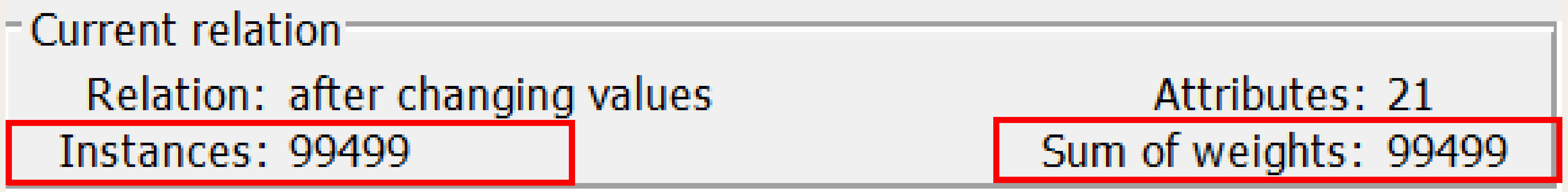
How Apply filter

- In Weka, go to **preprocess** and choose **supervised › instance › resample**. Then change the setting:
- **biasToUniformClass= 0**
- **debug = “false”**
- **doNotCheckCapabilities = “false”**
- **invertSelection = “False”**
- **noReplacement = “False”**
- **randomSeed = 1**
- **sampleSizePercent = 4**

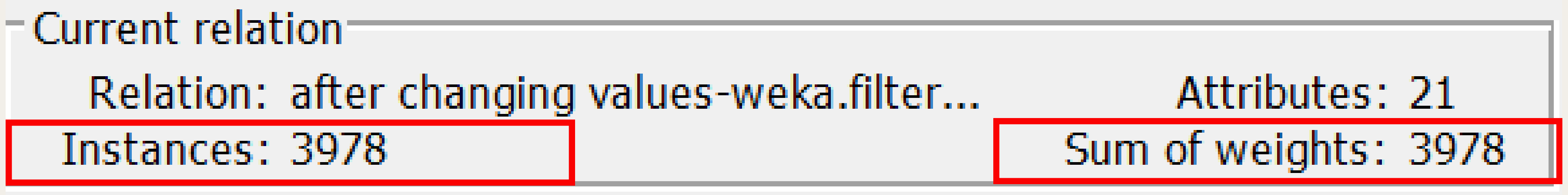


Dataset Before and After Resample

Before



After



Result after

- The result after resampling shows the data has been reduced to **3978**, making the dataset **less heavy** and **readable**.
- The data more likely to be **balance** after the resampling process
- sales category (**low sales (1335), medium sales (1325), high sales (1318)**) with is lower compare to before.

B. MODEL DEVELOPMENT AND EVALUATION

J48

WHY WE NEED TO APPLY J48 CLASSIFY METHOD?

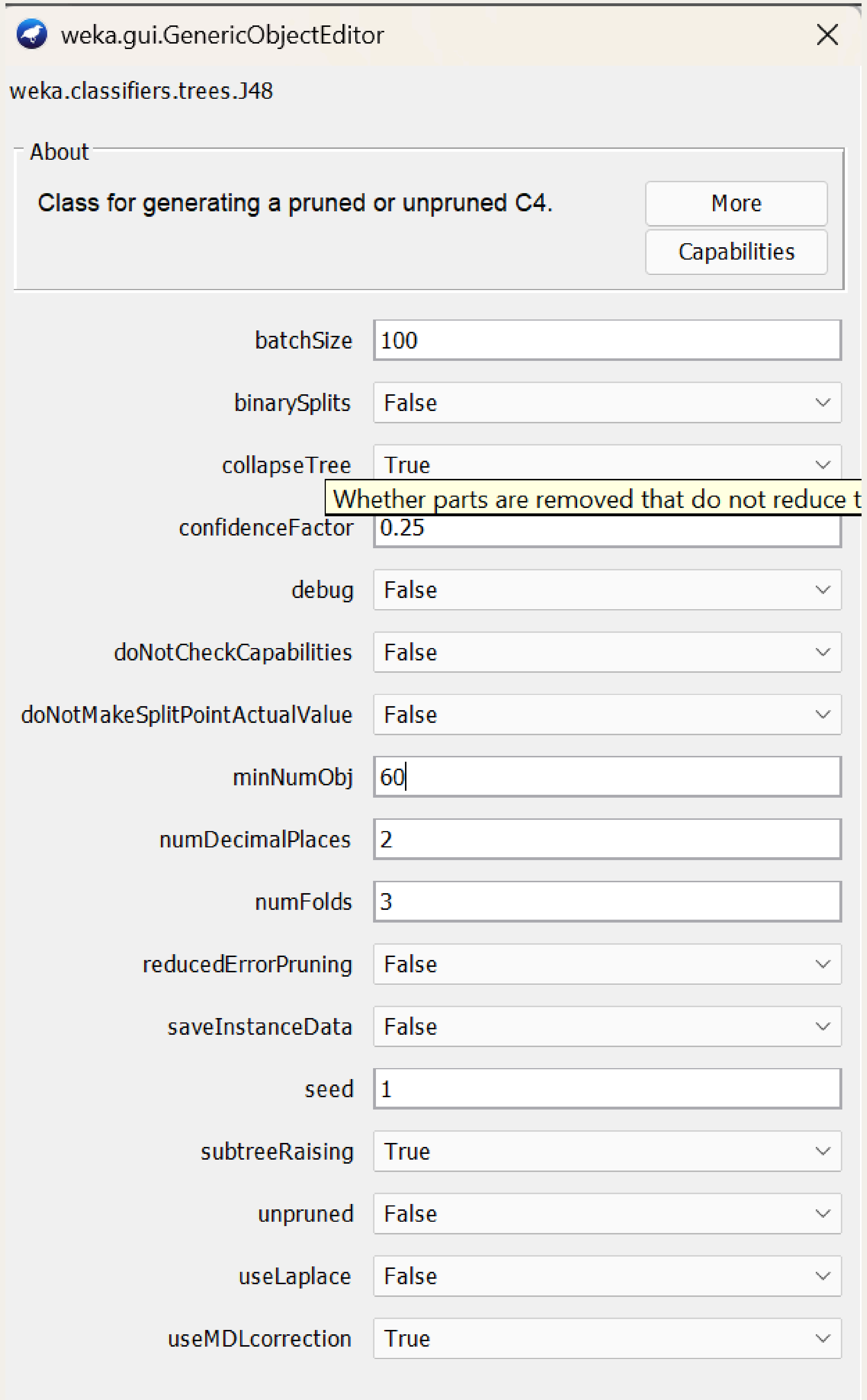
- We use **J48 classification method** for our dataset because it can generate an **interpretable decision** and **clear data**. This can help to achieve our goal of **understanding** and **visualize** the **different attributes** lead to specific outcomes
- also, it handles **numerical** and **categorical data**, that make it suitable for the wide **variety** of **attributes** in the dataset
- select the **most important features** and create a **decision rules**, that easy to follow and explain. It useful for supporting **business decisions** or **justifying pattern** that has been found in the data.

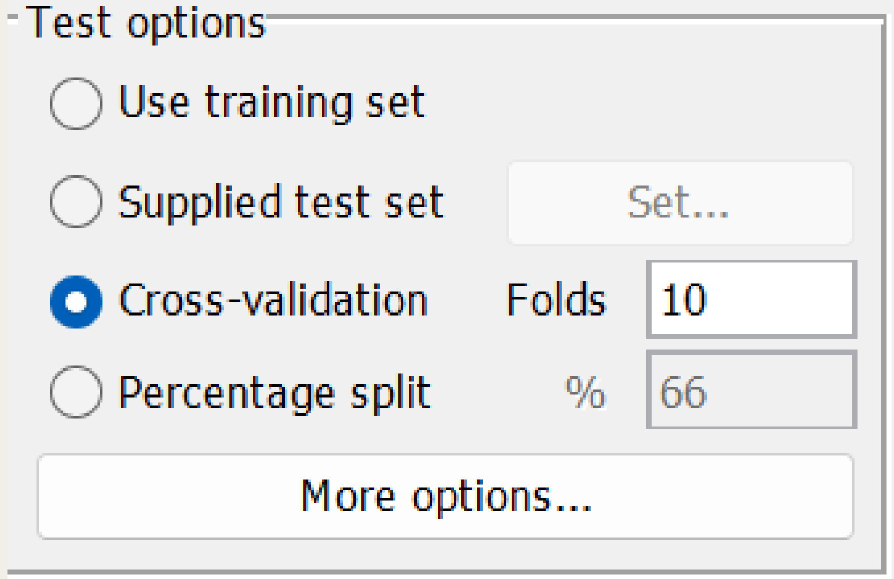
What is fold method/Cross-validation?

- the dataset is seperated into **k** parts or **"folds"**.
- Using a **different fold** for **testing** and the **remaining fold** for training, the **J48 algorithm** is trained and tested **k times**.
- This method ensures that every data point is used for both training and testing, giving a **more accurate** and **unbiased estimation** of the **model's performance**.

How Apply Classify for J48 10 fold method?

- In Weka, go to **classify** and choose **classfier › trees ›J48**. Then all setting default only changes the setting for:
- **minNumObj = 60**
- **test option › cross - validation › fold 10**





RESULT AFTER APPLY J48 10 FOLD

```
=== Run information ===

Scheme:       weka.classifiers.trees.J48 -C 0.25 -M 60
Relation:     after changing values-weka.filters.supervised.instance.Resample-B0.0-S1-Z100
Instances:    3978
Attributes:   21
              Product_Category
              Brand
              Product_Name
              Gender
              Size
              Color
              Season
              Payment_Method
              Customer_Type
              Selling_Price
              Cost_Price
              Discount_Percentage
              Quantity_Sold
              Total_Sales
              Stock_Availability
              Customer_Age
              Purchase_Frequency
              Store_Rating
              Return_Rate
              Markup
              Sales_Category
Test mode:    10-fold cross-validation
```

```
=== Classifier model (full training set) ===

J48 pruned tree
------------------

Cost_Price = Medium
|   Brand = Forever 21
|   |   Total_Sales = Medium: High Sales (87.0/55.0)
|   |   Total_Sales = High: Medium Sales (77.0/47.0)
|   |   Total_Sales = Very High: Medium Sales (0.0)
|   |   Total_Sales = Low: Medium Sales (4.0/2.0)
|   Brand = Wrangler
|   |   Customer_Type = New: High Sales (85.0/51.0)
|   |   Customer_Type = Returning: Low Sales (91.0/52.0)
|   Brand = Uniqlo: Medium Sales (168.0/100.0)
|   Brand = Reebok: High Sales (130.0/80.0)
|   Brand = Gucci
|   |   Gender = Women: Low Sales (90.0/55.0)
|   |   Gender = Men: Medium Sales (78.0/46.0)
|   Brand = Louis Vuitton
|   |   Customer_Type = New: High Sales (77.0/43.0)
|   |   Customer_Type = Returning: Low Sales (88.0/55.0)
|   Brand = Calvin Klein: Low Sales (194.0/119.0)
|   Brand = Zara
|   |   Total_Sales = Medium: Medium Sales (110.0/64.0)
|   |   Total_Sales = High: Low Sales (61.0/36.0)
|   |   Total_Sales = Very High: Low Sales (1.0)
|   |   Total_Sales = Low: High Sales (2.0/1.0)
|   Brand = Diesel
|   |   Total_Sales = Medium: Medium Sales (82.0/53.0)
|   |   Total_Sales = High: Low Sales (67.0/40.0)
|   |   Total_Sales = Very High: High Sales (1.0)
|   |   Total_Sales = Low: Medium Sales (2.0)
|   Brand = Versace
|   |   Gender = Women: High Sales (93.0/52.0)
|   |   Gender = Men: Low Sales (96.0/51.0)
```

```
|   Brand = Leviâ€™s: High Sales (178.0/109.0)
|   Brand = Balenciaga
|   |   Discount_Percentage = Moderate: Low Sales (62.0/35.0)
|   |   Discount_Percentage = Large: Medium Sales (70.0/39.0)
|   |   Discount_Percentage = Small: Medium Sales (43.0/23.0)
|   Brand = Puma: Medium Sales (173.0/108.0)
|   Brand = Under Armour
|   |   Total_Sales = Medium: High Sales (78.0/46.0)
|   |   Total_Sales = High: Medium Sales (74.0/44.0)
|   |   Total_Sales = Very High: Low Sales (2.0/1.0)
|   |   Total_Sales = Low: Medium Sales (1.0)
|   Brand = Adidas
|   |   Total_Sales = Medium: Low Sales (89.0/53.0)
|   |   Total_Sales = High: High Sales (88.0/54.0)
|   |   Total_Sales = Very High: Low Sales (1.0)
|   |   Total_Sales = Low: Low Sales (5.0/3.0)
|   Brand = Ralph Lauren
|   |   Total_Sales = Medium: Medium Sales (81.0/49.0)
|   |   Total_Sales = High: Low Sales (74.0/47.0)
|   |   Total_Sales = Very High: High Sales (2.0/1.0)
|   |   Total_Sales = Low: High Sales (3.0)
|   Brand = GAP: Medium Sales (172.0/105.0)
|   Brand = Nike
|   |   Season = Winter: Low Sales (60.0/33.0)
|   |   Season = All-Season: High Sales (55.0/32.0)
|   |   Season = Summer: High Sales (61.0/35.0)
|   Brand = Tommy Hilfiger: Low Sales (168.0/97.0)
|   Brand = H&M: Medium Sales (189.0/110.0)
Cost_Price = Low
|   Gender = Women: High Sales (176.0/96.0)
|   Gender = Men: Low Sales (170.0/106.0)
Cost_Price = High
|   Total_Sales = Medium: Medium Sales (120.0/69.0)
|   Total_Sales = High: High Sales (96.0/58.0)
|   Total_Sales = Very High: Medium Sales (2.0)
|   Total_Sales = Low: Medium Sales (1.0)
```

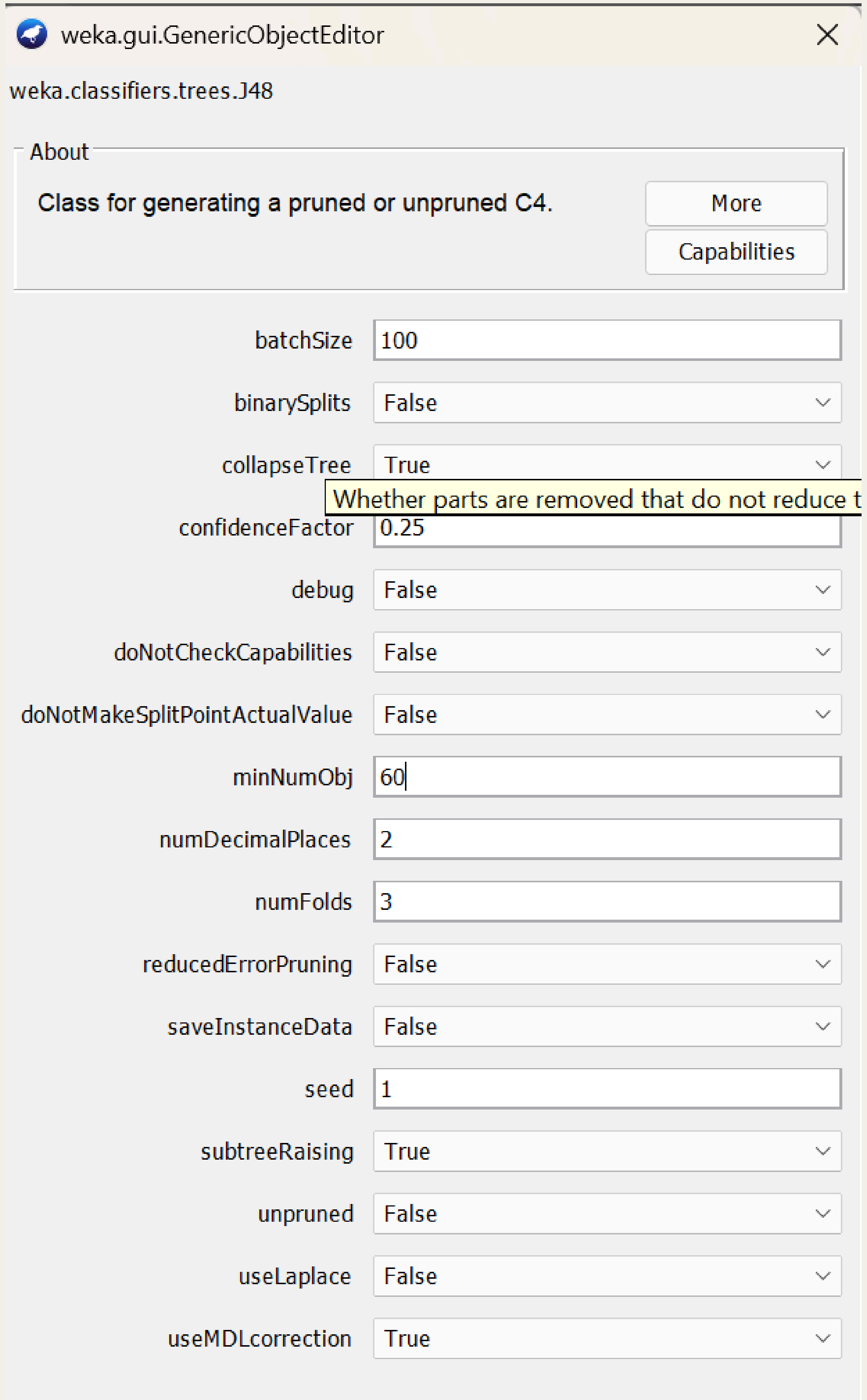
```
Number of Leaves  :      52

Size of the tree :       68


Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        1406               35.3444 %
Incorrectly Classified Instances      2572               64.6556 %
Kappa statistic                          0.0303
Mean absolute error                      0.4431
Root mean squared error                  0.4747
Relative absolute error                 99.6907 %
Root relative squared error            100.6921 %
Total Number of Instances             3978

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.323    0.321    0.337      0.323    0.330      0.002    0.499     0.332     High Sales
                 0.346    0.312    0.357      0.346    0.352      0.035    0.519     0.345     Low Sales
                 0.392    0.337    0.365      0.392    0.378      0.053    0.521     0.350     Medium Sales
Weighted Avg.    0.353    0.323    0.353      0.353    0.353      0.030    0.513     0.343

=== Confusion Matrix ===

   a   b   c   <-- classified as
 431 441 463 |   a = High Sales
 432 459 434 |   b = Low Sales
 416 386 516 |   c = Medium Sales
```

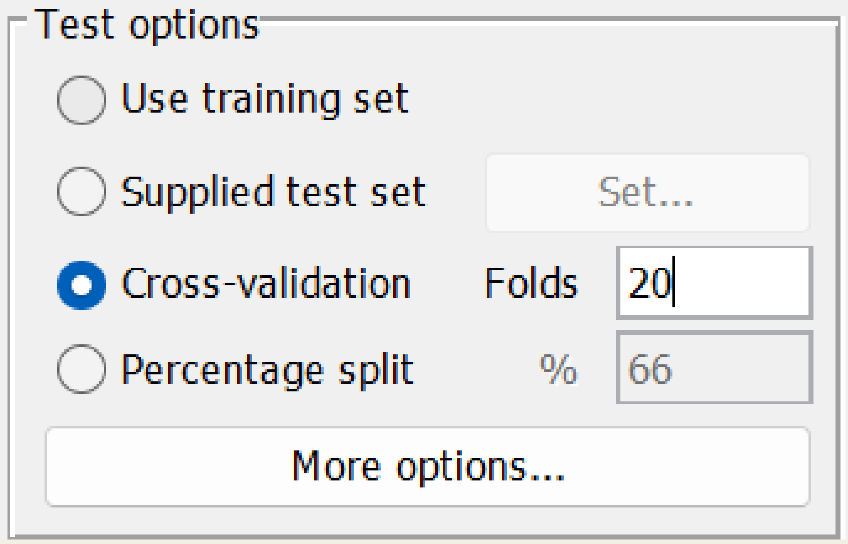
EXPLANATION

- Only about **1 out of 3 predictions** were **correct**. The data is **difficult** for the model to **classify appropriately**.
- A **significant percentage** of cases are **incorrectly classified**.
- **Poor agreement** between the **expected** and **actual classes** is indicated by a number **near 0**. This is only **slightly better** than **random guessing**.
- On this dataset, the **J48 decision** tree performs **poorly**.

How Apply Classify for J48 20 fold method?

- In Weka, go to **classify** and choose **classfier › trees ›J48**. Then all setting default only change the setting for:
- **minNumObj = 60**
- **test option › cross - validation › fold 20**





RESULT AFTER APPLY J48 20 FOLD

```
=== Run information ===

Scheme:       weka.classifiers.trees.J48 -C 0.25 -M 60
Relation:     after changing values-weka.filters.supervised.instance.Resample-B0.0-S1-Z100
Instances:    3978
Attributes:   21
              Product_Category
              Brand
              Product_Name
              Gender
              Size
              Color
              Season
              Payment_Method
              Customer_Type
              Selling_Price
              Cost_Price
              Discount_Percentage
              Quantity_Sold
              Total_Sales
              Stock_Availability
              Customer_Age
              Purchase_Frequency
              Store_Rating
              Return_Rate
              Markup
              Sales_Category
Test mode:    20-fold cross-validation
```

```
=== Classifier model (full training set) ===

J48 pruned tree
------------------

Cost_Price = Medium
|   Brand = Forever 21
|   |   Total_Sales = Medium: High Sales (87.0/55.0)
|   |   Total_Sales = High: Medium Sales (77.0/47.0)
|   |   Total_Sales = Very High: Medium Sales (0.0)
|   |   Total_Sales = Low: Medium Sales (4.0/2.0)
|   Brand = Wrangler
|   |   Customer_Type = New: High Sales (85.0/51.0)
|   |   Customer_Type = Returning: Low Sales (91.0/52.0)
|   Brand = Uniqlo: Medium Sales (168.0/100.0)
|   Brand = Reebok: High Sales (130.0/80.0)
|   Brand = Gucci
|   |   Gender = Women: Low Sales (90.0/55.0)
|   |   Gender = Men: Medium Sales (78.0/46.0)
|   Brand = Louis Vuitton
|   |   Customer_Type = New: High Sales (77.0/43.0)
|   |   Customer_Type = Returning: Low Sales (88.0/55.0)
|   Brand = Calvin Klein: Low Sales (194.0/119.0)
|   Brand = Zara
|   |   Total_Sales = Medium: Medium Sales (110.0/64.0)
|   |   Total_Sales = High: Low Sales (61.0/36.0)
|   |   Total_Sales = Very High: Low Sales (1.0)
|   |   Total_Sales = Low: High Sales (2.0/1.0)
|   Brand = Diesel
|   |   Total_Sales = Medium: Medium Sales (82.0/53.0)
|   |   Total_Sales = High: Low Sales (67.0/40.0)
|   |   Total_Sales = Very High: High Sales (1.0)
|   |   Total_Sales = Low: Medium Sales (2.0)
|   Brand = Versace
|   |   Gender = Women: High Sales (93.0/52.0)
|   |   Gender = Men: Low Sales (96.0/51.0)
```

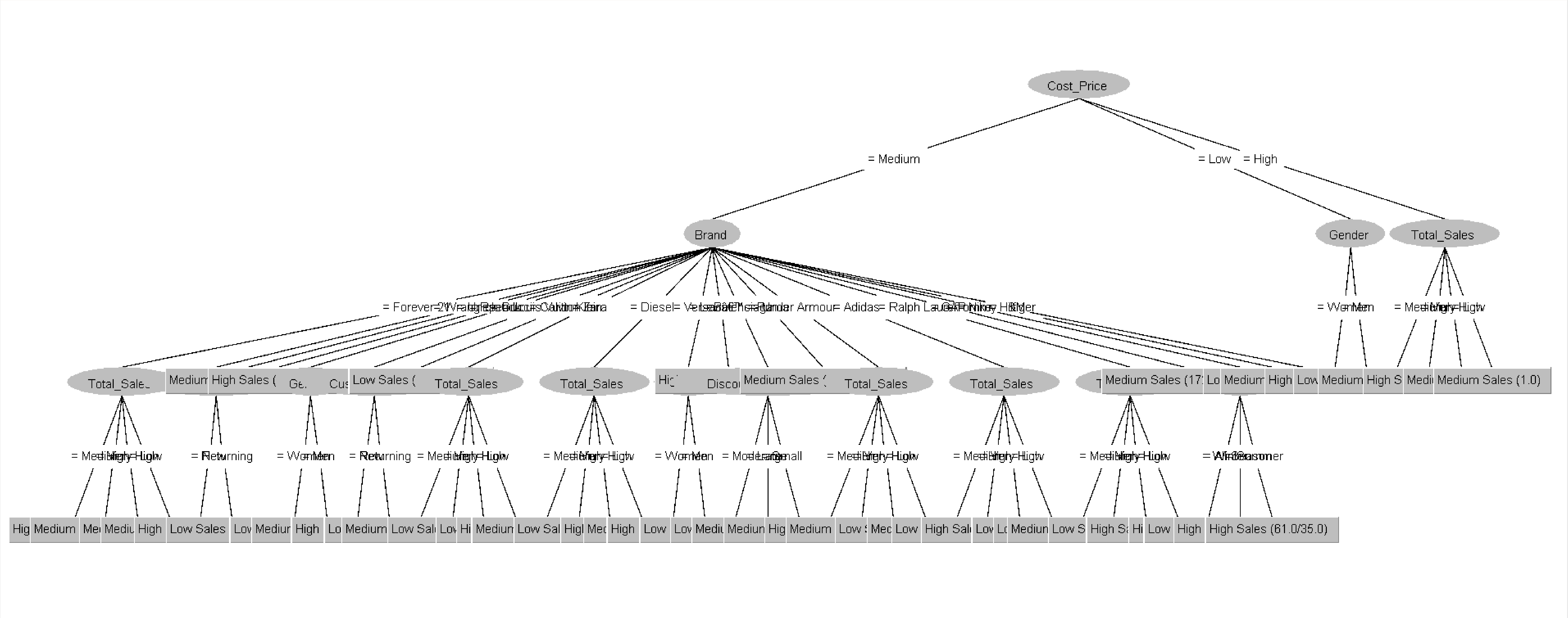
```
|   Brand = Leviâ€™s: High Sales (178.0/109.0)
|   Brand = Balenciaga
|   |   Discount_Percentage = Moderate: Low Sales (62.0/35.0)
|   |   Discount_Percentage = Large: Medium Sales (70.0/39.0)
|   |   Discount_Percentage = Small: Medium Sales (43.0/23.0)
|   Brand = Puma: Medium Sales (173.0/108.0)
|   Brand = Under Armour
|   |   Total_Sales = Medium: High Sales (78.0/46.0)
|   |   Total_Sales = High: Medium Sales (74.0/44.0)
|   |   Total_Sales = Very High: Low Sales (2.0/1.0)
|   |   Total_Sales = Low: Medium Sales (1.0)
|   Brand = Adidas
|   |   Total_Sales = Medium: Low Sales (89.0/53.0)
|   |   Total_Sales = High: High Sales (88.0/54.0)
|   |   Total_Sales = Very High: Low Sales (1.0)
|   |   Total_Sales = Low: Low Sales (5.0/3.0)
|   Brand = Ralph Lauren
|   |   Total_Sales = Medium: Medium Sales (81.0/49.0)
|   |   Total_Sales = High: Low Sales (74.0/47.0)
|   |   Total_Sales = Very High: High Sales (2.0/1.0)
|   |   Total_Sales = Low: High Sales (3.0)
|   Brand = GAP: Medium Sales (172.0/105.0)
|   Brand = Nike
|   |   Season = Winter: Low Sales (60.0/33.0)
|   |   Season = All-Season: High Sales (55.0/32.0)
|   |   Season = Summer: High Sales (61.0/35.0)
|   Brand = Tommy Hilfiger: Low Sales (168.0/97.0)
|   Brand = H&M: Medium Sales (189.0/110.0)
Cost_Price = Low
|   Gender = Women: High Sales (176.0/96.0)
|   Gender = Men: Low Sales (170.0/106.0)
Cost_Price = High
|   Total_Sales = Medium: Medium Sales (120.0/69.0)
|   Total_Sales = High: High Sales (96.0/58.0)
|   Total_Sales = Very High: Medium Sales (2.0)
|   Total_Sales = Low: Medium Sales (1.0)
```

```
Number of Leaves  :     52

Size of the tree :      68


Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        1363               34.2634 %
Incorrectly Classified Instances      2615               65.7366 %
Kappa statistic                          0.0141
Mean absolute error                      0.4439
Root mean squared error                  0.4756
Relative absolute error                 99.8749 %
Root relative squared error            100.8901 %
Total Number of Instances             3978

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.311    0.311    0.336      0.311    0.323      0.000    0.498     0.332     High Sales
                 0.323    0.325    0.332      0.323    0.327      -0.002   0.503     0.332     Low Sales
                 0.395    0.351    0.358      0.395    0.375      0.043    0.513     0.343     Medium Sales
Weighted Avg.    0.343    0.329    0.342      0.343    0.342      0.014    0.505     0.336

=== Confusion Matrix ===

   a   b   c   <-- classified as
 415 437 483 |   a = High Sales
 447 428 450 |   b = Low Sales
 374 424 520 |   c = Medium Sales
```

EXPLANATION

- The accuracy **remains low** (somewhat lower than previously).
- **2/3** of the **predictions** were **incorrect**.
- Nearly **0** is an even **worse agreement** than **previously**.

TREE CROSS VALIDATION

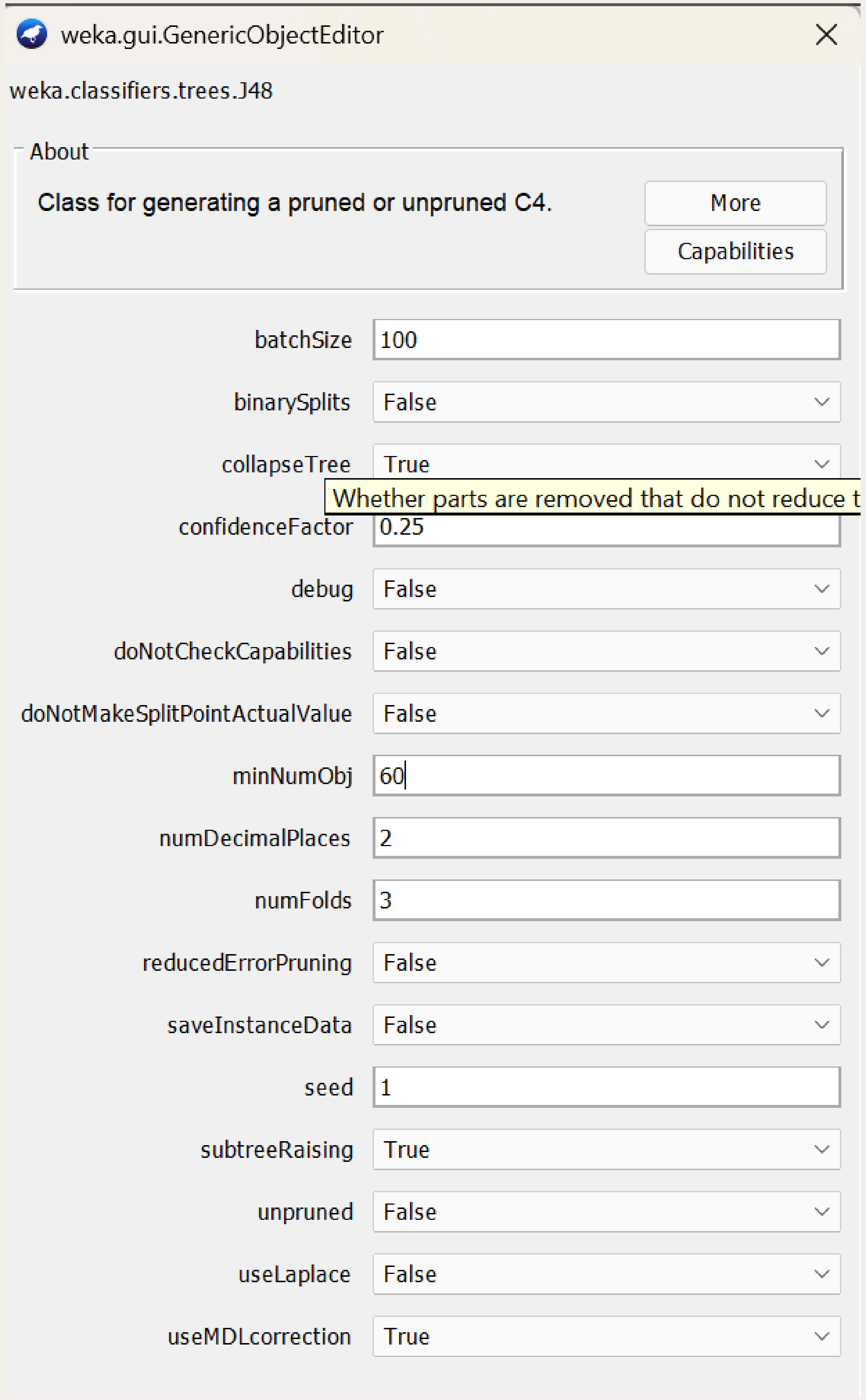


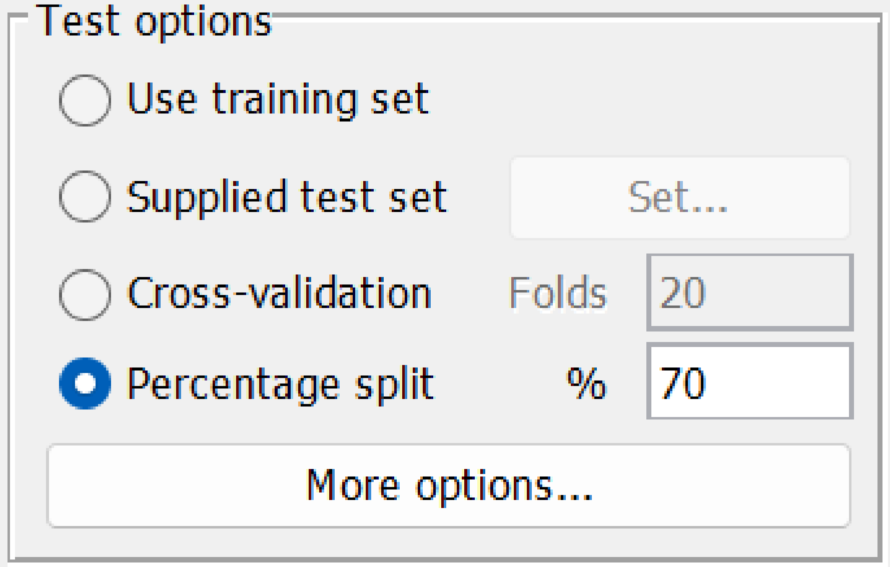
What is percentage split?

- You indicate a percentage, for example, 70%.
- **70%** of the data will be used by Weka to **train** the **model**.
- The **remaining 30%** should be used to **test** the **model**. This is referred to as a holdout evaluation method.
- A **quick model evaluation**

How Apply Classify for J48 percentage split 70:30%?

- In Weka, go to classify and choose classfier > trees >J48. Then all setting default only change the setting for:
- minNumObj = 60
- test option > percentage split > 70%





RESULT AFTER APPLY SPLIT PERCENTAGE 70:30

```
=== Run information ===

Scheme:       weka.classifiers.trees.J48 -C 0.25 -M 60
Relation:     after changing values-weka.filters.supervised.instance.Resample-B0.0-S1-Z100.
Instances:    3978
Attributes:   21
              Product_Category
              Brand
              Product_Name
              Gender
              Size
              Color
              Season
              Payment_Method
              Customer_Type
              Selling_Price
              Cost_Price
              Discount_Percentage
              Quantity_Sold
              Total_Sales
              Stock_Availability
              Customer_Age
              Purchase_Frequency
              Store_Rating
              Return_Rate
              Markup
              Sales_Category
Test mode:    split 70.0% train, remainder test
```

```
=== Classifier model (full training set) ===

J48 pruned tree
------------------

Cost_Price = Medium
|   Brand = Forever 21
|   |   Total_Sales = Medium: High Sales (87.0/55.0)
|   |   Total_Sales = High: Medium Sales (77.0/47.0)
|   |   Total_Sales = Very High: Medium Sales (0.0)
|   |   Total_Sales = Low: Medium Sales (4.0/2.0)
|   Brand = Wrangler
|   |   Customer_Type = New: High Sales (85.0/51.0)
|   |   Customer_Type = Returning: Low Sales (91.0/52.0)
|   Brand = Uniqlo: Medium Sales (168.0/100.0)
|   Brand = Reebok: High Sales (130.0/80.0)
|   Brand = Gucci
|   |   Gender = Women: Low Sales (90.0/55.0)
|   |   Gender = Men: Medium Sales (78.0/46.0)
|   Brand = Louis Vuitton
|   |   Customer_Type = New: High Sales (77.0/43.0)
|   |   Customer_Type = Returning: Low Sales (88.0/55.0)
|   Brand = Calvin Klein: Low Sales (194.0/119.0)
|   Brand = Zara
|   |   Total_Sales = Medium: Medium Sales (110.0/64.0)
|   |   Total_Sales = High: Low Sales (61.0/36.0)
|   |   Total_Sales = Very High: Low Sales (1.0)
|   |   Total_Sales = Low: High Sales (2.0/1.0)
|   Brand = Diesel
|   |   Total_Sales = Medium: Medium Sales (82.0/53.0)
|   |   Total_Sales = High: Low Sales (67.0/40.0)
|   |   Total_Sales = Very High: High Sales (1.0)
|   |   Total_Sales = Low: Medium Sales (2.0)
|   Brand = Versace
|   |   Gender = Women: High Sales (93.0/52.0)
|   |   Gender = Men: Low Sales (96.0/51.0)
```

```
|   Brand = Leviâ€™s: High Sales (178.0/109.0)
|   Brand = Balenciaga
|   |   Discount_Percentage = Moderate: Low Sales (62.0/35.0)
|   |   Discount_Percentage = Large: Medium Sales (70.0/39.0)
|   |   Discount_Percentage = Small: Medium Sales (43.0/23.0)
|   Brand = Puma: Medium Sales (173.0/108.0)
|   Brand = Under Armour
|   |   Total_Sales = Medium: High Sales (78.0/46.0)
|   |   Total_Sales = High: Medium Sales (74.0/44.0)
|   |   Total_Sales = Very High: Low Sales (2.0/1.0)
|   |   Total_Sales = Low: Medium Sales (1.0)
|   Brand = Adidas
|   |   Total_Sales = Medium: Low Sales (89.0/53.0)
|   |   Total_Sales = High: High Sales (88.0/54.0)
|   |   Total_Sales = Very High: Low Sales (1.0)
|   |   Total_Sales = Low: Low Sales (5.0/3.0)
|   Brand = Ralph Lauren
|   |   Total_Sales = Medium: Medium Sales (81.0/49.0)
|   |   Total_Sales = High: Low Sales (74.0/47.0)
|   |   Total_Sales = Very High: High Sales (2.0/1.0)
|   |   Total_Sales = Low: High Sales (3.0)
|   Brand = GAP: Medium Sales (172.0/105.0)
|   Brand = Nike
|   |   Season = Winter: Low Sales (60.0/33.0)
|   |   Season = All-Season: High Sales (55.0/32.0)
|   |   Season = Summer: High Sales (61.0/35.0)
|   Brand = Tommy Hilfiger: Low Sales (168.0/97.0)
|   Brand = H&M: Medium Sales (189.0/110.0)
Cost_Price = Low
|   Gender = Women: High Sales (176.0/96.0)
|   Gender = Men: Low Sales (170.0/106.0)
Cost_Price = High
|   Total_Sales = Medium: Medium Sales (120.0/69.0)
|   Total_Sales = High: High Sales (96.0/58.0)
|   Total_Sales = Very High: Medium Sales (2.0)
|   Total_Sales = Low: Medium Sales (1.0)
```

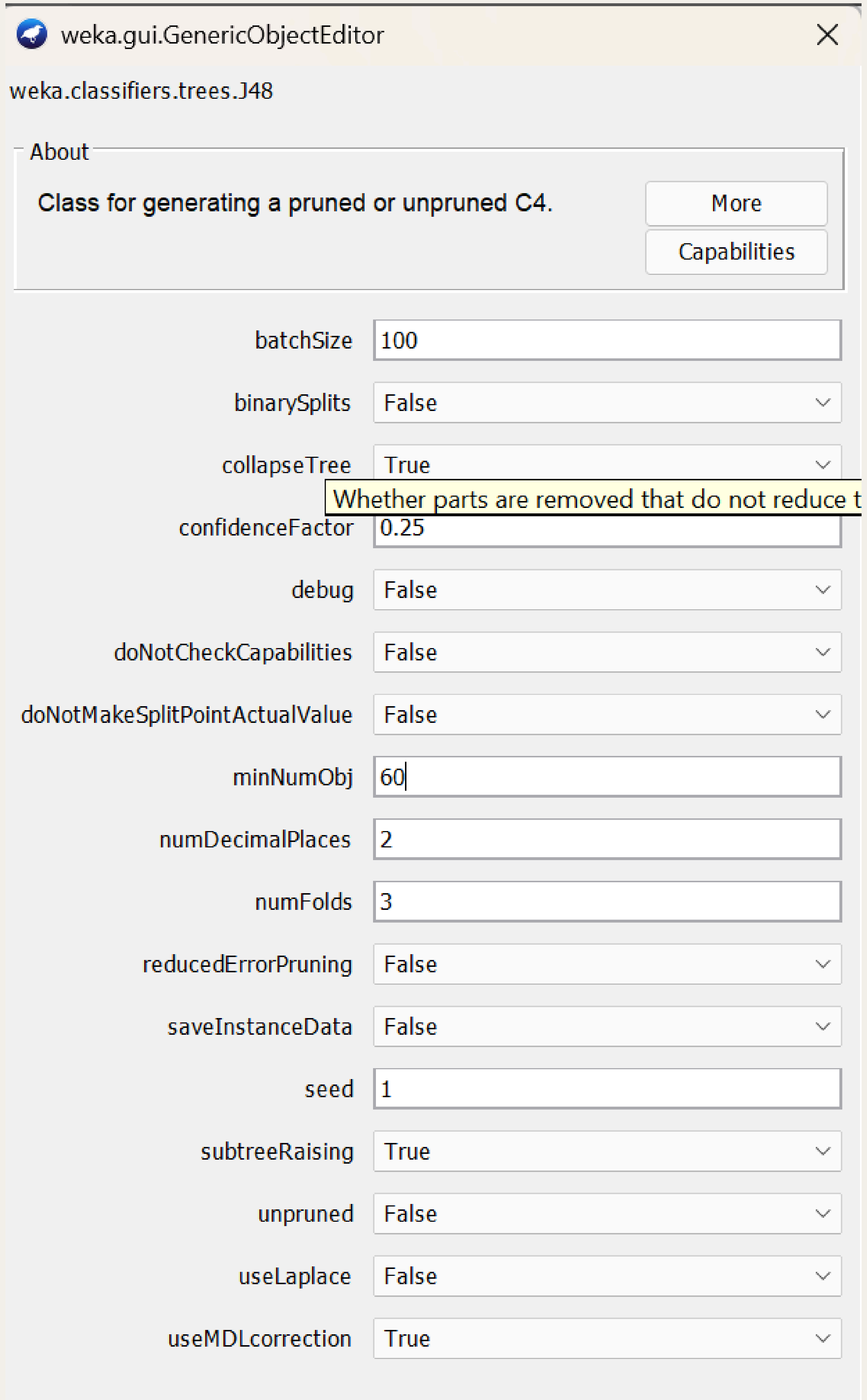
```
Number of Leaves  :     52

Size of the tree :      68


Time taken to build model: 0.01 seconds

=== Evaluation on test split ===

Time taken to test model on test split: 0 seconds

=== Summary ===

Correctly Classified Instances         420               35.2054 %
Incorrectly Classified Instances       773               64.7946 %
Kappa statistic                          0.0312
Mean absolute error                      0.4429
Root mean squared error                  0.4733
Relative absolute error                 99.6114 %
Root relative squared error            100.3366 %
Total Number of Instances             1193

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.391    0.362    0.334      0.391    0.360      0.027    0.523     0.339     High Sales
                 0.274    0.242    0.381      0.274    0.319      0.036    0.517     0.362     Low Sales
                 0.397    0.365    0.350      0.397    0.372      0.032    0.518     0.346     Medium Sales
Weighted Avg.    0.352    0.321    0.356      0.352    0.350      0.032    0.520     0.349

=== Confusion Matrix ===

   a   b   c   <-- classified as
 148  90 141 |   a = High Sales
 154 115 150 |   b = Low Sales
 141  97 157 |   c = Medium Sales
```

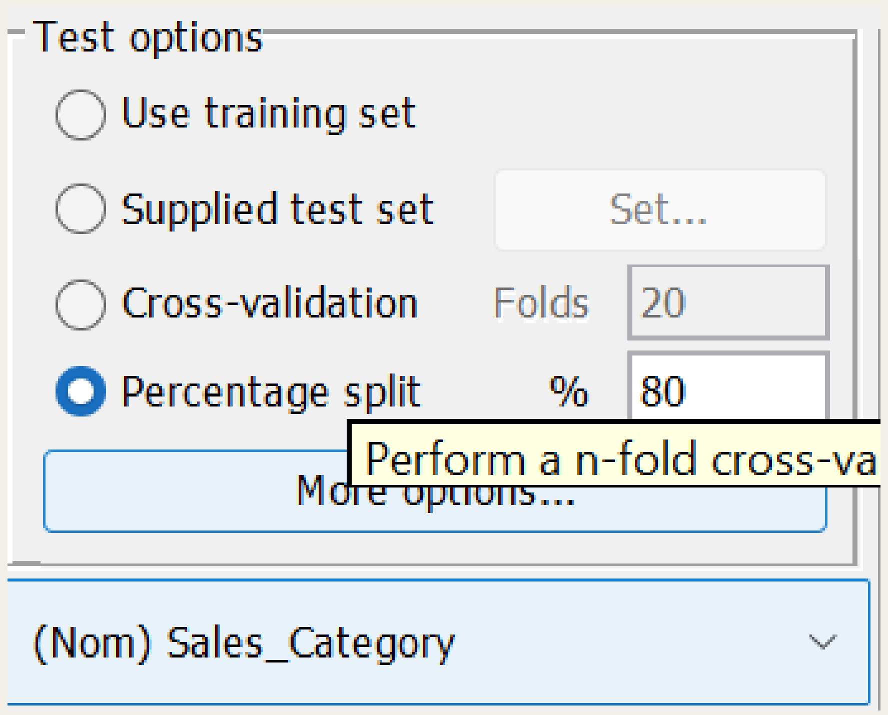
EXPLANATION

- **Accuracy** that is nearly **identical** to k-fold (k=10) results.
- Nearly **2/3** of **predictions** are **inaccurate**.
- Very **weak** agreement, still **close** to **random**.

How Apply Classify for J48 percentage split 80:20%?

- In Weka, go to **classify** and choose **classfier › trees ›J48.** Then all setting default only change the setting for:
- **minNumObj = 60**
- **test option › percentage split › 80%**





RESULT AFTER APPLY SPLIT PERCENTAGE 80:20

```
=== Run information ===

Scheme:       weka.classifiers.trees.J48 -C 0.25 -M 60
Relation:     after changing values-weka.filters.supervised.instance.Resample-B0.0-S1-Z100
Instances:    3978
Attributes:   21
              Product_Category
              Brand
              Product_Name
              Gender
              Size
              Color
              Season
              Payment_Method
              Customer_Type
              Selling_Price
              Cost_Price
              Discount_Percentage
              Quantity_Sold
              Total_Sales
              Stock_Availability
              Customer_Age
              Purchase_Frequency
              Store_Rating
              Return_Rate
              Markup
              Sales_Category
Test mode:    split 80.0% train, remainder test
```

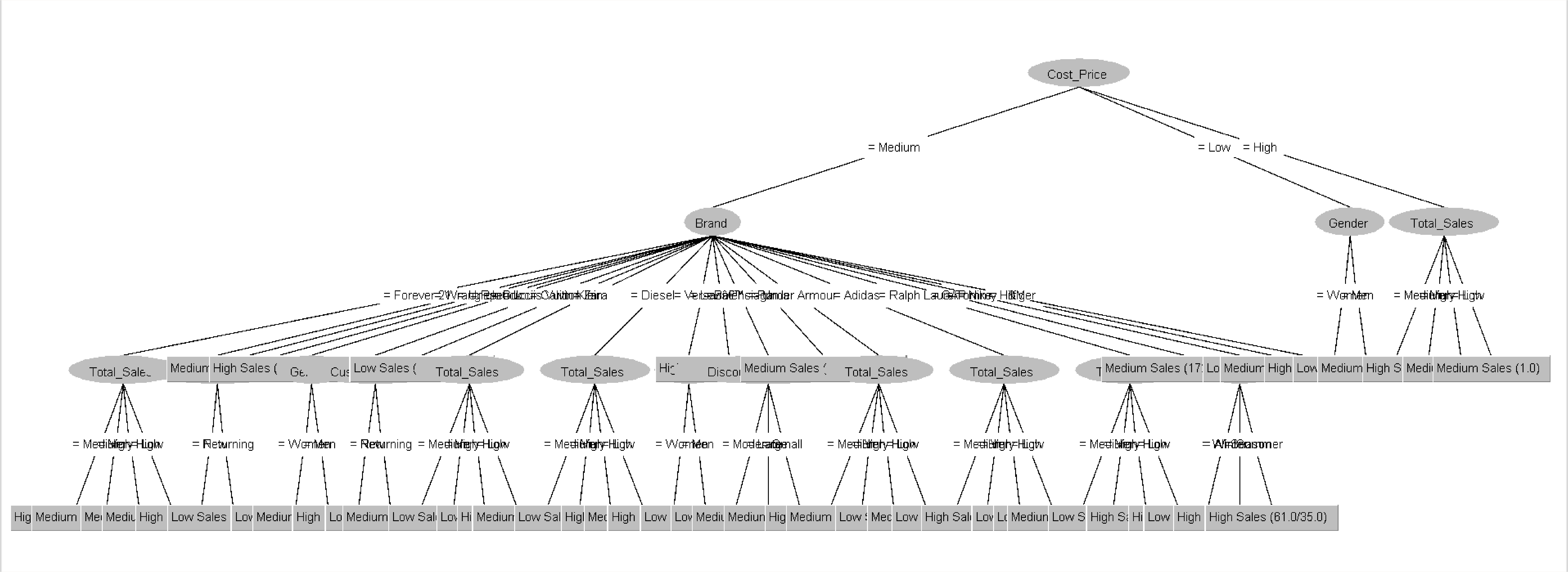
```
=== Classifier model (full training set) ===

J48 pruned tree
------------------

Cost_Price = Medium
|   Brand = Forever 21
|   |   Total_Sales = Medium: High Sales (87.0/55.0)
|   |   Total_Sales = High: Medium Sales (77.0/47.0)
|   |   Total_Sales = Very High: Medium Sales (0.0)
|   |   Total_Sales = Low: Medium Sales (4.0/2.0)
|   Brand = Wrangler
|   |   Customer_Type = New: High Sales (85.0/51.0)
|   |   Customer_Type = Returning: Low Sales (91.0/52.0)
|   Brand = Uniqlo: Medium Sales (168.0/100.0)
|   Brand = Reebok: High Sales (130.0/80.0)
|   Brand = Gucci
|   |   Gender = Women: Low Sales (90.0/55.0)
|   |   Gender = Men: Medium Sales (78.0/46.0)
|   Brand = Louis Vuitton
|   |   Customer_Type = New: High Sales (77.0/43.0)
|   |   Customer_Type = Returning: Low Sales (88.0/55.0)
|   Brand = Calvin Klein: Low Sales (194.0/119.0)
|   Brand = Zara
|   |   Total_Sales = Medium: Medium Sales (110.0/64.0)
|   |   Total_Sales = High: Low Sales (61.0/36.0)
|   |   Total_Sales = Very High: Low Sales (1.0)
|   |   Total_Sales = Low: High Sales (2.0/1.0)
|   Brand = Diesel
|   |   Total_Sales = Medium: Medium Sales (82.0/53.0)
|   |   Total_Sales = High: Low Sales (67.0/40.0)
|   |   Total_Sales = Very High: High Sales (1.0)
|   |   Total_Sales = Low: Medium Sales (2.0)
|   Brand = Versace
|   |   Gender = Women: High Sales (93.0/52.0)
|   |   Gender = Men: Low Sales (96.0/51.0)
|   Brand = Leviâ€™s: High Sales (178.0/109.0)
|   Brand = Balenciaga
|   |   Discount_Percentage = Moderate: Low Sales (62.0/35.0)
|   |   Discount_Percentage = Large: Medium Sales (70.0/39.0)
|   |   Discount_Percentage = Small: Medium Sales (43.0/23.0)
|   Brand = Puma: Medium Sales (173.0/108.0)
|   Brand = Under Armour
|   |   Total_Sales = Medium: High Sales (78.0/46.0)
|   |   Total_Sales = High: Medium Sales (74.0/44.0)
|   |   Total_Sales = Very High: Low Sales (2.0/1.0)
|   |   Total_Sales = Low: Medium Sales (1.0)
|   Brand = Adidas
|   |   Total_Sales = Medium: Low Sales (89.0/53.0)
|   |   Total_Sales = High: High Sales (88.0/54.0)
|   |   Total_Sales = Very High: Low Sales (1.0)
|   |   Total_Sales = Low: Low Sales (5.0/3.0)
|   Brand = Ralph Lauren
|   |   Total_Sales = Medium: Medium Sales (81.0/49.0)
|   |   Total_Sales = High: Low Sales (74.0/47.0)
|   |   Total_Sales = Very High: High Sales (2.0/1.0)
|   |   Total_Sales = Low: High Sales (3.0)
|   Brand = GAP: Medium Sales (172.0/105.0)
|   Brand = Nike
|   |   Season = Winter: Low Sales (60.0/33.0)
|   |   Season = All-Season: High Sales (55.0/32.0)
|   |   Season = Summer: High Sales (61.0/35.0)
|   Brand = Tommy Hilfiger: Low Sales (168.0/97.0)
|   Brand = H&M: Medium Sales (189.0/110.0)
Cost_Price = Low
|   Gender = Women: High Sales (176.0/96.0)
|   Gender = Men: Low Sales (170.0/106.0)
Cost_Price = High
|   Total_Sales = Medium: Medium Sales (120.0/69.0)
|   Total_Sales = High: High Sales (96.0/58.0)
|   Total_Sales = Very High: Medium Sales (2.0)
|   Total_Sales = Low: Medium Sales (1.0)
```

```
Number of Leaves  :     52

Size of the tree :      68


Time taken to build model: 0.01 seconds

=== Evaluation on test split ===

Time taken to test model on test split: 0 seconds

=== Summary ===

Correctly Classified Instances         278               34.9246 %
Incorrectly Classified Instances       518               65.0754 %
Kappa statistic                          0.0233
Mean absolute error                      0.4412
Root mean squared error                  0.4724
Relative absolute error                 99.25   %
Root relative squared error            100.1994 %
Total Number of Instances              796

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.310    0.301    0.331      0.310    0.320      0.009    0.520     0.344     High Sales
                 0.424    0.364    0.373      0.424    0.397      0.058    0.519     0.360     Low Sales
                 0.312    0.311    0.339      0.312    0.325      0.001    0.510     0.352     Medium Sales
Weighted Avg.    0.349    0.326    0.348      0.349    0.348      0.023    0.517     0.352

=== Confusion Matrix ===

   a   b   c   <-- classified as
  80  97  81 |   a = High Sales
  72 114  83 |   b = Low Sales
  90  95  84 |   c = Medium Sales
```

EXPLANATION

- The **accuracy** remains **low** (somewhat lower than previously).
- About 2/3 predictions were incorrect.
- The agreement is nearly **0**, which is far **worse** than before.

TREE PERCENTAGE SPLIT



ACCURACY AND KAPPA COMPARISON

- The Classification results by using the J48 Decision Tree Algorithm across different parameter settings and data splits consistently show low accuracy, approximately 34-35% with Kappa statistic close to zero.

- Adjust the train-test split from 70% to 80% and vary the parameters such as prunning or confidence factors(if applicable), seems to have no effect on enhancing the model's functionality.

- The decision tree model only slightly outperforms random chance in accurately identifying the sales categories, according to the low Kappa values.

- The outcome will suggest that either the features used are lack sufficient predictive power or the decision tree is incapable of successfully capturing the underlying patterns with the available data.

A4. DATA PREPARATION-REDUCTION

INFOGAINATTRIBUTE EVAL + RANKER

INFOGAINATTRIBUTE EVAL

THE EVALUATOR IS BASICALLY THE PROCESS OF MEASURING HOW BEST THE ATTRIBUTE THAT IS HELPFUL TO PREDICTING THE FINAL RESULT.

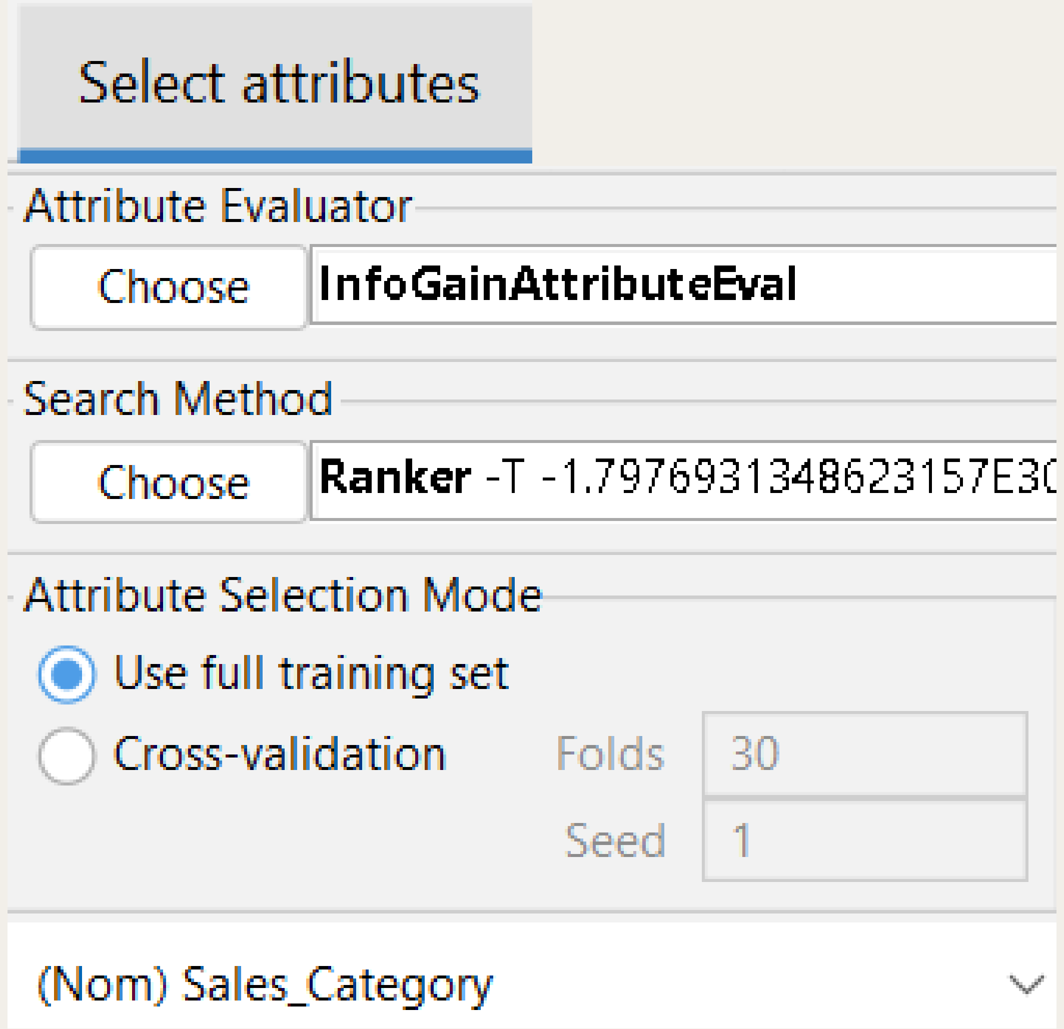
RANKER

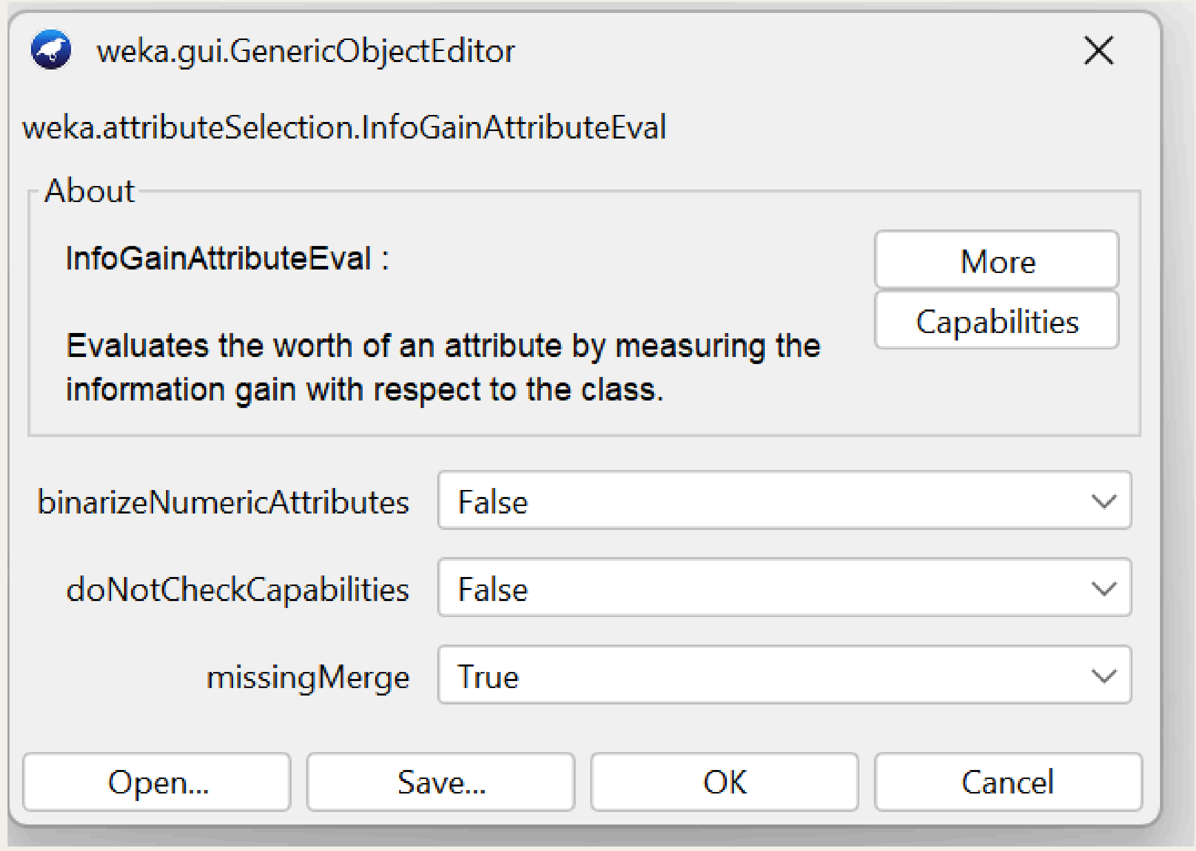
A SEARCH FILTER-LIKE STRATEGY FOR SEARCHING ALL THE BEST POSSIBLE BLENDS FOR EVERY ATTRIBUTE TO FIND THE GOOD SET FOR THE EVALUATOR

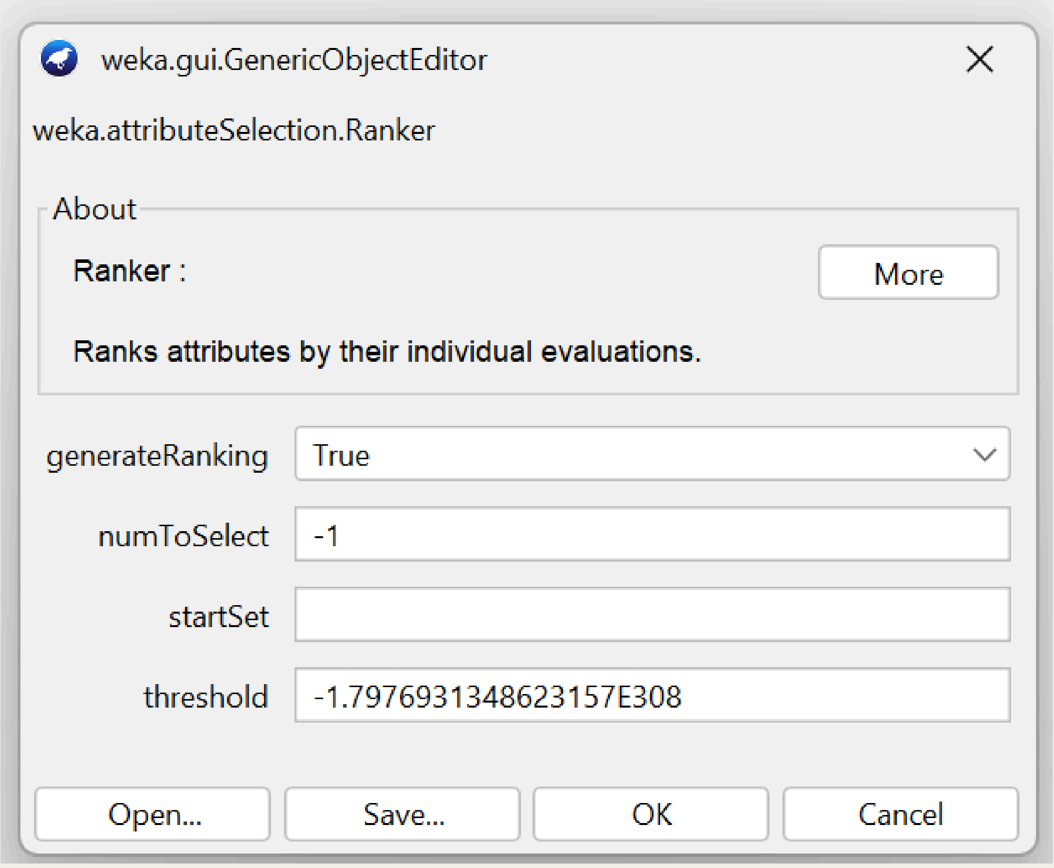
WHY WE NEED TO APPLY THIS INFOGAINATTRIBUTEEVAL?

- To choose the best and worst attributes for the dataset training classification method

HOW TO APPLY THIS FILTER INFOGAINATTRIBUTEEVAL?







- Select Attribute > choose attribute evaluator = InfoGainAttributeEval > choose search method = ranker > attribute selection mode = use full training set > set (nom) Sales_category

- Use default setting for InfoGainAttributeEval
- binarizeNumericAttribute= “False”
- doNotCheckCapabilities = “False”
- missingmerge = “True”

- Use default setting for Ranker
- generateRanking = “True”
- numToSelect = -1
- startset = blank
- threehold= -1.7976931348623157E308

AFTER APPLY THIS FILTER, INFOGAINATTRIBUTEEVAL

Attribute selection output

```
=== Attribute Selection on all input data ===

Search Method:
	Attribute ranking.

Attribute Evaluator (supervised, Class (nominal): 21 Sales_Category):
	Information Gain Ranking Filter

Ranked attributes:
 0.009182    2 Brand
 0.006299    3 Product_Name
 0.002294    6 Color
 0.001799   16 Customer_Age
 0.001723    1 Product_Category
 0.001611   11 Cost_Price
 0.001355    5 Size
 0.001152   14 Total_Sales
 0.001057   19 Return_Rate
 0.001044    7 Season
 0.001014   15 Stock_Availability
 0.000954   12 Discount_Percentage
 0.000859   13 Quantity_Sold
 0.000842   17 Purchase_Frequency
 0.000719   20 Markup
 0.000718   18 Store_Rating
 0.000506    8 Payment_Method
 0.000356    9 Customer_Type
 0.000195    4 Gender
 0.000192   10 Selling_Price

Selected attributes: 2,3,6,16,1,11,5,14,19,7,15,12,13,17,20,18,8,9,4,10 : 20
```

→ **11 best Attribute been choosen for Dataset 1**

→ **11 worst attribute has been choose for Dataset 2**

- The result shows it sorts based on the rank of the attribute and display value.
- Based on the value of the rank, we know which is the best, and until a lower attribute can be useful for predicting later
- Moving to next step, which is remove the attribute or data reduction

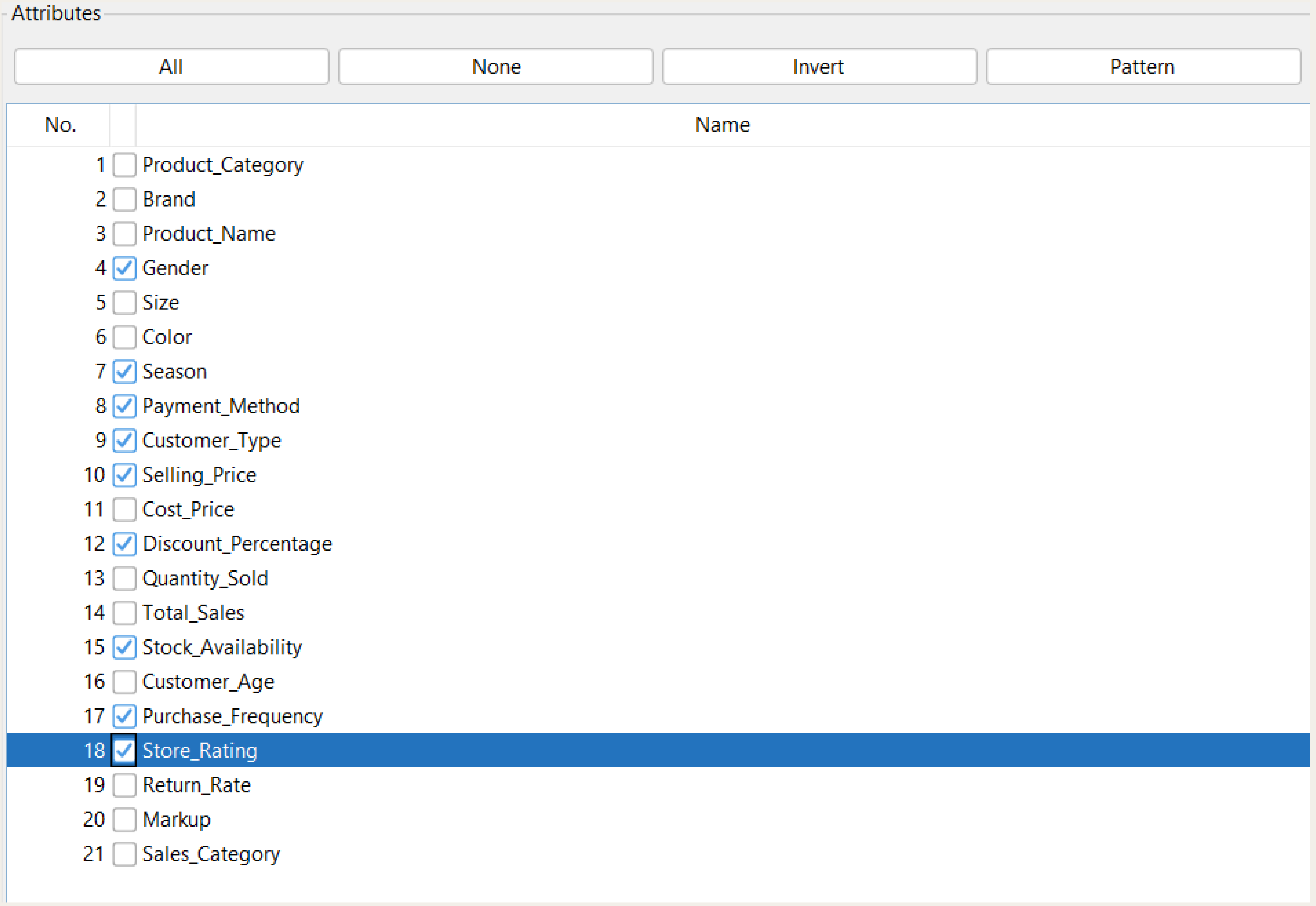
WHY DO WE NEED TO DO DATA REDUCTION/REMOVE ATTRIBUTES?

- Too much overfitting of data, which is when classified J48, it generates a complicated tree.
- The training time consumes a lot of time when evaluating the method.
- The accuracy of the model
- To compare the result between datasets 1 and 2 and before reduction

HOW TO REMOVE ATTRIBUTE BASED ON INFOGAINATTRBUTEEVAL

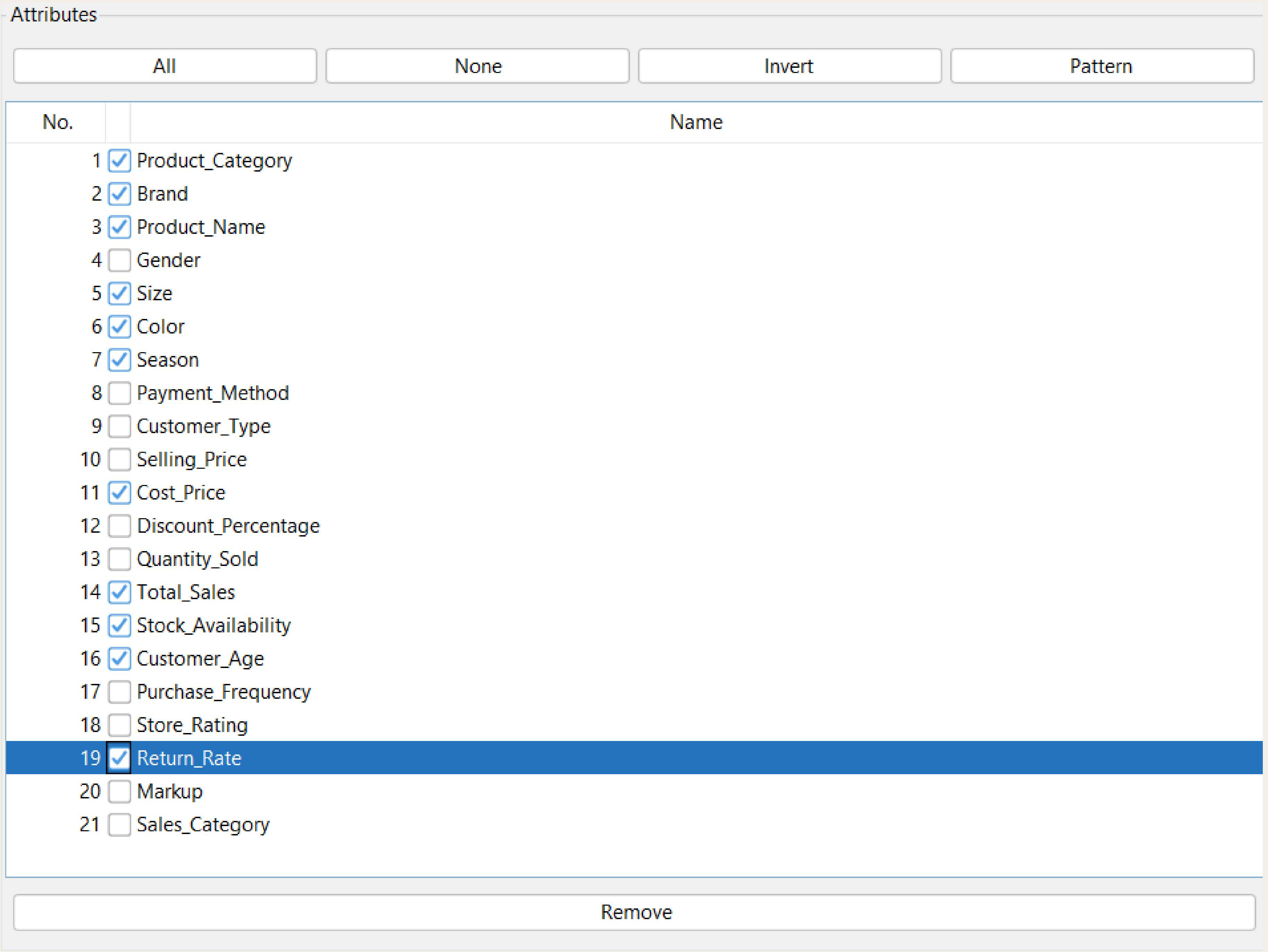
DATASET 1

- Based on the selected attribute on infogainattributeeval that has been sorted > Go to preprocess > choose 11 best attributes:
- Gender, season, payment_method, customer_type, selling_price, discount_percentage, stock_availability, purchase_frequency, store rating, don't remove class > click Remove.



DATASET 2

- Based on the selected attribute on infogainattributeeval that has been sorted > Go to preprocess > choose 11 worst attributes:
- Product_Category, brand, product_name, size, color, season, cost_price, total_sales, stock_availability, customer_age, return_rate, don't remove class > click Remove



AFTER REMOVE ATTRIBUTE

DATASET 1

Viewer

Relation: after changing values-weka.filters.supervised.instance.Resample-B0.0-S1-Z100.0-weka.filters.supervised.instance.Resample-B0.0-S1-Z4.0-weka.filters.unsupervised.attribute.Remove-R4,8-10,12-13,17-18,20

No.	1: Product_Category Nominal	2: Brand Nominal	3: Product_Name Nominal	4: Size Nominal	5: Color Nominal	6: Season Nominal	7: Cost_Price Nominal	8: Total_Sales Nominal	9: Stock_Availability Nominal	10: Customer_Age Nominal	11: Return_Rate Nominal	12: **Sales_Category** Nominal
1	Bottoms	Balenci...	Kurta	XXL	Red	All-Season	Medium	Medium	Low	Adult	High	High Sales
2	Athleisure	Nike	Leggings	XL	Purple	Summer	Medium	Medium	Medium	Elder	Medium	High Sales
3	Tops	Ralph L...	Skirt	XXL	Purple	All-Season	Medium	High	High	Teen	High	High Sales
4	Traditional Wear	Ralph L...	Palazzo	M	Red	Winter	Medium	High	High	Adult	High	High Sales
5	Traditional Wear	Versace	Dress	XL	Purple	All-Season	Medium	Medium	High	Elder	Low	High Sales
6	Traditional Wear	Forever...	Polo Shirt	S	Black	Winter	Medium	Medium	Low	Teen	Low	High Sales
7	Bottoms	Puma	Leggings	XL	Green	Winter	Medium	Medium	Low	Senior	High	High Sales
8	Traditional Wear	Gucci	Palazzo	XL	Pink	All-Season	Medium	Medium	Low	Adult	Medium	High Sales
9	Bottoms	Forever...	Blazer	XXL	Purple	Winter	Medium	Medium	Low	Elder	High	High Sales
10	Athleisure	H&M	Sweatshirt	L	Red	Winter	Medium	Medium	Low	Senior	High	High Sales
11	Tops	Balenci...	Casual Shirt	XXL	Yellow	Winter	Medium	High	Medium	Elder	High	High Sales
12	Tops	Wrangler	Jumpsuit	S	Yellow	Winter	Medium	Medium	High	Teen	Low	High Sales
13	Athleisure	Nike	Jacket	XL	Pink	Summer	Medium	Medium	Low	Teen	High	High Sales
14	Tops	Gucci	Jeans	XL	Blue	Winter	Medium	Medium	Low	Teen	Medium	High Sales
15	Athleisure	GAP	T-Shirt	XL	White	All-Season	Medium	Medium	Low	Elder	Low	High Sales
16	Outerwear	Diesel	Tops	S	White	Winter	Medium	Medium	Medium	Teen	High	High Sales
17	Athleisure	Versace	Formal Shirt	L	Red	Summer	Medium	High	Medium	Senior	High	High Sales
18	Tops	Calvin ...	Trousers	L	Green	Winter	Medium	Low	Medium	Adult	High	High Sales
19	Traditional Wear	Ralph L...	T-Shirt	XL	Red	Summer	High	Medium	High	Senior	Medium	High Sales
20	Traditional Wear	Calvin ...	Jeans	L	Blue	All-Season	Medium	High	High	Elder	High	High Sales

- Some unuseful attributes have been reduced until 12 attribute left

DATASET 2

Viewer

Relation: after changing values-weka.filters.supervised.instance.Resample-B0.0-S1-Z100.0-weka.filters.supervised.instance.Resample-B0.0-S1-Z4.0-weka.filters.unsupervised.attribute.Remove-R1-3,5-6,11,14,16,19

No.	1: Gender Nominal	2: Season Nominal	3: Payment_Method Nominal	4: Customer_Type Nominal	5: Selling_Price Nominal	6: Discount_Percentage Nominal	7: Quantity_Sold Nominal	8: Stock_Availability Nominal	9: Purchase_Frequency Nominal	10: Store_Rating Nominal	11: Markup Nominal	12: **Sales_Category** Nominal
1	Men	All-Season	Cash	Returning	Medium	Large	High	Low	Moderate	Average	Medium	High Sales
2	Women	Summer	Card	Returning	Medium	Moderate	Low	Medium	Moderate	Poor	Medium	High Sales
3	Men	All-Season	Cash	New	Medium	Small	Medium	High	Moderate	Average	Medium	High Sales
4	Women	Winter	Cash	Returning	Medium	Small	Very Low	High	Moderate	Average	Medium	High Sales
5	Men	All-Season	UPI	Returning	Medium	Large	Medium	High	Moderate	Average	Medium	High Sales
6	Women	Winter	Card	Returning	Medium	Large	High	Low	Moderate	Excellent	Medium	High Sales
7	Men	Winter	Card	New	Medium	Moderate	High	Low	Moderate	Average	Medium	High Sales
8	Men	All-Season	Cash	Returning	Medium	Small	Very Low	Low	Moderate	Average	Medium	High Sales
9	Men	Winter	UPI	New	Medium	Small	Medium	Low	Moderate	Average	Medium	High Sales
10	Women	Winter	Cash	New	Medium	Moderate	High	Low	Moderate	Average	Medium	High Sales
11	Women	Winter	UPI	New	Medium	Large	Very Low	Medium	Moderate	Excellent	Medium	High Sales
12	Men	Winter	Card	New	Medium	Small	Very Low	High	Moderate	Poor	Medium	High Sales
13	Women	Summer	Cash	Returning	Medium	Small	High	Low	Moderate	Average	Medium	High Sales
14	Men	Winter	Net Banking	Returning	Medium	Large	Medium	Low	Moderate	Excellent	Medium	High Sales
15	Women	All-Season	Net Banking	Returning	Medium	Moderate	Low	Low	Moderate	Average	Medium	High Sales
16	Men	Winter	Net Banking	New	High	Moderate	Very Low	Medium	Moderate	Average	Medium	High Sales
17	Women	Summer	Cash	New	Medium	Large	Very Low	Medium	Moderate	Excellent	Medium	High Sales
18	Women	Winter	Card	New	Medium	Small	High	Medium	Moderate	Excellent	Medium	High Sales
19	Men	Summer	Net Banking	Returning	Medium	Small	High	High	Moderate	Excellent	Low	High Sales
20	Men	All-Season	UPI	Returning	Medium	Moderate	Very Low	High	Moderate	Poor	Medium	High Sales

- Some useful attributes have been reduced until 12 attribute left

B.MODEL DEVELOPMENT AND EVALUATION

FIRST REDUCED

J48

RESULT AFTER APPLY J48 10 FOLD

```
=== Run information ===

Scheme:       weka.classifiers.trees.J48 -C 0.25 -M 60
Relation:     after changing values-weka.filters.supervised.instance.Resample-B0.0-S1-Z100
Instances:    3978
Attributes:   12
              Product_Category
              Brand
              Product_Name
              Size
              Color
              Season
              Cost_Price
              Total_Sales
              Stock_Availability
              Customer_Age
              Return_Rate
              Sales_Category
Test mode:    10-fold cross-validation
```

```
=== Classifier model (full training set) ===

J48 pruned tree
------------------

Cost_Price = Medium
|   Brand = Forever 21
|   |   Total_Sales = Medium: High Sales (87.0/55.0)
|   |   Total_Sales = High: Medium Sales (77.0/47.0)
|   |   Total_Sales = Very High: Medium Sales (0.0)
|   |   Total_Sales = Low: Medium Sales (4.0/2.0)
|   Brand = Wrangler: High Sales (176.0/113.0)
|   Brand = Uniqlo: Medium Sales (168.0/100.0)
|   Brand = Reebok: High Sales (130.0/80.0)
|   Brand = Gucci: High Sales (168.0/110.0)
|   Brand = Louis Vuitton
|   |   Total_Sales = Medium: Low Sales (75.0/47.0)
|   |   Total_Sales = High: High Sales (84.0/52.0)
|   |   Total_Sales = Very High: High Sales (2.0/1.0)
|   |   Total_Sales = Low: High Sales (4.0/2.0)
|   Brand = Calvin Klein: Low Sales (194.0/119.0)
|   Brand = Zara
|   |   Total_Sales = Medium: Medium Sales (110.0/64.0)
|   |   Total_Sales = High: Low Sales (61.0/36.0)
|   |   Total_Sales = Very High: Low Sales (1.0)
|   |   Total_Sales = Low: High Sales (2.0/1.0)
|   Brand = Diesel
|   |   Total_Sales = Medium: Medium Sales (82.0/53.0)
|   |   Total_Sales = High: Low Sales (67.0/40.0)
|   |   Total_Sales = Very High: High Sales (1.0)
|   |   Total_Sales = Low: Medium Sales (2.0)
|   Brand = Versace
|   |   Total_Sales = Medium: Low Sales (105.0/62.0)
|   |   Total_Sales = High: High Sales (80.0/51.0)
|   |   Total_Sales = Very High: High Sales (1.0)
|   |   Total_Sales = Low: Medium Sales (3.0/1.0)
|   Brand = Leviâ€™s: High Sales (178.0/109.0)
|   Brand = Balenciaga: Medium Sales (175.0/106.0)
|   Brand = Puma: Medium Sales (173.0/108.0)
|   Brand = Under Armour
|   |   Total_Sales = Medium: High Sales (78.0/46.0)
|   |   Total_Sales = High: Medium Sales (74.0/44.0)
|   |   Total_Sales = Very High: Low Sales (2.0/1.0)
|   |   Total_Sales = Low: Medium Sales (1.0)
|   Brand = Adidas
|   |   Total_Sales = Medium: Low Sales (89.0/53.0)
|   |   Total_Sales = High: High Sales (88.0/54.0)
|   |   Total_Sales = Very High: Low Sales (1.0)
|   |   Total_Sales = Low: Low Sales (5.0/3.0)
|   Brand = Ralph Lauren
|   |   Total_Sales = Medium: Medium Sales (81.0/49.0)
|   |   Total_Sales = High: Low Sales (74.0/47.0)
|   |   Total_Sales = Very High: High Sales (2.0/1.0)
|   |   Total_Sales = Low: High Sales (3.0)
|   Brand = GAP: Medium Sales (172.0/105.0)
|   Brand = Nike
|   |   Season = Winter: Low Sales (60.0/33.0)
|   |   Season = All-Season: High Sales (55.0/32.0)
|   |   Season = Summer: High Sales (61.0/35.0)
|   Brand = Tommy Hilfiger: Low Sales (168.0/97.0)
|   Brand = H&M: Medium Sales (189.0/110.0)
Cost_Price = Low: High Sales (346.0/212.0)
Cost_Price = High
|   Total_Sales = Medium: Medium Sales (120.0/69.0)
|   Total_Sales = High: High Sales (96.0/58.0)
|   Total_Sales = Very High: Medium Sales (2.0)
|   Total_Sales = Low: Medium Sales (1.0)

Number of Leaves  :     51

Size of the tree :      63
```

```
Time taken to build model: 0.02 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        1282               32.2272 %
Incorrectly Classified Instances      2696               67.7728 %
Kappa statistic                         -0.0165
Mean absolute error                      0.4452
Root mean squared error                  0.4751
Relative absolute error                100.1654 %
Root relative squared error            100.7862 %
Total Number of Instances             3978

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.291    0.303    0.326      0.291    0.307      -0.013   0.488     0.334     High Sales
                 0.370    0.395    0.319      0.370    0.342      -0.024   0.494     0.324     Low Sales
                 0.307    0.318    0.323      0.307    0.315      -0.012   0.495     0.326     Medium Sales
Weighted Avg.    0.322    0.339    0.323      0.322    0.321      -0.016   0.492     0.328

=== Confusion Matrix ===

   a   b   c   <-- classified as
 388 544 403 |   a = High Sales
 392 490 443 |   b = Low Sales
 410 504 404 |   c = Medium Sales
```

EXPLANATION

- similar to using all features, showing **no significant improvement.**
- most of the instances is **incorrectly classified**
- **Kappa** statistic near zero indicates performance barely better than random.

RESULT AFTER APPLY J48 20 FOLD

```
=== Run information ===

Scheme:       weka.classifiers.trees.J48 -C 0.25 -M 60
Relation:     after changing values-weka.filters.supervised.instance.Resample-B0.0-S1-Z100
Instances:    3978
Attributes:   12
              Product_Category
              Brand
              Product_Name
              Size
              Color
              Season
              Cost_Price
              Total_Sales
              Stock_Availability
              Customer_Age
              Return_Rate
              Sales_Category
Test mode:    20-fold cross-validation
```

```
=== Classifier model (full training set) ===

J48 pruned tree
------------------

Cost_Price = Medium
|   Brand = Forever 21
|   |   Total_Sales = Medium: High Sales (87.0/55.0)
|   |   Total_Sales = High: Medium Sales (77.0/47.0)
|   |   Total_Sales = Very High: Medium Sales (0.0)
|   |   Total_Sales = Low: Medium Sales (4.0/2.0)
|   Brand = Wrangler: High Sales (176.0/113.0)
|   Brand = Uniqlo: Medium Sales (168.0/100.0)
|   Brand = Reebok: High Sales (130.0/80.0)
|   Brand = Gucci: High Sales (168.0/110.0)
|   Brand = Louis Vuitton
|   |   Total_Sales = Medium: Low Sales (75.0/47.0)
|   |   Total_Sales = High: High Sales (84.0/52.0)
|   |   Total_Sales = Very High: High Sales (2.0/1.0)
|   |   Total_Sales = Low: High Sales (4.0/2.0)
|   Brand = Calvin Klein: Low Sales (194.0/119.0)
|   Brand = Zara
|   |   Total_Sales = Medium: Medium Sales (110.0/64.0)
|   |   Total_Sales = High: Low Sales (61.0/36.0)
|   |   Total_Sales = Very High: Low Sales (1.0)
|   |   Total_Sales = Low: High Sales (2.0/1.0)
|   Brand = Diesel
|   |   Total_Sales = Medium: Medium Sales (82.0/53.0)
|   |   Total_Sales = High: Low Sales (67.0/40.0)
|   |   Total_Sales = Very High: High Sales (1.0)
|   |   Total_Sales = Low: Medium Sales (2.0)
|   Brand = Versace
|   |   Total_Sales = Medium: Low Sales (105.0/62.0)
|   |   Total_Sales = High: High Sales (80.0/51.0)
|   |   Total_Sales = Very High: High Sales (1.0)
|   |   Total_Sales = Low: Medium Sales (3.0/1.0)
```

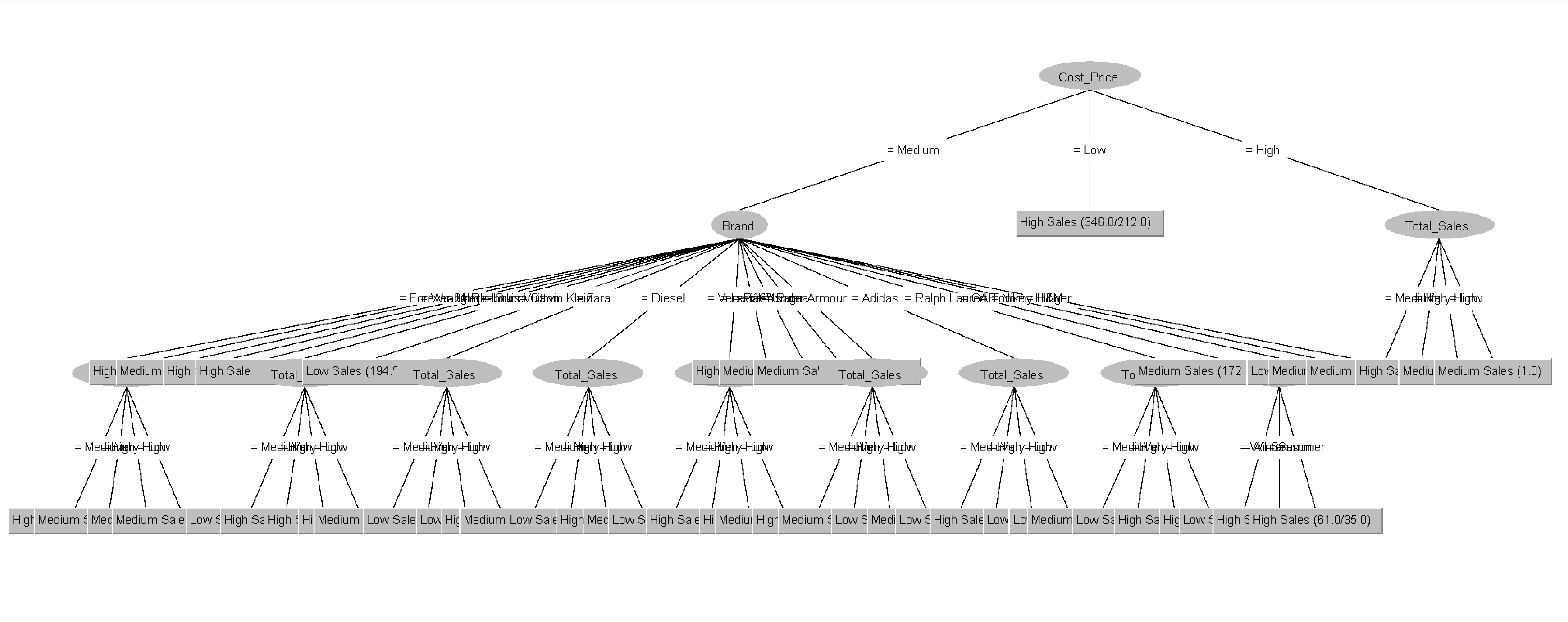
```
|   Brand = Leviâ€™s: High Sales (178.0/109.0)
|   Brand = Balenciaga: Medium Sales (175.0/106.0)
|   Brand = Puma: Medium Sales (173.0/108.0)
|   Brand = Under Armour
|   |   Total_Sales = Medium: High Sales (78.0/46.0)
|   |   Total_Sales = High: Medium Sales (74.0/44.0)
|   |   Total_Sales = Very High: Low Sales (2.0/1.0)
|   |   Total_Sales = Low: Medium Sales (1.0)
|   Brand = Adidas
|   |   Total_Sales = Medium: Low Sales (89.0/53.0)
|   |   Total_Sales = High: High Sales (88.0/54.0)
|   |   Total_Sales = Very High: Low Sales (1.0)
|   |   Total_Sales = Low: Low Sales (5.0/3.0)
|   Brand = Ralph Lauren
|   |   Total_Sales = Medium: Medium Sales (81.0/49.0)
|   |   Total_Sales = High: Low Sales (74.0/47.0)
|   |   Total_Sales = Very High: High Sales (2.0/1.0)
|   |   Total_Sales = Low: High Sales (3.0)
|   Brand = GAP: Medium Sales (172.0/105.0)
|   Brand = Nike
|   |   Season = Winter: Low Sales (60.0/33.0)
|   |   Season = All-Season: High Sales (55.0/32.0)
|   |   Season = Summer: High Sales (61.0/35.0)
|   Brand = Tommy Hilfiger: Low Sales (168.0/97.0)
|   Brand = H&M: Medium Sales (189.0/110.0)
Cost_Price = Low: High Sales (346.0/212.0)
Cost_Price = High
|   Total_Sales = Medium: Medium Sales (120.0/69.0)
|   Total_Sales = High: High Sales (96.0/58.0)
|   Total_Sales = Very High: Medium Sales (2.0)
|   Total_Sales = Low: Medium Sales (1.0)

Number of Leaves  :     51

Size of the tree :      63
```

```
Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        1274               32.0261 %
Incorrectly Classified Instances      2704               67.9739 %
Kappa statistic                         -0.0196
Mean absolute error                      0.445
Root mean squared error                  0.4763
Relative absolute error                100.137  %
Root relative squared error            101.0489 %
Total Number of Instances             3978

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.304    0.347    0.307      0.304    0.306      -0.043   0.474     0.316     High Sales
                 0.310    0.344    0.310      0.310    0.310      -0.034   0.494     0.322     Low Sales
                 0.347    0.329    0.343      0.347    0.345      0.018    0.511     0.338     Medium Sales
Weighted Avg.    0.320    0.340    0.320      0.320    0.320      -0.020   0.493     0.325

=== Confusion Matrix ===

   a   b   c   <-- classified as
 406 477 452 |   a = High Sales
 491 411 423 |   b = Low Sales
 425 436 457 |   c = Medium Sales
```

EXPLANATION

- Accuracy **dropped** to about **32.0%**, which is **lower** than the **34.4% accuracy** at **k=10.**
- The **Kappa** statistic is **negative (-0.0196),** meaning the model performs worse than random guessing.

TREE CROSS VALIDATION



RESULT AFTER APPLY J48 SPLIT PERCENTAGE 70:30

```
=== Run information ===

Scheme:       weka.classifiers.trees.J48 -C 0.25 -M 60
Relation:     after changing values-weka.filters.supervised.instance.Resample-B0.0-S1-Z100
Instances:    3978
Attributes:   12
              Product_Category
              Brand
              Product_Name
              Size
              Color
              Season
              Cost_Price
              Total_Sales
              Stock_Availability
              Customer_Age
              Return_Rate
              Sales_Category
Test mode:    split 70.0% train, remainder test

=== Classifier model (full training set) ===
```

```
=== Classifier model (full training set) ===

J48 pruned tree
------------------

Cost_Price = Medium
|   Brand = Forever 21
|   |   Total_Sales = Medium: High Sales (87.0/55.0)
|   |   Total_Sales = High: Medium Sales (77.0/47.0)
|   |   Total_Sales = Very High: Medium Sales (0.0)
|   |   Total_Sales = Low: Medium Sales (4.0/2.0)
|   Brand = Wrangler: High Sales (176.0/113.0)
|   Brand = Uniqlo: Medium Sales (168.0/100.0)
|   Brand = Reebok: High Sales (130.0/80.0)
|   Brand = Gucci: High Sales (168.0/110.0)
|   Brand = Louis Vuitton
|   |   Total_Sales = Medium: Low Sales (75.0/47.0)
|   |   Total_Sales = High: High Sales (84.0/52.0)
|   |   Total_Sales = Very High: High Sales (2.0/1.0)
|   |   Total_Sales = Low: High Sales (4.0/2.0)
|   Brand = Calvin Klein: Low Sales (194.0/119.0)
|   Brand = Zara
|   |   Total_Sales = Medium: Medium Sales (110.0/64.0)
|   |   Total_Sales = High: Low Sales (61.0/36.0)
|   |   Total_Sales = Very High: Low Sales (1.0)
|   |   Total_Sales = Low: High Sales (2.0/1.0)
|   Brand = Diesel
|   |   Total_Sales = Medium: Medium Sales (82.0/53.0)
|   |   Total_Sales = High: Low Sales (67.0/40.0)
|   |   Total_Sales = Very High: High Sales (1.0)
|   |   Total_Sales = Low: Medium Sales (2.0)
|   Brand = Versace
|   |   Total_Sales = Medium: Low Sales (105.0/62.0)
|   |   Total_Sales = High: High Sales (80.0/51.0)
|   |   Total_Sales = Very High: High Sales (1.0)
|   |   Total_Sales = Low: Medium Sales (3.0/1.0)
|   Brand = Leviâ€™s: High Sales (178.0/109.0)
|   Brand = Balenciaga: Medium Sales (175.0/106.0)
|   Brand = Puma: Medium Sales (173.0/108.0)
|   Brand = Under Armour
|   |   Total_Sales = Medium: High Sales (78.0/46.0)
|   |   Total_Sales = High: Medium Sales (74.0/44.0)
|   |   Total_Sales = Very High: Low Sales (2.0/1.0)
|   |   Total_Sales = Low: Medium Sales (1.0)
|   Brand = Adidas
|   |   Total_Sales = Medium: Low Sales (89.0/53.0)
|   |   Total_Sales = High: High Sales (88.0/54.0)
|   |   Total_Sales = Very High: Low Sales (1.0)
|   |   Total_Sales = Low: Low Sales (5.0/3.0)
|   Brand = Ralph Lauren
|   |   Total_Sales = Medium: Medium Sales (81.0/49.0)
|   |   Total_Sales = High: Low Sales (74.0/47.0)
|   |   Total_Sales = Very High: High Sales (2.0/1.0)
|   |   Total_Sales = Low: High Sales (3.0)
|   Brand = GAP: Medium Sales (172.0/105.0)
|   Brand = Nike
|   |   Season = Winter: Low Sales (60.0/33.0)
|   |   Season = All-Season: High Sales (55.0/32.0)
|   |   Season = Summer: High Sales (61.0/35.0)
|   Brand = Tommy Hilfiger: Low Sales (168.0/97.0)
|   Brand = H&M: Medium Sales (189.0/110.0)
Cost_Price = Low: High Sales (346.0/212.0)
Cost_Price = High
|   Total_Sales = Medium: Medium Sales (120.0/69.0)
|   Total_Sales = High: High Sales (96.0/58.0)
|   Total_Sales = Very High: Medium Sales (2.0)
|   Total_Sales = Low: Medium Sales (1.0)

Number of Leaves  :     51

Size of the tree :      63
```

```
Time taken to build model: 0 seconds

=== Evaluation on test split ===

Time taken to test model on test split: 0 seconds

=== Summary ===

Correctly Classified Instances         404               33.8642 %
Incorrectly Classified Instances       789               66.1358 %
Kappa statistic                          0.0115
Mean absolute error                      0.4439
Root mean squared error                  0.4737
Relative absolute error                 99.8387 %
Root relative squared error            100.4254 %
Total Number of Instances             1193

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.380    0.376    0.320      0.380    0.347      0.004    0.509     0.327     High Sales
                 0.267    0.234    0.382      0.267    0.315      0.037    0.517     0.369     Low Sales
                 0.375    0.378    0.329      0.375    0.350      -0.004   0.504     0.335     Medium Sales
Weighted Avg.    0.339    0.327    0.345      0.339    0.337      0.013    0.510     0.345

=== Confusion Matrix ===

   a   b   c   <-- classified as
 144  83 152 |   a = High Sales
 157 112 150 |   b = Low Sales
 149  98 148 |   c = Medium Sales
```

EXPLANATION

- This means the model **correctly predicted** the sales category roughly **one-third** of the **time**.
- The **Kappa statistic** is **very low (0.0115)**, indicating the model performs only **slightly better** than **random guessing**.

RESULT AFTER APPLY J48 SPLIT PERCENTAGE 80:20

```
=== Run information ===

Scheme:       weka.classifiers.trees.J48 -C 0.25 -M 60
Relation:     after changing values-weka.filters.supervised.instance.Resample-B0.0-S1-Z100
Instances:    3978
Attributes:   12
              Product_Category
              Brand
              Product_Name
              Size
              Color
              Season
              Cost_Price
              Total_Sales
              Stock_Availability
              Customer_Age
              Return_Rate
              Sales_Category
Test mode:    split 80.0% train, remainder test
```

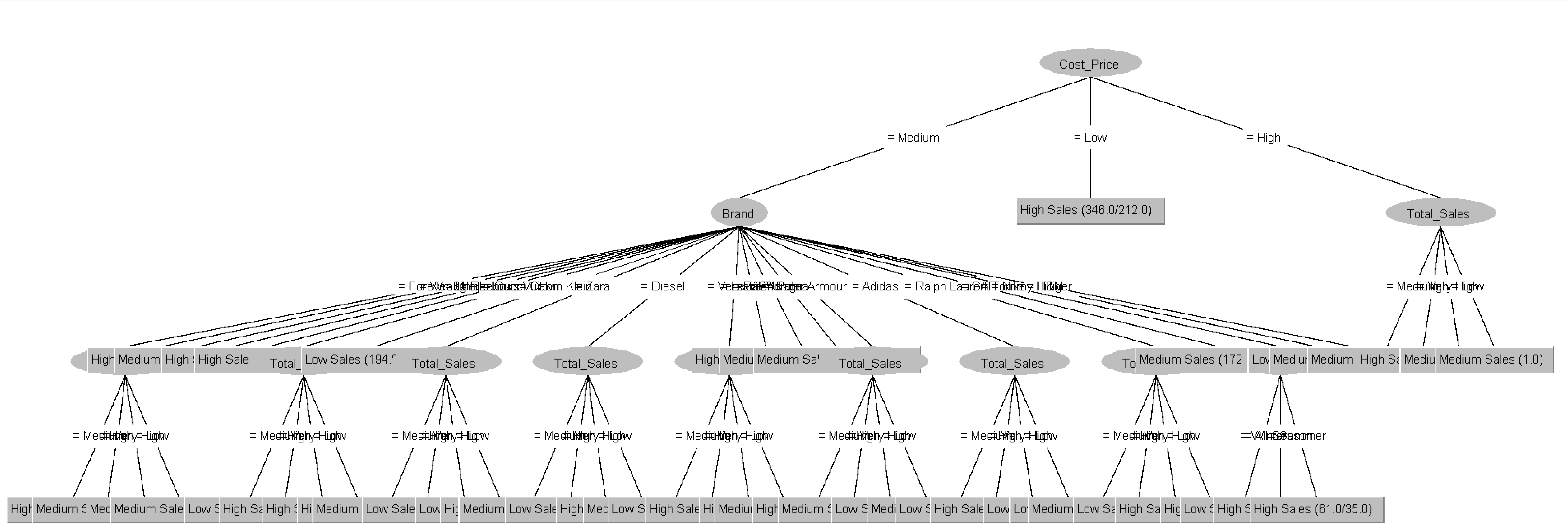
```
=== Classifier model (full training set) ===

J48 pruned tree
------------------

Cost_Price = Medium
|   Brand = Forever 21
|   |   Total_Sales = Medium: High Sales (87.0/55.0)
|   |   Total_Sales = High: Medium Sales (77.0/47.0)
|   |   Total_Sales = Very High: Medium Sales (0.0)
|   |   Total_Sales = Low: Medium Sales (4.0/2.0)
|   Brand = Wrangler: High Sales (176.0/113.0)
|   Brand = Uniqlo: Medium Sales (168.0/100.0)
|   Brand = Reebok: High Sales (130.0/80.0)
|   Brand = Gucci: High Sales (168.0/110.0)
|   Brand = Louis Vuitton
|   |   Total_Sales = Medium: Low Sales (75.0/47.0)
|   |   Total_Sales = High: High Sales (84.0/52.0)
|   |   Total_Sales = Very High: High Sales (2.0/1.0)
|   |   Total_Sales = Low: High Sales (4.0/2.0)
|   Brand = Calvin Klein: Low Sales (194.0/119.0)
|   Brand = Zara
|   |   Total_Sales = Medium: Medium Sales (110.0/64.0)
|   |   Total_Sales = High: Low Sales (61.0/36.0)
|   |   Total_Sales = Very High: Low Sales (1.0)
|   |   Total_Sales = Low: High Sales (2.0/1.0)
|   Brand = Diesel
|   |   Total_Sales = Medium: Medium Sales (82.0/53.0)
|   |   Total_Sales = High: Low Sales (67.0/40.0)
|   |   Total_Sales = Very High: High Sales (1.0)
|   |   Total_Sales = Low: Medium Sales (2.0)
|   Brand = Versace
|   |   Total_Sales = Medium: Low Sales (105.0/62.0)
|   |   Total_Sales = High: High Sales (80.0/51.0)
|   |   Total_Sales = Very High: High Sales (1.0)
|   |   Total_Sales = Low: Medium Sales (3.0/1.0)
|   Brand = Leviâ€™s: High Sales (178.0/109.0)
|   Brand = Balenciaga: Medium Sales (175.0/106.0)
|   Brand = Puma: Medium Sales (173.0/108.0)
|   Brand = Under Armour
|   |   Total_Sales = Medium: High Sales (78.0/46.0)
|   |   Total_Sales = High: Medium Sales (74.0/44.0)
|   |   Total_Sales = Very High: Low Sales (2.0/1.0)
|   |   Total_Sales = Low: Medium Sales (1.0)
|   Brand = Adidas
|   |   Total_Sales = Medium: Low Sales (89.0/53.0)
|   |   Total_Sales = High: High Sales (88.0/54.0)
|   |   Total_Sales = Very High: Low Sales (1.0)
|   |   Total_Sales = Low: Low Sales (5.0/3.0)
|   Brand = Ralph Lauren
|   |   Total_Sales = Medium: Medium Sales (81.0/49.0)
|   |   Total_Sales = High: Low Sales (74.0/47.0)
|   |   Total_Sales = Very High: High Sales (2.0/1.0)
|   |   Total_Sales = Low: High Sales (3.0)
|   Brand = GAP: Medium Sales (172.0/105.0)
|   Brand = Nike
|   |   Season = Winter: Low Sales (60.0/33.0)
|   |   Season = All-Season: High Sales (55.0/32.0)
|   |   Season = Summer: High Sales (61.0/35.0)
|   Brand = Tommy Hilfiger: Low Sales (168.0/97.0)
|   Brand = H&M: Medium Sales (189.0/110.0)
Cost_Price = Low: High Sales (346.0/212.0)
Cost_Price = High
|   Total_Sales = Medium: Medium Sales (120.0/69.0)
|   Total_Sales = High: High Sales (96.0/58.0)
|   Total_Sales = Very High: Medium Sales (2.0)
|   Total_Sales = Low: Medium Sales (1.0)

Number of Leaves  :     51

Size of the tree :      63
```

```
Time taken to build model: 0.01 seconds

=== Evaluation on test split ===

Time taken to test model on test split: 0 seconds

=== Summary ===

Correctly Classified Instances         276               34.6734 %
Incorrectly Classified Instances       520               65.3266 %
Kappa statistic                          0.0199
Mean absolute error                      0.4413
Root mean squared error                  0.4717
Relative absolute error                 99.2878 %
Root relative squared error            100.0435 %
Total Number of Instances              796

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.333    0.316    0.336      0.333    0.335      0.017    0.531     0.351     High Sales
                 0.357    0.287    0.389      0.357    0.372      0.072    0.527     0.360     Low Sales
                 0.349    0.378    0.321      0.349    0.335      -0.028   0.499     0.344     Medium Sales
Weighted Avg.    0.347    0.327    0.349      0.347    0.347      0.021    0.519     0.352

=== Confusion Matrix ===

   a   b   c   <-- classified as
  86  68 104 |   a = High Sales
  78  96  95 |   b = Low Sales
  92  83  94 |   c = Medium Sales
```

EXPLANATION

- This means the model **correctly classified** the sales category roughly **one-third** of the **time**.
- The **Kappa statistic** is **very low (0.0199),** indicating the **model** is only **slightly better** than **random guessing**.

TREE PERCENTAGE SPLIT



ACCURACY AND KAPPA COMPARISON

- From all of these test, the accuracy of the model hovered around 33% to 35%, which, in a three-class scenario, is only marginally superior to random guessing.

- In some cases kappa statistic remained very low or negative, demonstrating that the model has little predictive ability beyond chance.

- increasing the k value from 10 to 20 will decrease the performance, possibly due to over-smoothing or underfitting. Changing the train-test split from 70% to 80% only affect the marginal

- The original dataset outperformed the initial reduced dataset by a little margin. The differences between original data are not very large, but the original data constantly achieved slightly higher accuracy and kappa values, suggesting that it offers somewhat improved predictive performance for the J48 decision tree

SECOND REDUCED

J48

RESULT AFTER APPLY J48 10 FOLD

```
=== Run information ===

Scheme:       weka.classifiers.trees.J48 -C 0.25 -M 60
Relation:     after changing values-weka.filters.supervised.instance.Resample-B0.0-S1-Z100
Instances:    3978
Attributes:   12
              Gender
              Season
              Payment_Method
              Customer_Type
              Selling_Price
              Discount_Percentage
              Quantity_Sold
              Stock_Availability
              Purchase_Frequency
              Store_Rating
              Markup
              Sales_Category
Test mode:    10-fold cross-validation
```

```
=== Classifier model (full training set) ===

J48 pruned tree
------------------

Purchase_Frequency = Moderate
|   Markup = Medium
|   |   Store_Rating = Poor
|   |   |   Stock_Availability = Medium
|   |   |   |   Gender = Women: High Sales (173.0/104.0)
|   |   |   |   Gender = Men: Low Sales (147.0/91.0)
|   |   |   Stock_Availability = High: Low Sales (307.0/186.0)
|   |   |   Stock_Availability = Low: Low Sales (287.0/180.0)
|   |   Store_Rating = Average
|   |   |   Selling_Price = Medium
|   |   |   |   Stock_Availability = Medium
|   |   |   |   |   Season = Winter: Medium Sales (117.0/73.0)
|   |   |   |   |   Season = All-Season: Medium Sales (112.0/71.0)
|   |   |   |   |   Season = Summer: Low Sales (116.0/72.0)
|   |   |   |   Stock_Availability = High
|   |   |   |   |   Discount_Percentage = Moderate: Low Sales (133.0/79.0)
|   |   |   |   |   Discount_Percentage = Large: Medium Sales (119.0/62.0)
|   |   |   |   |   Discount_Percentage = Small: Low Sales (93.0/61.0)
|   |   |   |   Stock_Availability = Low
|   |   |   |   |   Season = Winter: High Sales (104.0/60.0)
|   |   |   |   |   Season = All-Season: Low Sales (78.0/50.0)
|   |   |   |   |   Season = Summer: Low Sales (99.0/56.0)
|   |   |   Selling_Price = Low: Low Sales (39.0/20.0)
|   |   |   Selling_Price = High: Medium Sales (64.0/40.0)
|   |   Store_Rating = Excellent
|   |   |   Payment_Method = Card
|   |   |   |   Season = Winter: High Sales (77.0/44.0)
|   |   |   |   Season = All-Season: High Sales (82.0/51.0)
|   |   |   |   Season = Summer: Medium Sales (66.0/35.0)
```

```
|   |   |   Payment_Method = UPI
|   |   |   |   Stock_Availability = Medium: Medium Sales (82.0/48.0)
|   |   |   |   Stock_Availability = High: High Sales (58.0/30.0)
|   |   |   |   Stock_Availability = Low: High Sales (69.0/35.0)
|   |   |   Payment_Method = Cash
|   |   |   |   Discount_Percentage = Moderate: High Sales (91.0/54.0)
|   |   |   |   Discount_Percentage = Large: Medium Sales (64.0/41.0)
|   |   |   |   Discount_Percentage = Small: Medium Sales (62.0/35.0)
|   |   |   Payment_Method = Net Banking
|   |   |   |   Season = Winter: Low Sales (72.0/38.0)
|   |   |   |   Season = All-Season: Medium Sales (75.0/46.0)
|   |   |   |   Season = Summer: Medium Sales (82.0/51.0)
|   Markup = Low
|   |   Store_Rating = Poor: High Sales (87.0/52.0)
|   |   Store_Rating = Average: High Sales (96.0/52.0)
|   |   Store_Rating = Excellent: Medium Sales (90.0/55.0)
|   Markup = High: Low Sales (178.0/109.0)
urchase_Frequency = Frequent: High Sales (171.0/102.0)
urchase_Frequency = Infrequent
|   Customer_Type = New
|   |   Stock_Availability = Medium: Low Sales (91.0/55.0)
|   |   Stock_Availability = High: High Sales (99.0/60.0)
|   |   Stock_Availability = Low: High Sales (65.0/37.0)
|   Customer_Type = Returning
|   |   Gender = Women: High Sales (109.0/66.0)
|   |   Gender = Men: Medium Sales (124.0/71.0)

umber of Leaves  :     37

ize of the tree :      56
```

```
Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        1328               33.3836 %
Incorrectly Classified Instances      2650               66.6164 %
Kappa statistic                          0.0005
Mean absolute error                      0.4437
Root mean squared error                  0.4752
Relative absolute error                 99.8351 %
Root relative squared error            100.7967 %
Total Number of Instances             3978

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.384    0.359    0.351      0.384    0.367      0.025    0.518     0.344     High Sales
                 0.339    0.339    0.333      0.339    0.336      0.000    0.512     0.338     Low Sales
                 0.278    0.302    0.313      0.278    0.294      -0.025   0.491     0.323     Medium Sales
Weighted Avg.    0.334    0.333    0.333      0.334    0.333      0.000    0.507     0.335

=== Confusion Matrix ===

   a   b   c   <-- classified as
 513 438 384 |   a = High Sales
 458 449 418 |   b = Low Sales
 491 461 366 |   c = Medium Sales
```

EXPLANATION

- The model achieved **33.38% accuracy** meaning it **correctly classified** about one-third of the instances.
- The **Kappa statistic** is near zero **(0.0005)**, which indicates performance is only **marginally better** than **random guessing**.

RESULT AFTER APPLY J48 20 FOLD

```
=== Run information ===

Scheme:       weka.classifiers.trees.J48 -C 0.25 -M 60
Relation:     after changing values-weka.filters.supervised.instance.Resample-B0.0-S1-Z1
Instances:    3978
Attributes:   12
              Gender
              Season
              Payment_Method
              Customer_Type
              Selling_Price
              Discount_Percentage
              Quantity_Sold
              Stock_Availability
              Purchase_Frequency
              Store_Rating
              Markup
              Sales_Category
Test mode:    20-fold cross-validation
```

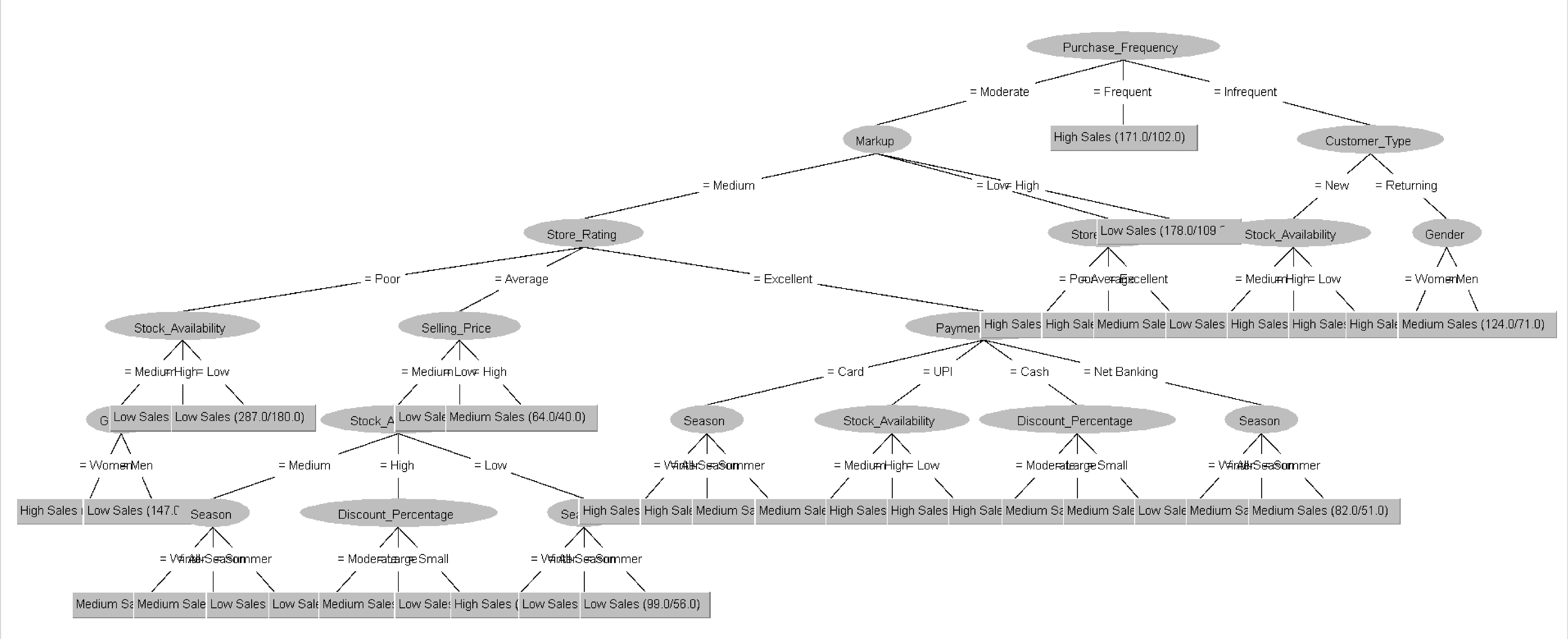
```
=== Classifier model (full training set) ===

J48 pruned tree
------------------

Purchase_Frequency = Moderate
|   Markup = Medium
|   |   Store_Rating = Poor
|   |   |   Stock_Availability = Medium
|   |   |   |   Gender = Women: High Sales (173.0/104.0)
|   |   |   |   Gender = Men: Low Sales (147.0/91.0)
|   |   |   Stock_Availability = High: Low Sales (307.0/186.0)
|   |   |   Stock_Availability = Low: Low Sales (287.0/180.0)
|   |   Store_Rating = Average
|   |   |   Selling_Price = Medium
|   |   |   |   Stock_Availability = Medium
|   |   |   |   |   Season = Winter: Medium Sales (117.0/73.0)
|   |   |   |   |   Season = All-Season: Medium Sales (112.0/71.0)
|   |   |   |   |   Season = Summer: Low Sales (116.0/72.0)
|   |   |   |   Stock_Availability = High
|   |   |   |   |   Discount_Percentage = Moderate: Low Sales (133.0/79.0)
|   |   |   |   |   Discount_Percentage = Large: Medium Sales (119.0/62.0)
|   |   |   |   |   Discount_Percentage = Small: Low Sales (93.0/61.0)
|   |   |   |   Stock_Availability = Low
|   |   |   |   |   Season = Winter: High Sales (104.0/60.0)
|   |   |   |   |   Season = All-Season: Low Sales (78.0/50.0)
|   |   |   |   |   Season = Summer: Low Sales (99.0/56.0)
|   |   |   Selling_Price = Low: Low Sales (39.0/20.0)
|   |   |   Selling_Price = High: Medium Sales (64.0/40.0)
|   |   Store_Rating = Excellent
|   |   |   Payment_Method = Card
|   |   |   |   Season = Winter: High Sales (77.0/44.0)
|   |   |   |   Season = All-Season: High Sales (82.0/51.0)
|   |   |   |   Season = Summer: Medium Sales (66.0/35.0)
|   |   |   Payment_Method = UPI
|   |   |   |   Stock_Availability = Medium: Medium Sales (82.0/48.0)
|   |   |   |   Stock_Availability = High: High Sales (58.0/30.0)
|   |   |   |   Stock_Availability = Low: High Sales (69.0/35.0)
|   |   |   Payment_Method = Cash
|   |   |   |   Discount_Percentage = Moderate: High Sales (91.0/54.0)
|   |   |   |   Discount_Percentage = Large: Medium Sales (64.0/41.0)
|   |   |   |   Discount_Percentage = Small: Medium Sales (62.0/35.0)
|   |   |   Payment_Method = Net Banking
|   |   |   |   Season = Winter: Low Sales (72.0/38.0)
|   |   |   |   Season = All-Season: Medium Sales (75.0/46.0)
|   |   |   |   Season = Summer: Medium Sales (82.0/51.0)
|   Markup = Low
|   |   Store_Rating = Poor: High Sales (87.0/52.0)
|   |   Store_Rating = Average: High Sales (96.0/52.0)
|   |   Store_Rating = Excellent: Medium Sales (90.0/55.0)
|   Markup = High: Low Sales (178.0/109.0)
Purchase_Frequency = Frequent: High Sales (171.0/102.0)
Purchase_Frequency = Infrequent
|   Customer_Type = New
|   |   Stock_Availability = Medium: Low Sales (91.0/55.0)
|   |   Stock_Availability = High: High Sales (99.0/60.0)
|   |   Stock_Availability = Low: High Sales (65.0/37.0)
|   Customer_Type = Returning
|   |   Gender = Women: High Sales (109.0/66.0)
|   |   Gender = Men: Medium Sales (124.0/71.0)

Number of Leaves  :     37

Size of the tree :      56
```

EXPLANATION

```
Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        1313               33.0065 %
Incorrectly Classified Instances      2665               66.9935 %
Kappa statistic                         -0.0051
Mean absolute error                      0.4442
Root mean squared error                  0.4756
Relative absolute error                 99.9466 %
Root relative squared error            100.8871 %
Total Number of Instances             3978

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.355    0.353    0.337      0.355    0.346      0.002    0.501     0.337     High Sales
                 0.377    0.376    0.334      0.377    0.354      0.001    0.509     0.337     Low Sales
                 0.257    0.276    0.316      0.257    0.284      -0.020   0.493     0.326     Medium Sales
Weighted Avg.    0.330    0.335    0.329      0.330    0.328      -0.006   0.501     0.334

=== Confusion Matrix ===

   a   b   c   <-- classified as
 474 502 359 |   a = High Sales
 450 500 375 |   b = Low Sales
 483 496 339 |   c = Medium Sales
```

- The model get the accuracy **33.01%,** which is **slightly low** than the **10-fold** setup.
- The **Kappa statistic** value is **-0.0051**, that mean the model performance very close or **slightly worse** than **random guessing**.

TREE CROSS VALIDATION



RESULT AFTER APPLY J48 SPLIT PERCENTAGE 70:30

```
== Run information ===

cheme:       weka.classifiers.trees.J48 -C 0.25 -M 60
elation:     after changing values-weka.filters.supervised.instance.Resample-B0.0-S1-Z100
nstances:    3978
ttributes:   12
             Gender
             Season
             Payment_Method
             Customer_Type
             Selling_Price
             Discount_Percentage
             Quantity_Sold
             Stock_Availability
             Purchase_Frequency
             Store_Rating
             Markup
             Sales_Category
est mode:    split 70.0% train, remainder test
```

```
=== Classifier model (full training set) ===

J48 pruned tree
------------------

Purchase_Frequency = Moderate
|   Markup = Medium
|   |   Store_Rating = Poor
|   |   |   Stock_Availability = Medium
|   |   |   |   Gender = Women: High Sales (173.0/104.0)
|   |   |   |   Gender = Men: Low Sales (147.0/91.0)
|   |   |   Stock_Availability = High: Low Sales (307.0/186.0)
|   |   |   Stock_Availability = Low: Low Sales (287.0/180.0)
|   |   Store_Rating = Average
|   |   |   Selling_Price = Medium
|   |   |   |   Stock_Availability = Medium
|   |   |   |   |   Season = Winter: Medium Sales (117.0/73.0)
|   |   |   |   |   Season = All-Season: Medium Sales (112.0/71.0)
|   |   |   |   |   Season = Summer: Low Sales (116.0/72.0)
|   |   |   |   Stock_Availability = High
|   |   |   |   |   Discount_Percentage = Moderate: Low Sales (133.0/79.0)
|   |   |   |   |   Discount_Percentage = Large: Medium Sales (119.0/62.0)
|   |   |   |   |   Discount_Percentage = Small: Low Sales (93.0/61.0)
|   |   |   |   Stock_Availability = Low
|   |   |   |   |   Season = Winter: High Sales (104.0/60.0)
|   |   |   |   |   Season = All-Season: Low Sales (78.0/50.0)
|   |   |   |   |   Season = Summer: Low Sales (99.0/56.0)
|   |   |   Selling_Price = Low: Low Sales (39.0/20.0)
|   |   |   Selling_Price = High: Medium Sales (64.0/40.0)
|   |   Store_Rating = Excellent
|   |   |   Payment_Method = Card
|   |   |   |   Season = Winter: High Sales (77.0/44.0)
|   |   |   |   Season = All-Season: High Sales (82.0/51.0)
|   |   |   |   Season = Summer: Medium Sales (66.0/35.0)
```

```
|   |   |   Payment_Method = UPI
|   |   |   |   Stock_Availability = Medium: Medium Sales (82.0/48.0)
|   |   |   |   Stock_Availability = High: High Sales (58.0/30.0)
|   |   |   |   Stock_Availability = Low: High Sales (69.0/35.0)
|   |   |   Payment_Method = Cash
|   |   |   |   Discount_Percentage = Moderate: High Sales (91.0/54.0)
|   |   |   |   Discount_Percentage = Large: Medium Sales (64.0/41.0)
|   |   |   |   Discount_Percentage = Small: Medium Sales (62.0/35.0)
|   |   |   Payment_Method = Net Banking
|   |   |   |   Season = Winter: Low Sales (72.0/38.0)
|   |   |   |   Season = All-Season: Medium Sales (75.0/46.0)
|   |   |   |   Season = Summer: Medium Sales (82.0/51.0)
|   Markup = Low
|   |   Store_Rating = Poor: High Sales (87.0/52.0)
|   |   Store_Rating = Average: High Sales (96.0/52.0)
|   |   Store_Rating = Excellent: Medium Sales (90.0/55.0)
|   Markup = High: Low Sales (178.0/109.0)
Purchase_Frequency = Frequent: High Sales (171.0/102.0)
Purchase_Frequency = Infrequent
|   Customer_Type = New
|   |   Stock_Availability = Medium: Low Sales (91.0/55.0)
|   |   Stock_Availability = High: High Sales (99.0/60.0)
|   |   Stock_Availability = Low: High Sales (65.0/37.0)
|   Customer_Type = Returning
|   |   Gender = Women: High Sales (109.0/66.0)
|   |   Gender = Men: Medium Sales (124.0/71.0)

Number of Leaves  :     37

Size of the tree :      56
```

```
Time taken to build model: 0.01 seconds

=== Evaluation on test split ===

Time taken to test model on test split: 0 seconds

=== Summary ===

Correctly Classified Instances         382               32.0201 %
Incorrectly Classified Instances       811               67.9799 %
Kappa statistic                          -0.0171
Mean absolute error                       0.4459
Root mean squared error                   0.4778
Relative absolute error                 100.2858 %
Root relative squared error             101.3076 %
Total Number of Instances              1193

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.433    0.397    0.337      0.433    0.379      0.034    0.527     0.340     High Sales
                 0.255    0.320    0.301      0.255    0.276      -0.068   0.464     0.327     Low Sales
                 0.281    0.301    0.316      0.281    0.298      -0.020   0.485     0.317     Medium Sales
Weighted Avg.    0.320    0.338    0.318      0.320    0.316      -0.020   0.491     0.328

=== Confusion Matrix ===

   a   b   c   <-- classified as
 164 122  93 |   a = High Sales
 165 107 147 |   b = Low Sales
 158 126 111 |   c = Medium Sales
```

EXPLANATION

- The model classified the instance **1 in 3 instances correctly,** which is low.
- The **negative value** give a meaning that model performed **slighlty worse** than **random guessing**.
- worse overall accuracy

RESULT AFTER APPLY J48 SPLIT PERCENTAGE 80:20

```
=== Run information ===

Scheme:       weka.classifiers.trees.J48 -C 0.25 -M 60
Relation:     after changing values-weka.filters.supervised.instance.Resample-B0.0-S1-Z100
Instances:    3978
Attributes:   12
              Gender
              Season
              Payment_Method
              Customer_Type
              Selling_Price
              Discount_Percentage
              Quantity_Sold
              Stock_Availability
              Purchase_Frequency
              Store_Rating
              Markup
              Sales_Category
Test mode:    split 80.0% train, remainder test
```

```
=== Classifier model (full training set) ===

J48 pruned tree
------------------

Purchase_Frequency = Moderate
|   Markup = Medium
|   |   Store_Rating = Poor
|   |   |   Stock_Availability = Medium
|   |   |   |   Gender = Women: High Sales (173.0/104.0)
|   |   |   |   Gender = Men: Low Sales (147.0/91.0)
|   |   |   Stock_Availability = High: Low Sales (307.0/186.0)
|   |   |   Stock_Availability = Low: Low Sales (287.0/180.0)
|   |   Store_Rating = Average
|   |   |   Selling_Price = Medium
|   |   |   |   Stock_Availability = Medium
|   |   |   |   |   Season = Winter: Medium Sales (117.0/73.0)
|   |   |   |   |   Season = All-Season: Medium Sales (112.0/71.0)
|   |   |   |   |   Season = Summer: Low Sales (116.0/72.0)
|   |   |   |   Stock_Availability = High
|   |   |   |   |   Discount_Percentage = Moderate: Low Sales (133.0/79.0)
|   |   |   |   |   Discount_Percentage = Large: Medium Sales (119.0/62.0)
|   |   |   |   |   Discount_Percentage = Small: Low Sales (93.0/61.0)
|   |   |   |   Stock_Availability = Low
|   |   |   |   |   Season = Winter: High Sales (104.0/60.0)
|   |   |   |   |   Season = All-Season: Low Sales (78.0/50.0)
|   |   |   |   |   Season = Summer: Low Sales (99.0/56.0)
|   |   |   Selling_Price = Low: Low Sales (39.0/20.0)
|   |   |   Selling_Price = High: Medium Sales (64.0/40.0)
|   |   Store_Rating = Excellent
|   |   |   Payment_Method = Card
|   |   |   |   Season = Winter: High Sales (77.0/44.0)
|   |   |   |   Season = All-Season: High Sales (82.0/51.0)
|   |   |   |   Season = Summer: Medium Sales (66.0/35.0)
|   |   |   Payment_Method = UPI
|   |   |   |   Stock_Availability = Medium: Medium Sales (82.0/48.0)
|   |   |   |   Stock_Availability = High: High Sales (58.0/30.0)
|   |   |   |   Stock_Availability = Low: High Sales (69.0/35.0)
|   |   |   Payment_Method = Cash
|   |   |   |   Discount_Percentage = Moderate: High Sales (91.0/54.0)
|   |   |   |   Discount_Percentage = Large: Medium Sales (64.0/41.0)
|   |   |   |   Discount_Percentage = Small: Medium Sales (62.0/35.0)
|   |   |   Payment_Method = Net Banking
|   |   |   |   Season = Winter: Low Sales (72.0/38.0)
|   |   |   |   Season = All-Season: Medium Sales (75.0/46.0)
|   |   |   |   Season = Summer: Medium Sales (82.0/51.0)
|   Markup = Low
|   |   Store_Rating = Poor: High Sales (87.0/52.0)
|   |   Store_Rating = Average: High Sales (96.0/52.0)
|   |   Store_Rating = Excellent: Medium Sales (90.0/55.0)
|   Markup = High: Low Sales (178.0/109.0)
Purchase_Frequency = Frequent: High Sales (171.0/102.0)
Purchase_Frequency = Infrequent
|   Customer_Type = New
|   |   Stock_Availability = Medium: Low Sales (91.0/55.0)
|   |   Stock_Availability = High: High Sales (99.0/60.0)
|   |   Stock_Availability = Low: High Sales (65.0/37.0)
|   Customer_Type = Returning
|   |   Gender = Women: High Sales (109.0/66.0)
|   |   Gender = Men: Medium Sales (124.0/71.0)

Number of Leaves  :     37

Size of the tree :      56
```

```
Time taken to build model: 0.01 seconds

=== Evaluation on test split ===

Time taken to test model on test split: 0 seconds

=== Summary ===

Correctly Classified Instances         275               34.5477 %
Incorrectly Classified Instances       521               65.4523 %
Kappa statistic                          0.0177
Mean absolute error                      0.4442
Root mean squared error                  0.4754
Relative absolute error                 99.9444 %
Root relative squared error            100.8308 %
Total Number of Instances              796

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.337    0.297    0.352      0.337    0.345      0.040    0.507     0.334     High Sales
                 0.323    0.304    0.352      0.323    0.337      0.020    0.515     0.356     Low Sales
                 0.375    0.381    0.334      0.375    0.354      -0.006   0.473     0.327     Medium Sales
Weighted Avg.    0.345    0.328    0.346      0.345    0.345      0.018    0.498     0.339

=== Confusion Matrix ===

   a   b   c   <-- classified as
  87  73  98 |   a = High Sales
  79  87 103 |   b = Low Sales
  81  87 101 |   c = Medium Sales
```

EXPLANATION

- The model classified the **highest** among the **second reduced dataset correctly**.
- The **Kappa statistic** is **near** to **zero** which denoting performance is only **marginally better** than **random guessing**.

ACCURACY AND KAPPA COMPARISON

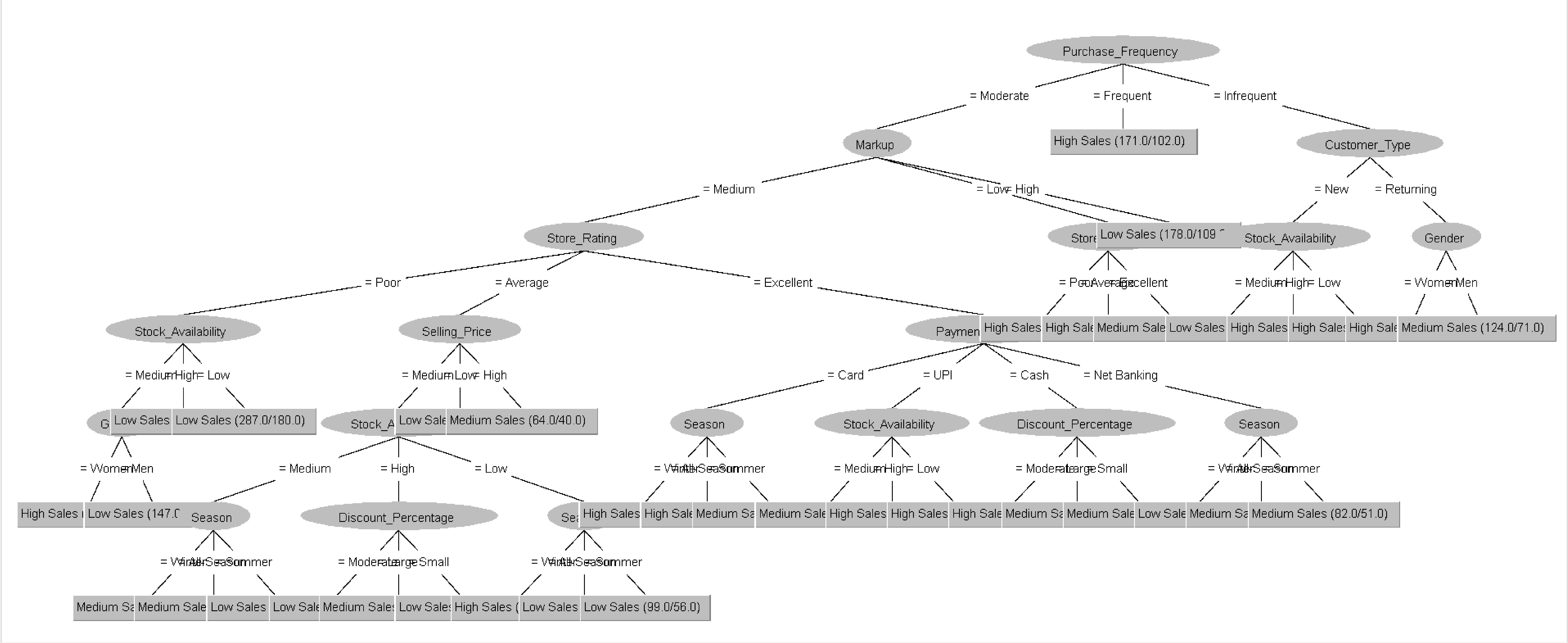
- The classification performance on the second reduced dataset normally show the accuracy between 32% to 34.5%, which remains only slightly better than random guessing in a three-class problem

- The Kappa statistic mainly hovered near to zero or were slightly negative, it means the model ability to predict beyond chance was minimal, and goes same to the original dataset but generally weaker.

- By comparing different values of K=10 and K=20 in KNN, the increasing K to 20 is typically lowered or did not improve the performance. It suggesting over-smoothing or underfitting, constantly with observations on the original dataset.

- In some conditions, the second reduced dataset did not perform as well as the first and original reduced dataset.

TREE PERCENTAGE SPLIT



ALL TREE COMPARISON

BEFORE REDUCE

- The tree result of using all dataset has been process and resemple still generate a complicated tree due to noisy and unrelevant features

AFTER REDUCE (DATASET 1)

- The tree result generated by using 11 best attributes results in the tree still likely being oversized and overfit but improved a little.

AFTER REDUCE (DATASET 2)

- The tree result generated by using 11 worst attributes results in the tree still likely being a better version of the tree because it doesn’t include a lot of nominal data.

.

ALL COMPARISON EVALUATION RESULT

J48	fold=10	fold=20	70:30	80:20
Before reduced	35.3444	34.2634	35.2054	34.9246
First reduced	34.3891	32.0261	33.8642	34.6734
Second reduced	33.3836	33.0065	32.0201	34.5477


J48 Correctly Classified Instances
36
35
34
33
32
31
30
fold=10
fold=20
percentage split=70
percentage split=80
Before reduced
First reduced
Second reduced

OVERALL ACCURACY AND KAPPA COMPARISON

- The most important discovery is that the accuracy of the model was **continuously decreased** by the feature reduction procedure. On the original, "Before reduced" dataset, the **J48 classifier** outperformed the other **four assessment techniques**. This implies that important information necessary for the model to provide precise **predictions** was present in the **features eliminated** during the **First** and **Second reduction rounds**.

- The **"Before reduced"** dataset, which was assessed using **10-fold cross-validation**, gave the **highest** overall accuracy **(35.3444%)**. This suggests that for this particular problem, this combination **worked well**.

- Except for the **20-fold test**, the **second reduced dataset** often did **worse** than the **first reduced dataset**. This suggests that the **predictive power** of the model **decreased** even more when **more and more** important information was eliminated in each subsequent reduction stage.

- The **worst performance** was shown with the **second reduced dataset**, which had a **70:30 percentage split** and an **accuracy** of only **32.0201%.**

Conclusion

In conclusion, our dataset, the J48, cannot predict the outcome clearly. Besides, the data doesn't have a stronger correlation between sales attributes.